Scalar inversion and geodesic counts by winding on SE(2)

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September 2026

Abstract

We count normal geodesics of the standard sub-Riemannian problem on SE(2), retaining those that continue past their cut time and distinguishing them by heading winding. The rotating endpoint map admits an exact scalar inverse: the modulus and amplitudes are algebraic functions of one coordinate, and an elliptic level equation selects the actual periods. At strict headings every rotating fiber is finite, with locally bounded cardinality even across the target seam. An exact separatrix surface partitions the zero-winding fiber into one oscillating source, one separatrix source, or a finite rotating fiber; the last has odd cardinality at regular targets. Fixed-heading convexity gives sharp radial existence intervals, and attained angular thresholds classify the exterior rays supporting each positive period. At $|\tan(\theta/2)| = 3/4$, a computer-assisted proof identifies a unique first threshold fold and an explicit radial count window with ordered post-cut travel times. A separate target has exactly nine regular rotating sources. Along a specified target segment the complete count is 7/8/9/10/11, with two ordinary fold values and strict time ordering. For general $V > 1, W > 0$, an analytical comparison orders momenta within a signed period family and clocks across a nondegenerate connected sublevel component. Under simultaneous endpoint and effective-period scaling, an explicit limiting modulus determines eventual counts and clock offsets for two selected families. On the critical surface a separate expansion classifies the counts unless both leading coefficients vanish. Certified examples realize all four nondegenerate outcomes and show that joint initial existence need not imply persistence. On the half-heading ray the rotating count is exact at every positive radius and grows as the inverse square of a vanishing heading, with an explicit coefficient and bounded remainder. At fixed strict heading, the zero-winding count along a spatial ray has three large-distance regimes, proportional to R, to $R/\log R$, or equal to one. Every nonzero winding class has exactly one oscillating source at every radius, so the full fiber is countably infinite. Headings zero and π have separate exact catalogues, including an infinite rotating fiber on the nonzero vertical axis. The minimizing conclusions use the known optimal synthesis. General eventual scale thresholds remain unquantified.

Keywords: sub-Riemannian geometry, group of motions of the plane, exponential map, geodesic counting, Jacobi elliptic functions.
MSC 2020: 53C17, 49K15, 33E05.

1 Problem, scope, and main statement

We declare the left-invariant vector fields

$$X_1 = \cos\theta\,\partial_x + \sin\theta\,\partial_y, \qquad X_2 = \partial_\theta$$

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orthonormal, and start trajectories at $(0,0,0)$. Thus $\dot{x} = u_1 \cos\theta$, $\dot{y} = u_1 \sin\theta$, $\dot{\theta} = u_2$, with unit-speed normalization $u_1^2 + u_2^2 = 1$. The normal Hamiltonian system reduces to a pendulum. Its rotating stratum is denoted by C_2, following [1, 2]. A source means an initial normal covector of Hamiltonian energy $1/2$ together with a positive travel time; C_2 is specified when restricting to rotating sources. We count distinct sources, without identifying reflection pairs or discarding trajectories after their cut time. This convention is also equivalent to counting the corresponding constant-speed parametrized normal geodesics on a fixed time interval.

Throughout, elliptic integrals use the *parameter* $m = k^2$:

$$K(m) = \int_0^{\pi/2} \frac{d\phi}{\sqrt{1 - m\sin^2\phi}}, \quad E(m) = \int_0^{\pi/2} \sqrt{1 - m\sin^2\phi}\, d\phi.$$

We write $L(m) = 4(K(m) - E(m))$ for the longitudinal drift per period. The corresponding incomplete integrals are $F(\phi \mid m)$ and $E(\phi \mid m)$, continued as real integrals in ϕ. Set $\mathbf{e}(a) = (\sin a, -\cos a)$ and $\mathbf{e}_\perp(a) = (\cos a, \sin a)$.

Theorem 1.1 (Far-field rotating count)**.** *Fix $a_0, b_0 \in (-\pi/2, \pi/2)$ with $\theta = a_0 - b_0 \neq 0$, and put*

$$q_R = (R\mathbf{e}(a_0), \theta), \qquad m_* = \max(\sin^2 a_0, \sin^2 b_0), \qquad L_* = L(m_*).$$

There is $R_0 = R_0(a_0, b_0)$ such that, for $R > R_0$, the number $\nu_2(q_R)$ of positive-time C_2 sources is finite and

$$\nu_2(q_R) = \frac{4R}{L_*} + O(1), \qquad R \longrightarrow \infty. \tag{1}$$

The same statement holds for $(-R\mathbf{e}(a_0), \theta)$ by spatial negation. The constants depend on the fixed angles. No effective value of R_0 or uniform estimate toward the excluded angle boundaries is asserted.

The proof distinguishes $\sin^2 a_0 \neq \sin^2 b_0$ from $b_0 = -a_0$. In the first case an exact eventual floor formula is given in Theorem 3.3. In the second, omitting the zero-transverse fiber would lose half of the coefficient in (1). The finite quantity in Theorem 1.1 is the rotating count. The next theorem describes its place in the full fiber. Neither theorem includes $\theta = 0$.

Theorem 1.2 (Normal fibers by heading winding)**.** *For the angles and targets of Theorem 1.1, for all sufficiently large R, the full positive-time normal fiber has the following form. With continuous heading $\widetilde{\theta}(0) = 0$, define $\ell \in \mathbb{Z}$ by $\widetilde{\theta}(t) = \theta + 2\pi\ell$. The class $\ell = 0$ consists precisely of the finitely many C_2 sources, with count (1). Each class $\ell \neq 0$ contains exactly one source, belonging to C_1. There are no C_3, C_4, C_5 sources. The full fiber is countably infinite; every C_1 source is regular. Finiteness of the class $\ell = 0$ and the description of the classes $\ell \neq 0$ hold in fact at every radius $R > 0$ (Corollary 4.13, Propositions 4.14 and 6.2); the count (1) and the exclusion of C_3 are the eventual clauses.*

If $a_0 + b_0 \neq 0$, the unique minimizing source is strictly pre-cut and belongs to actual C_2 cell zero. If $a_0 + b_0 = 0$, exactly two sources minimize, both in actual C_2 cell zero at their first Maxwell time. The same conclusions hold for spatially negated targets.

The minimizing conclusions use the published optimal synthesis. The proof in Section 6 includes the oscillating covering and seam arguments and the exclusions of the remaining regimes. The eventual threshold for the combined theorem remains non-effective; regularity of every rotating source is not asserted.

Theorem 1.3 (All fixed spatial directions at strict heading)**.** *Fix a unit vector* $\mathbf{e} = (e_x, e_y)$ *and* $0 < |\theta| < \pi$*. For* $q_R = (R\mathbf{e}, \theta)$ *put*

$$\delta_0 = e_y, \qquad \delta_1 = e_y \cos\theta - e_x \sin\theta, \qquad \Delta = \delta_0 \delta_1.$$

For sufficiently large R*, the zero-winding fiber is*

condition	*stratum*	*number of sources*
$\Delta > 0$	C_2	$4R/L_* + O(1)$,
$\Delta = 0$	C_2	$R/[2(\log(4R/\lvert\sin\theta\rvert) - 1)] + O(1)$,
$\Delta < 0$	C_1	1,

where $L_* = L(1 - \min(\delta_0^2, \delta_1^2))$ *in the first row. In every row there are no* C_3, C_4, C_5 *sources eventually. Exactly one source minimizes, except on the interior line* $e_x \cos(\theta/2) + e_y \sin(\theta/2) = 0$*, where exactly two minimize. The interior and boundary thresholds are non-effective. For the exterior, an explicit sufficient threshold after spatial sign normalization is* (108)*.*

For every $X \neq 0$ *and* $0 < |\theta| < \pi$*, every nonzero winding class contains exactly one source, in* C_1*, and it is regular. This last assertion holds at every radius, including horizontal directions.*

The interior statement is Theorem 1.2; the nonhorizontal boundary and exterior proofs are in Sections 7 and 8. Section 9 gives the horizontal completion and the invariant sign formulation. Proposition 6.2 proves the all-radius winding assertion. The remainder constants and thresholds in these fixed-angle statements are not uniform when the angles vary with R. When the terminal angle approaches the rotating boundary together with the radius, the count passes through exponentially small transitions; that regime is treated in a separate note in preparation [13].

The remaining headings have exact catalogues at every radius: Section 10 treats heading π and pure orientation, and Section 11 treats heading zero, including its coordinate axes. These exact exceptional-heading results are separate from the eventual strict-heading table. The identity has no positive-time winding-zero source.

Relation to established results. The elliptic endpoint parametrization and reflection symmetries are standard inputs from Moiseev–Sachkov [1, Sections 3–5]. Sachkov's conjugate-time analysis [2, Section 2] concerns local optimality; the complete conjugate-time set of the rotating geodesics is determined in [12]. The optimal synthesis [3, Theorems 3.1–3.2] gives one or two minimizing trajectories on its stated target strata; its source domain is restricted by the cut time. It therefore does not count all positive-time sources considered here. Duits–Boscain–Rossi–Sachkov [4, Theorems 5–6 and 11] already reduce the cuspless boundary-value problem to a scalar zero, using spatial arclength and a source domain ending at the first cusp. That inverse concerns their cuspless domain, whereas the present components retain all positive-time rotating sources, including post-cut cells. Scalar reduction itself is therefore an established method. Lerario–Rizzi [5, Example 2, equation (2), and Theorems 2, 7–9] derive quantitative counts for contact Carnot groups and lower bounds under local nilpotent approximation. In particular, the Heisenberg count already has inverse-square growth in the horizontal displacement at fixed nonzero vertical coordinate. The exponent two in Theorem 4.11 thus has an established model precedent. Here the limit is taken in the fixed SE(2) geometry at fixed positive radius, and the theorem gives its explicit radius-dependent coefficient and bounded remainder. This limit, and the large-distance limit in (1), differ from their local nilpotent scaling. Wang–Ku–Bravo-Doddoli [6, Theorems A–B] classify periodic geodesics and metric lines. Those statements

address recurrence and global minimizing behavior, rather than counts of all sources of a specified endpoint.

The contributions of this paper are the complete post-cut scalar components of the rotating inverse (Proposition 2.1) and their global eventual control (Theorems 3.3 and 7.3), the seam and directional completion (Sections 4 and 9), the fixed-cell scale (Proposition 5.1), and the exact exceptional-heading counts (Sections 10 and 11). They also include finiteness of every nonzero strict-heading rotating fiber, uniformly on compact target families (Corollaries 2.5 and 4.13, Proposition 4.14), the all-radius half-heading ray count (Theorem 4.7), and its small-heading asymptotic (Theorem 4.11). The exact rotating image and odd regular-fiber parity are established in Theorem 6.6. The degree method is classical [9, Section 4]; the all-period image barrier and the properness needed for this application are proved in Section 6.1. Corollary 6.9 combines that barrier with the classical synthesis [3] to classify the entire zero-winding fiber by pendulum regime at every radius; it asserts no new optimal synthesis or exact arbitrary-target rotating count. Section 6.2 derives strict fixed-heading convexity and the resulting sharp radial existence classification from the same hyperbolic graph. The half-heading onset is already in [3, (2.13), Lemma 2.3]; these geometric consequences are distinguished from the all-time source-count results. Section 6.3 excludes all positive rotating periods in an explicit exterior sector, giving an exact count there by the classical pre-cut uniqueness theorem. Proposition 6.14 uses a complete interval certificate to count three regular rotating sources at a wider exterior target, including two post-cut sources. Theorem 6.15 characterizes the sharp angular support of each positive period by an attained minimum, with period one determining the all-radius uniqueness sector. It identifies conjugacy at the threshold without asserting a fold/cusp type or an explicit value of the minimum. These post-cut support statements use the classical endpoint formulas and minimizing synthesis as inputs. Corollary 6.16 encloses the first threshold at $w = 3/4$ in an interval of width $3 \cdot 10^{-15}$ using a complete interval certificate and separate coverage audit. Theorem 6.17 identifies a unique attaining fold for this heading and its local radial count window. Corollary 6.18 supplies an explicit slope range and certified radial endpoints at its upper cap. Theorem 6.19 then orders the two post-cut clocks throughout that window, using classical first variation and a separate momentum-sign certificate. Propositions 6.20 and 6.21 give a formal period-action identity and a complete nine-source rotating fiber with strictly ordered period groups, including a certified connected comparison between periods one and two. Proposition 6.22 continues that count along an explicit target segment with exactly two ordinary fold values. Corollary 6.24 orders all its zero-winding travel times throughout that segment. Theorem 6.23 supplies a general analytical within-family momentum comparison for $V > 1, W > 0$ and orders the clocks across a nondegenerate connected sublevel component. Neither result compares arbitrary branches in different signed period families. Section 6.9 studies simultaneous endpoint and period scaling. Its uniform source limit and strict-threshold theorem give eventual counts and clock offsets for two selected signed families; a critical-value expansion treats the equality surface except where both unfolding coefficients vanish. Certified examples show multiple critical transitions and failure of joint initial existence to guarantee persistence. The source formulas and elliptic derivative identities are classical [1, 2, 7]; the scaling is a formal period interpolation, not the nilpotent approximation used in the contact counting results of [5]. No new special-function identity is claimed. The heading-zero and heading-π source conditions, and the intervals containing the roots of f_1, are established in [1, Propositions 5.3–5.6 and Lemma 5.3]. The minimizing zero-heading branch and heading-π radial equation occur in [3, Lemmas 2.9–2.14]. The additional arguments here invert the modulus at a fixed target, order positive-cell branches, identify overlaps and count all-time sources. Source reversal and the Hamiltonian equations used for pure orientation are also known

inputs. In particular, the all-period Maxwell roots and their ray condition are already in [1, Lemma 5.3, Proposition 5.6 and Theorem 5.2]. The first ray threshold is Sachkov's curve $R_1^1(\theta)$, and its pair of minimizing sources is part of the optimal synthesis [3, Lemma 2.5, Table 2 and Theorems 3.1–3.3]. Theorem 4.7 uses those known Maxwell branches to invert the radius in every actual period and count the full rotating fiber. Corollary 4.8 identifies the established minimizing part inside this all-time count.

Notation. The following symbols recur; each is also introduced where it first enters.

- $\mathrm{Exp}(\lambda, t)$ is the exponential map, the endpoint at time t of the normal geodesic with initial covector λ; a source is a pair (λ, t) with $t > 0$. A source is *regular* if the differential of Exp is nonsingular there, that is, t is not a conjugate time. The cut time of a source is t_{cut} [3]; a source is strictly pre-cut if $t < t_{\mathrm{cut}}$ and post-cut if $t > t_{\mathrm{cut}}$.
- $\nu_2(q)$ is the number of positive-time C_2 sources of q, and $N_0(q)$ the number of sources of q, in all pendulum strata, whose heading winding (Theorem 1.2) is zero. By Corollary 6.9, $N_0 = \nu_2$ throughout the exact rotating image and $N_0 = 1$ on and outside its separatrix boundary.
- The *cell* of a rotating source is its period index $n = \lfloor t/(4kK) \rfloor$ in (4); "actual cell" stresses this integer as opposed to a chart label. For oscillating sources the cell is the integer n in $p = t/2 = 2nK + r$, $|r| < K$ (Lemma 6.1). In Section 11 a rotating source carries the label n of $p = t/(2k) = 2nK + F(A \mid k^2)$ (Lemma 11.3), which is its period index at the central sources $p = 2nK$.
- $s_1 = \operatorname{sgn}\cos(\gamma/2)$ and $s_2 = \operatorname{sgn} c$ are the sign parameters of [1, 2] in the pendulum coordinates (γ, c) of Section 6; the sign ε of the rotating decomposition (3) corresponds to s_2.
- $f_1(p) = \operatorname{cn} p\,(\mathcal{E}(p) - p) - \operatorname{dn} p \operatorname{sn} p$ is the function of [1, (5.12)]; its first positive root p_1 (written p_1^1 in Section 11, where p_n^1 is the n-th root) gives the rotating cut time $2kp_1$ [3]. It first enters in the proof of Theorem 1.2.
- $a_X \in (-\pi/2, \pi/2)$ is the angle with $X = R\,\mathbf{e}(a_X)$, where defined (Proposition 2.3); for the targets of Theorem 1.1 it is a_0.
- η is the scaled cell index n/R in the Remark following Proposition 2.3, and the amplitude $\operatorname{am}\tau$ of the midpoint phase τ in Sections 6 and 11.
- Travel times are t, or T where they are the unknown of a discrete equation: $T = |\Theta_\ell|$ in Lemma 10.1; $T_n(R) = 2(2n+1)K$ in Theorem 10.2 and Proposition 10.3, and $T = 4nK$ in Theorem 11.1, are the travel times of the oscillating sources of windings $n, -n-1$, respectively $\pm n$; in the proof of Theorem 11.1, $T_n = 4nkK$ is the travel time of the central rotating pair of cell n on the vertical axis.

2 Exact endpoint decomposition and scalar inverse

Write a rotating source as $(\psi, k, t, \varepsilon)$, where $0 < k < 1$, $\psi \in \mathbb{R}/(4K(k^2)\mathbb{Z})$, $t > 0$, and $\varepsilon \in \{1, -1\}$. With $v = \psi + t/k$, define

$$a = \arcsin(k \operatorname{sn}(\psi \mid m)), \quad b = \arcsin(k \operatorname{sn}(v \mid m)), \quad m = k^2. \tag{2}$$

Both angles lie strictly between $-\pi/2$ and $\pi/2$. The published rotating endpoint formulas [1, Section 3.3] give, by regrouping their two spatial components,

$$\begin{gathered} \theta = a - b, \qquad X = \varepsilon\{A\mathbf{e}(a) + B\mathbf{e}_\perp(a)\}, \\ A = \int_\psi^v m \operatorname{sn}^2(u \mid m)\, du > 0, \qquad B = k(\operatorname{cn}\psi - \operatorname{cn} v). \end{gathered} \tag{3}$$

Here and below Jacobi functions without a displayed parameter use m. Indeed $\operatorname{dn}\psi = \cos a$, $k\operatorname{sn}\psi = \sin a$, and the longitudinal integral equals $v - \psi - \mathcal{E}(v) + \mathcal{E}(\psi)$, where $\mathcal{E}$ is Jacobi's epsilon function, $\mathcal{E}'(u) = \operatorname{dn}^2 u$ [7, Section 22.16]. This verifies the decomposition directly. The heading difference is in $(-\pi, \pi)$ and represents the group heading. In particular

$$|B| \le 2k < 2, \qquad A = nL(m) + Q_0, \quad 0 \le Q_0 < L(m), \quad n = \left\lfloor \frac{t}{4kK(m)} \right\rfloor. \tag{4}$$

For nonzero heading, $0 < Q_0 < L(m)$. The phase circle here is $4K$, even when scalar conjugate-point expressions have a $2K$ symmetry [12]. Changing ε negates X at unchanged heading.

Proposition 2.1 (Scalar inverse off the seam)**.** *Fix $X = (x, y) \neq 0$ and $0 < |\theta| < \pi$. For the sign $\varepsilon = 1$, put $z = \tan a$ and define*

$$\begin{aligned} H &= 1 + z^2, & C &= x + yz, & D &= z\cos\theta - \sin\theta, \\ d &= D^2 - z^2, & U &= C^2 + d, & A &= \frac{xz - y}{\sqrt{H}}, \quad B = \frac{C}{\sqrt{H}}, \\ f &= \frac{U}{2C\sqrt{H}}, & g &= \frac{d - C^2}{2C\sqrt{H}}, & m &= \frac{z^2 + (U/(2C))^2}{H}. \end{aligned} \tag{5}$$

Its admissible domain is exactly

$$C \neq 0, \quad \cos\theta + z\sin\theta > 0, \quad xz - y > 0, \quad 0 < m < 1. \tag{6}$$

At such a point let

$$\begin{aligned} \phi_0 &= \operatorname{atan2}(z/\sqrt{H}, f) \pmod{2\pi}, & \phi_1 &= \operatorname{atan2}(D/\sqrt{H}, g) \pmod{2\pi}, \\ \psi &= F(\phi_0 \mid m), & v_0 &= F(\phi_1 \mid m), \end{aligned}$$

using representatives in $[0, 2\pi)$ for the amplitudes. Set

$$\begin{aligned} w &= \mathbf{1}_{\{v_0 < \psi\}}, \qquad \delta = v_0 - \psi + 4K(m)w, \\ Q_0 &= \delta - E(\phi_1 \mid m) + E(\phi_0 \mid m) - 4E(m)w, \\ N(z) &= \frac{A - Q_0}{L(m)}. \end{aligned} \tag{7}$$

Every intersection $N(z) = n \in \mathbb{Z}_{\ge 0}$ gives exactly one source

$$(\psi, \sqrt{m}, \sqrt{m}\,\delta + 4n\sqrt{m}K(m), 1).$$

These exhaust the $B \neq 0$ sources of that sign. For $\varepsilon = -1$, apply the same construction to $-X$.

Proof. The transverse and heading equations imply $f - g = B$ and $f^2 - g^2 = \sin^2(a - \theta) - \sin^2 a$ for $f = k\operatorname{cn}\psi$, $g = k\operatorname{cn} v$. Solving gives (5). Conversely those formulas satisfy $f^2 + \sin^2 a = g^2 + \sin^2(a - \theta) = m$. The second inequality in (6) says $\cos(a - \theta) > 0$ and selects the required principal terminal angle. Thus both signed sine–cosine pairs determine unique phases modulo $4K$. Nonzero heading excludes coincident phases, so $0 < \delta < 4K$ and $0 < Q_0 < L$. The remaining longitudinal equation is exactly $N = n$. Its reconstructed time is positive and has the stated actual period index. No division by either cosine occurred, so their zeros are included. Local phase lifts remove the artificial discontinuities of the chosen amplitude representatives. □

The admissible domain also has a useful algebraic description:

$$m - 1 = \frac{(U - 2C)(U + 2C)}{4C^2 H}. \tag{8}$$

Both numerator factors are quadratic in z. Their roots and the linear boundaries in (6) partition the domain exactly. The condition $m = 0$ cannot occur under the other hypotheses: it would force $z = 0$ and $x^2 + \sin^2\theta = 0$.

Proposition 2.2 (Regularity and folds in this chart)**.** *At a scalar integer intersection with $B \neq 0$, the full exponential map is regular if $N_z \neq 0$. If $N_z = 0$, it has corank one and is a fold if and only if $N_{zz} \neq 0$.*

Proof. The reconstruction in Proposition 2.1, now with independent variables (a, B, θ), gives local analytic source coordinates. Write $Z = nL + Q_0$. In the oriented spatial basis $(\mathbf{e}, \mathbf{e}_\perp)$, the determinant of $X = Z\mathbf{e} + B\mathbf{e}_\perp$ at fixed θ is

$$J = Z_a - B - ZZ_B.$$

The B column is $Z_B\mathbf{e}+\mathbf{e}_\perp$, and the heading column has last component one, so the full rank is at least two. For a fixed target put $F(a) = A(a) - Z(a, B(a), \theta)$. Since $A' = B$, $B' = -A$, at a root $J = -F' = -L(1 + z^2)N_z$. At a stationary root the kernel is $V = \partial_a - Z\partial_B$; differentiating the displayed determinant gives $VJ = -F'' = -L(1 + z^2)^2 N_{zz}$ there. The corank-one fold criterion is precisely nonvanishing of this kernel derivative. This argument does not extend the chart to $B = 0$. □

The decomposition already bounds where sources can lie.

Proposition 2.3 (Exact cell bounds)**.** *Let $R = |X| > 2$, and let $n \geq 1$ be the period count of a rotating source of (X, θ) with $\varepsilon = 1$. Then*

$$\frac{\sqrt{R^2 - 4}}{n + 1} \leq L(m) \leq \frac{R}{n}. \tag{9}$$

If moreover $X = R\mathbf{e}(a_X)$ with $a_X \in (-\pi/2, \pi/2)$ and $\varepsilon_R = \arcsin(2/R)$, then

$$|a - a_X| \leq \varepsilon_R, \qquad k \geq k_* - \varepsilon_R, \qquad k_* = \max\{|\sin a_X|, |\sin(a_X - \theta)|\}, \tag{10}$$

so that, when $k_* > \varepsilon_R$, only the cells $n \leq R/L\big((k_* - \varepsilon_R)^2\big)$ can contain sources. For $\varepsilon = -1$ replace X by $-X$.

Proof. Orthogonality of $\mathbf{e}(a)$ and $\mathbf{e}_\perp(a)$ in (3) gives $R^2 = A^2 + B^2$, and (4) gives $nL \leq A < (n + 1)L$, $|B| < 2$; this is (9). Scalar products with $\mathbf{e}(a)$ and $\mathbf{e}_\perp(a)$ give $A = R\cos(a - a_X) > 0$ and $B = R\sin(a_X - a)$; both angles lie in $(-\pi/2, \pi/2)$ and the cosine is positive, so $|a - a_X| < \pi/2$ and $|B| < 2$ gives the first bound in (10). Since $k \geq |\sin a|$, $k \geq |\sin b|$ and $b = a - \theta$, the Lipschitz bound of the sine gives the second; the cell bound follows from $nL \leq A \leq R$ and $L' > 0$. □

Proposition 2.4 (Finite positive-period fibers)**.** *Let*

$$\Omega = \{(x, y, \theta) : (x, y) \neq 0, \quad 0 < |\theta| < \pi, \quad x\cos(\theta/2) + y\sin(\theta/2) \neq 0\}.$$

For every target in Ω, the number of rotating sources with actual period index $n \geq 1$, across both rotation signs, is finite. These numbers are uniformly bounded on every compact subset of Ω. Every such source satisfies the explicit index restriction

$$1 \leq n < \frac{|X|}{L(\sin^2(\theta/2))}. \tag{11}$$

In particular, no such source exists if $|X| \le L(\sin^2(\theta/2))$. *The uniform root-count bound is not effective. These assertions do not include period index zero or targets on the excluded seam.*

Proof. Work first with $\varepsilon = 1$ and $a = \arctan z$ on the admissible open set of Proposition 2.1; put $b = a - \theta$ and $R = |X|$. The circle equations give

$$m \ge \max\{\sin^2 a, \sin^2 b\} \ge \sin^2(\theta/2) =: m_h > 0. \tag{12}$$

Indeed, at least one of $|a|, |b|$ is at least $|\theta|/2$, and both angles are principal. At a point with $N \ge \ell > 0$, $0 < Q_0 < L$ gives

$$\ell L(m) \le A - Q_0 < A \le R, \qquad m < L^{-1}(R/\ell) < 1, \qquad A > \ell L(m_h) > 0. \tag{13}$$

Here L is strictly increasing from zero to infinity. In particular, (11) follows at $N = n \ge 1$.

The positive superlevel $N \ge \ell$ on any maximal admissible component is compact. To see this, take a sequence in that superlevel and a limit of its angles in the closure of the principal-angle interval. An endpoint with $|a| = \pi/2$ or $|b| = \pi/2$ would force $m \to 1$, contrary to (13). The same bounds exclude $A \to 0$ and $m \to 0, 1$. Finally the exact circle identity

$$\sin^2 a - \sin^2 b = -B(f + g) = \sin\theta \sin(2a - \theta)$$

excludes $B \to 0$: the boundedness of f, g would force $a = \theta/2$, since $2a - \theta \in (-\pi, \pi)$ even at the principal interval's endpoints for this fixed strict heading. The limiting equation $B = 0$ would then be the excluded target seam. Thus the limit remains admissible and, by continuity, in the superlevel. It lies in the same component, since components are relatively closed in the admissible set.

For completeness, N is jointly real analytic in (a, x, y, θ) on this open set, including the apparent amplitude cuts and the zeros of f or g. The nonzero unit-circle pairs $(\sin a, f)/\sqrt{m}$ and $(\sin b, g)/\sqrt{m}$ admit analytic local angle lifts. Choose them with $0 < \phi_1 - \phi_0 < 2\pi$; equality modulo 2π would imply $a = b$, contrary to the nonzero heading. Then

$$Q_0 = \int_{\phi_0}^{\phi_1} \frac{m \sin^2 u}{\sqrt{1 - m \sin^2 u}}\, du$$

is jointly analytic, as follows by changing to a fixed integration interval. Simultaneous 2π shifts of the lifts leave it unchanged. This also proves the continuity used above without imposing artificial phase cuts on the domain.

On a maximal component, N cannot be identically a positive constant: the corresponding superlevel would be the whole nonempty open interval, contradicting compactness. The analytic identity theorem therefore makes each positive level set discrete, and compactness makes it finite. There are finitely many components by (8) and the linear domain boundaries. The index bound leaves finitely many positive integers. Proposition 2.1 gives exactly one source per integer intersection, with residual time in $(0, 4kK)$ and actual index n. A source with $B = 0$ would lie on the excluded target seam by the same circle identity. Replacing X by $-X$ covers the other rotation sign and preserves all these restrictions.

To prove uniformity, let $\mathcal{Q} \subset \Omega$ be nonempty and compact, set $R_{\max} = \max_{\mathcal{Q}} |X|$ and $m_* = \min_{\mathcal{Q}} \sin^2(\theta/2) > 0$. At every positive integer intersection,

$$n < R_{\max}/L(m_*), \qquad m_* \le m < L^{-1}(R_{\max}) < 1, \qquad A > L(m_*) > 0.$$

The preceding boundary argument, now allowing the target to converge inside $\mathcal{Q}$, proves compactness of the joint root set in (a, q, n, ε). At each of its points some finite derivative

$\partial_a^j(N-n)$, $j \geq 1$, is nonzero; otherwise analyticity would make N identically n on that component. Joint analyticity keeps this derivative nonzero on a product neighborhood in angle and target. Repeated Rolle's theorem bounds the number of roots there for each fixed target by j. A finite cover of the compact root set supplies a uniform bound by summing these orders. This argument assumes neither simple roots nor a sign predicate for critical points. □

Corollary 2.5 (Period zero and the full rotating fiber)**.** *For every $q \in \Omega$ of Proposition 2.4, the entire rotating fiber is finite, including actual period index zero. Its cardinality is uniformly bounded on each compact subset of Ω. Every period-zero source satisfies*

$$|X| \leq t < |X| + 2\pi. \tag{14}$$

The finiteness bound is not effective and gives no exact count.

Proof. Only period zero remains after Proposition 2.4. For such a source $t = k\delta$ and

$$\delta = A + \int_{\psi}^{\psi+\delta} \mathrm{dn}^2\, u\, du, \qquad 0 < \delta < 4K.$$

The last integral is strictly between zero and $4E(m) \leq 2\pi$. Since $A \leq |X|$ and $k < 1$, this gives the upper bound in (14); the lower bound follows from unit speed.

Use the original normalized initial covector (γ, c), with γ taken modulo 4π. Its pendulum energy obeys

$$\mathsf{E} = c^2/2 - \cos\gamma = 2/m - 1, \qquad |c| \leq 2/\sqrt{m} \leq 2/|\sin(\theta/2)|$$

by (12). Thus all period-zero sources of a fixed target lie in a compact set of initial covectors and positive times after allowing the boundary $\mathsf{E} = 1$. The original Hamiltonian flow, unlike elliptic phase representatives at $m = 1$, is analytic there. Every limiting source of energy one belongs to $C_3 \cup C_5$ and is a regular point of the full exponential map at positive time [2, Theorem 2.4]. The inverse function theorem therefore excludes an accumulation of distinct sources of the same target at that boundary.

The scalar function N cannot be identically zero on a maximal admissible component. Otherwise take angles approaching an endpoint of that open component and the corresponding distinct period-zero sources. The bounds just proved supply a convergent subsequence in the original source coordinates. If its limit has energy greater than one, Proposition 2.1 applies at that source: the target is off the seam, both angles are strict principal angles, and the limiting angle would be in the same admissible component, contrary to its being an endpoint. An energy-one limit contradicts the preceding local injectivity. Hence $N \not\equiv 0$ on every component. Analyticity and the exact inverse now make each period-zero source isolated. Infinitely many such sources would accumulate in the compact set above, either at an isolated rotating source or at a regular energy-one source; both alternatives are impossible.

Finally let the target range over a compact subset of Ω. The time and covector bounds are uniform, with a positive lower time bound, so the closure of the joint period-zero source set is compact in the original coordinates. A rotating limit still has period zero: the only possible upper period boundary would have $t = 4kK$ and heading zero. At such a rotating limit, $N \not\equiv 0$ supplies a nonzero finite angle derivative. The product-neighborhood and Rolle argument in Proposition 2.4 bounds the nearby roots uniformly. At an energy-one limit the inverse function theorem gives at most one source per nearby target. A finite cover of the compact closure yields a uniform bound for period zero. Adding the positive-period bound proves the assertion for the full rotating fiber. □

Remark 2.6. The bounds are necessary conditions, not a count. On the cells with n/R in a compact subinterval of $(0, 1/L_*)$ the exact system, divided by R, is a regular perturbation of $\eta\, L(m)\mathbf{e}(a) = \mathbf{e}(a_X)$, $\eta = n/R$, $\theta = a - b$, whose four solutions (two initial and two terminal cosine signs) continue by the implicit function theorem, and a compactness argument excludes others: exactly four regular sources per bulk cell. Theorem 3.3 below gives the complete eventual count, of which this is the local mechanism; the compactness argument is that of Theorem 3.3 localised to one cell.

3 The generic ray: global control of the scalar graph

Assume $d_0 = \sin^2 b_0 - \sin^2 a_0 \neq 0$. For large R, the sign $\varepsilon = -1$ is impossible: (3) would give $y = A\cos a - B\sin a > -2$, whereas q_R has $y = -R\cos a_0 < -2$. For the other sign, $|B| < 2$, $A > 0$ uniquely determine

$$\begin{aligned} &a = a_0 - \arcsin(B/R), \quad b = a - \theta, \quad A = \sqrt{R^2 - B^2}, \\ &d = \sin^2 b - \sin^2 a, \qquad f = \frac{B^2 + d}{2B}, \quad g = \frac{d - B^2}{2B}, \quad m_R(B) = \sin^2 a + f^2. \end{aligned} \tag{15}$$

All angles remain principal uniformly for $|B| < 2$ and large R. Moreover $d \to d_0$ uniformly, so the condition $m_R < 1$ excludes a fixed neighborhood of $B = 0$. Proposition 2.1 gives $N_R(B) = (A - Q_0)/L(m_R(B))$ and the exact equation $N_R = n$.

Lemma 3.1 (Domain and limiting profile)**.** *For sufficiently large R, the admissible B domain consists of two intervals. Their four endpoints, indexed by cosine signs $\sigma_0, \sigma_1 \in \{1, -1\}$, are*

$$B_{\sigma_0\sigma_1}(R) = \frac{Ru}{\sqrt{(R - v)^2 + u^2}}, \qquad u = \sigma_0\cos a_0 - \sigma_1\cos b_0, \quad v = \sigma_0\sin a_0 - \sigma_1\sin b_0. \tag{16}$$

On compact subsets of their limiting interiors,

$$\frac{N_R(B)}{R} \longrightarrow \frac{1}{L(m_0(B))} \quad \text{in } C^2, \qquad m_0(B) = \frac{B^2 + 2(\sin^2 a_0 + \sin^2 b_0) + d_0^2/B^2}{4}. \tag{17}$$

Each limiting component has one nondegenerate maximum of this profile, at $B_*^{\pm} = \pm\sqrt{|d_0|}$, with value $1/L_*$.

Proof. At $m = 1$, $f = \sigma_0\cos a$ and $g = \sigma_1\cos b$. The equation $B = f - g$ becomes $B(1 - v/R) = u\sqrt{1 - B^2/R^2}$. For $R > 2$ the first factor is positive, so its unique solution with the prescribed sign is (16). Substitution proves sufficiency. The four limits are $\pm(\cos a_0 + \cos b_0)$ and $\pm|\cos a_0 - \cos b_0|$, distinct by $d_0 \neq 0$. They are all roots of $m_R = 1$. The limiting sign chart, the excluded neighborhood of zero, and $|B| < 2$ therefore give exactly two components. The modulus is positive because $m_R = 0$ would force $a = b = 0$, contrary to the nonzero heading.

Direct differentiation gives

$$m_0' = \frac{B}{2} - \frac{d_0^2}{2B^3}, \qquad m_0'' = \frac{1}{2} + \frac{3d_0^2}{2B^4}.$$

Its two minima have value m_* and curvature 2. The standard derivative identity [7, Section 19.4] is

$$L_m(m) = \frac{2E(m)}{1 - m} > 0. \tag{18}$$

Hence the profile curvature at either maximum is $-2L_m(m_*)/L_*^2$. On compact subsets the reconstructed phases have analytic local lifts, even when a cosine vanishes. Nonzero heading keeps the residual time away from both period boundaries. Thus Q_0 and its first two derivatives remain bounded there, proving (17). □

Compact convergence alone cannot exclude extra critical points close to $m = 1$. The next estimate supplies this missing endpoint control. Define the signed short integral

$$q(s,m) = \int_0^s \frac{u^2\,du}{\sqrt{1-u^2}\sqrt{m-u^2}}, \qquad |s| < \sqrt{m}. \tag{19}$$

It is odd in s. Near a domain endpoint, both cosine signs are fixed and

$$\begin{aligned} Q_0 &= cL + \sigma_1 q(\sin b, m) - \sigma_0 q(\sin a, m), \\ P_R &= A + \sigma_0 q(\sin a, m) - \sigma_1 q(\sin b, m), \qquad N_R = \frac{P_R}{L} - c, \end{aligned} \tag{20}$$

where $c = 1/2$ for opposite signs and $c = \mathbf{1}_{\{\sigma_0\theta>0\}}$ for equal signs. To check the period coefficient, a phase of cosine sign σ has a lift $(1-\sigma)K + \sigma F(\arcsin(s/\sqrt{m}) \mid m)$. Equal signs need a full wrap exactly when $\sigma_0\theta > 0$; opposite signs contribute one half-period. Subtracting the incomplete second-kind integral proves (20) for either sign of s.

Lemma 3.2 (Uniform endpoint monotonicity)**.** *There are neighborhoods of the four moving endpoints and R_1 such that for every $R > R_1$ and every interior point of those neighborhoods,* $\operatorname{sgn} N_R' = -\operatorname{sgn} m_R'$. *The endpoint limits of N_R are $-c$ and are approached from above.*

Proof. The limiting endpoint roots are simple and the fixed angles are strict. We can therefore choose the neighborhoods uniformly so that $|\sin a|, |\sin b| \le s_{\max} < 1$, $m \ge m_c > s_{\max}^2$, and $|m_R'| \ge \gamma > 0$. Differentiation under (19) bounds its needed derivatives uniformly up to $m = 1$. Formula (15) consequently gives $P_R = R + O(1)$ and $|P_R'| \le C$ throughout these tails. By the usual complete-integral asymptotics [7, Section 19.12], L/L_m extends continuously to zero at $m = 1$ and is bounded, say by H. For sufficiently large R, with $P_R \ge R - C_0$,

$$|P_R'|\frac{L}{L_m} \le CH < (R - C_0)\gamma \le P_R|m_R'|.$$

The identity $N_R' = (P_R'L - P_R L_m m_R')/L^2$ proves the sign. Finally $P_R > 0$ is finite as $m \to 1$ at fixed R, while $L \to \infty$. Thus $N_R \to -c$ from above. □

Theorem 3.3 (Exact eventual generic count)**.** *When $d_0 \neq 0$, for sufficiently large R each admissible component has exactly one critical point, a nondegenerate maximum. If the heights are $M_-(R)$ and $M_+(R)$, then*

$$\nu_2(q_R) = 2\lfloor M_- \rfloor + 2\lfloor M_+ \rfloor + 3 - \mathbf{1}_{\{M_-\in\mathbb{Z}\}} - \mathbf{1}_{\{M_+\in\mathbb{Z}\}}. \tag{21}$$

Moreover $M_\pm = R/L_* + O(1)$. Every source is regular except an integer-maximum source, which is a fold.

Proof. Near either profile maximum in Lemma 3.1, strict negative second derivative and the implicit function theorem give exactly one nearby maximum. On the remaining compact interior, C^1 convergence excludes critical points. Lemma 3.2 excludes them on all four tails, exhausting the moving domain. Analytic dependence on $1/R$ near either maximum gives $M_\pm = R/L_* + O(1)$, so both heights are positive for large R.

The two opposite-cosine endpoints are the outer endpoints; each has level limit $-1/2$. The equal-cosine endpoints are the inner ones, with respective limits 0 and -1. Thus one

component has limits $0, -1/2$, and the other $-1, -1/2$. On the first, level zero has one root; on the second it has two. Every positive integer below a maximum has two roots in that component. At an integer maximum those two roots coalesce to one. Strict monotonicity on each arm proves (21). There is no $B = 0$ source: that condition in (3) would force $b_0 = -a_0$. The other sign was already excluded, so the count is complete. Proposition 2.2 applies because $dz/dB = -(1+z^2)/\sqrt{R^2 - B^2} \neq 0$. Equation (1) follows in this case. □

Figure 1 shows the level function N of (7), in the chart $z = \tan a$ of Proposition 2.1, for one target of this kind.

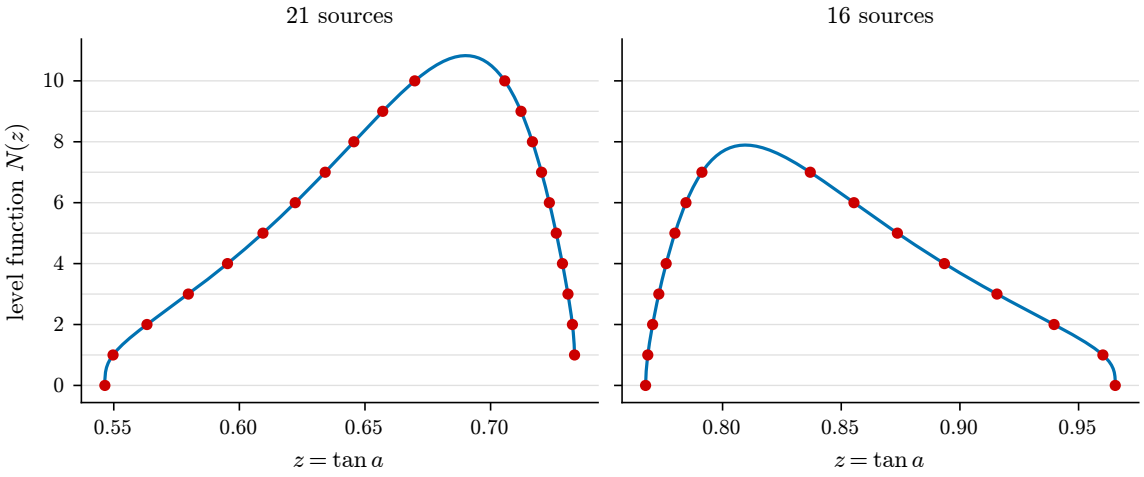


Figure 1: The level function $N(z)$ of (7) for the target $(39/5, -52/5, 3/10)$, that is $R = 13$ and $\theta = 3/10$, on its two admissible components, against $z = \tan a$. The horizontal lines are the integer levels and the dots the 37 integer crossings (21 and 16), one rotating source each, in agreement with (21). When a maximum crosses an integer level as the target varies, the corresponding sources merge in a fold (Proposition 2.2); a noninteger critical level alone does not represent a source of this target.

4 The half-heading ray and its separate seam

The only remaining case is $b_0 = -a_0$. First take $\alpha = a_0 \in (0, \pi/2)$ and abbreviate $s = \sin\alpha$, $c_\alpha = \cos\alpha$, $L_* = L(s^2)$. For $B \neq 0$, the exact formulas (15) reduce to

$$\begin{aligned} a &= \alpha - \arcsin(B/R), & b &= -\alpha - \arcsin(B/R), \\ f &= \frac{B}{2} + \frac{\sin(2\alpha)\sqrt{R^2 - B^2}}{R^2}, & g &= -\frac{B}{2} + \frac{\sin(2\alpha)\sqrt{R^2 - B^2}}{R^2}, \\ m_R(B) &= m_c + D_R B^2, & m_c &= s^2 + \frac{4s^2c_\alpha^2}{R^2}, \\ D_R &= \frac{(R^2 - 4s^2)(R^2 + 4c_\alpha^2)}{4R^4}. \end{aligned} \tag{22}$$

For $R > 2s$, this formula has admissible interval

$$|B| < B_{\max} = \frac{2Rc_\alpha}{\sqrt{R^2 + 4c_\alpha^2}}. \tag{23}$$

Indeed the largest possible absolute angle is less than $\alpha + \arctan(2c_\alpha/R) < \pi/2$, so no further angle boundary occurs.

Proposition 4.1 (Nonzero transverse sources)**.** *For sufficiently large R, the extended scalar N_R on (23) is even and has exactly one critical point, a nondegenerate maximum at zero. If $M_0 = N_R(0)$, its nonzero-B sources number $2\max(0, \lceil M_0 \rceil)$, and*

$$M_0 = \frac{R}{L(m_c)} - 1 + \frac{2q_c}{L(m_c)} = \frac{R}{L_*} - \frac{1}{2} + O(R^{-1}),$$

where m_c is the modulus selected at $B = 0$, $h_c = 2sc_\alpha/R$ and $q_c = F(\arctan(s/h_c) \mid m_c) - E(\arctan(s/h_c) \mid m_c)$; the counts use the exact expression, whose $O(R^{-1})$ term is not small at moderate R. The extension at zero selects only modulus m_c; it does not represent the full $B = 0$ fiber. The nonzero-B count formula extends to every $R > 2s$ by Theorem 4.7 below; the unimodality statement here remains restricted to sufficiently large R.

Proof. Replacing B by $-B$ sends (a, b, f, g) to $(-b, -a, g, f)$. The phases become $(-v, -\psi)$ modulo $4K$, preserving the positive residual time and the integral of the even function sn^2. Hence Q_0, m_R, and N_R are even. Their phase parametrization extends analytically at zero: its sine entries are nonzero and the residual time stays strictly between zero and $4K$. Let $h_c = 2sc_\alpha/R$. At fixed finite R the value at zero is the long arc $4K - 2u_c$, where $u_c = F(\arctan(s/h_c) \mid m_c) < K$; this approaches $2K$ as $R \to \infty$. The normalized profile converges in C^2 on compact interiors to $1/L(s^2 + B^2/4)$, whose unique maximum has curvature $-L_m(s^2)/(2L_*^2)$. The compact argument of Theorem 3.3 and Lemma 3.2 apply; the two endpoint signs are opposite and their limits are $-1/2$. Evenness places the sole maximum exactly at zero.

Let $q_c = F(\arctan(s/h_c) \mid m_c) - E(\arctan(s/h_c) \mid m_c)$. At zero the residual is the long arc, so $M_0 = R/L(m_c) - 1 + 2q_c/L(m_c)$. Here $m_c = s^2 + O(R^{-2})$ and $q_c = L_*/4 + O(R^{-1})$, which proves the height estimate. Every nonnegative integer strictly below the height contributes a pair. An integer maximum is excluded because it lies at $B = 0$, to be counted separately below. □

Proposition 4.2 (Complete zero-transverse fiber)**.** *Set $Q = \mathrm{atanh}(s) - s$. For each integer $n \geq 1$ there is a unique $m_n \in (s^2, 1)$ minimizing $R_n^{\mathrm{short}}(m) = nL(m) + 2q(s, m)$. Write its value as S_n. Then the number of $B = 0$ sources is exactly*

$$\mathbf{1}_{\{R > 2Q\}} + 2\#\{n \geq 1 : S_n < R\} + \#\{n \geq 1 : S_n = R\}, \tag{24}$$

and it equals $2R/L_* + O(1)$. In particular it is finite for every fixed positive R. The thresholds satisfy

$$nL_* + 2Q < S_n < (n + 1/2)L_*, \qquad S_{n+1} > S_n + L_*. \tag{25}$$

Proof. At $B = 0$, the equations force $a = \alpha$, $b = -\alpha$, and $f = g$. For $m \in [s^2, 1)$, put $u = F(\arcsin(s/\sqrt{m}) \mid m)$. The initial phases are $2K - u$ and u, with terminal phase $-\psi$. Their radii in actual period cell $n \geq 0$ are exactly

$$R_n^{\mathrm{short}}(m) = nL(m) + 2q(s, m), \qquad R_n^{\mathrm{long}}(m) = (n + 1)L(m) - 2q(s, m). \tag{26}$$

At $m = s^2$ the two phases coincide and both radii are $H_n = (n + 1/2)L_*$. This source must be counted once.

For $m > s^2$, differentiation under the short integral gives

$$q_m = -\frac{1}{2}\int_0^s \frac{u^2\, du}{\sqrt{1 - u^2}(m - u^2)^{3/2}} < 0.$$

Consequently every long branch is strictly increasing from H_n to infinity. The short branch for $n = 0$ decreases from H_0 to $2Q$, the latter limit being unattained. After deduplication at H_0 their union contributes exactly $\mathbf{1}_{\{R>2Q\}}$.

For $n \geq 1$ define

$$\mathcal{H}(m) = -\frac{2q_m}{L_m} = \frac{1-m}{2E(m)} \int_0^s \frac{u^2\, du}{\sqrt{1-u^2}(m-u^2)^{3/2}}.$$

It decreases strictly from infinity to zero. To see the derivative sign, its integral is positive and strictly decreasing, and its positive prefactor is strictly decreasing because

$$E + (1-m)E_m = \frac{(m+1)E - (1-m)K}{2m} > 0.$$

The last inequality follows directly from $E-(1-m)K = m\int_0^{\pi/2} \cos^2\phi/\sqrt{1-m\sin^2\phi}\, d\phi > 0$. The lower-endpoint divergence of the integral and its finite limit at $m = 1$ prove the asserted limits of $\mathcal{H}$. Thus $(R_n^{\text{short}})' = L_m(n - \mathcal{H})$ has exactly one zero, with second derivative $-L_m\mathcal{H}' > 0$.

The short branch decreases from H_n to S_n, then increases to infinity. Combining it with the long branch gives zero, one, or two distinct sources according as $R < S_n$, $R = S_n$, or $R > S_n$. At $R = H_n$, its lower-endpoint source is shared by the two branches; the other short-branch root remains distinct, giving two in total. This proves (24). Since $L(m) > L_*$ and $q(s, m) > Q$, the lower bound in (25) follows; the initial decrease gives the upper bound. Evaluating $R_n^{\text{short}} + L$ at the next minimizer gives $S_{n+1} > S_n + L_*$. These bounds prove finiteness and $\#\{n \geq 1 : S_n < R\} = R/L_* + O(1)$. □

Proposition 4.3 (Full-map singularities away from the branch join)**.** *Fix $0 < \alpha < \pi/2$ and $m \in (s^2, 1)$ in (26), and put $h = \sqrt{m - s^2}$ and $c_\alpha = \cos\alpha$. On a short branch with $n \geq 1$, the unique minimum S_n is a corank-one fold of the full exponential map, and all other points are regular. The short branch with $n = 0$ is regular throughout. Each long branch, including $n = 0$, has exactly one singular point, characterized by*

$$R_n^{\text{long}}(m) = \frac{2sc_\alpha}{\sqrt{m - s^2}}. \tag{27}$$

It has corank one and is a cusp, as proved in Corollary 4.4. The common branch endpoint $m = s^2$ is regular by Proposition 4.5.

Proof. Use local source coordinates (a, θ, m) with $b = a - \theta$. They are valid when the signed cosines f, g are nonzero: in the original phase and time variables, $a_\psi = f$ and $b_t = g/k$. Write $\sigma = -1$ on the short branch and $\sigma = 1$ on the long branch. Near the source in question,

$$f = \sigma\sqrt{m - \sin^2 a}, \quad g = \sigma\sqrt{m - \sin^2 b}, \quad B = f - g,$$
$$A = \left(n + \frac{1+\sigma}{2}\right) L - \sigma\{q(\sin a, m) - q(\sin b, m)\}.$$

Here the defining integral for q is extended to negative upper limits, and all angles remain principal. At fixed θ, the endpoint $X = A\mathbf{e}(a) + B\mathbf{e}^\perp(a)$ has planar Jacobian

$$J = (A_a - B)B_m - (A + B_a)A_m.$$

At the seam $a = \alpha$, $b = -\alpha$ one has $A_a = B = B_m = 0$ and $B_a = -2\sigma sc_\alpha/h$. Thus

$$J = -HA_m, \qquad H = A - \frac{2\sigma sc_\alpha}{h}, \qquad X_a = H\mathbf{e}^\perp(\alpha), \quad X_m = A_m\mathbf{e}(\alpha). \tag{28}$$

Since the third output is θ, these formulas also determine the rank of the full map.

For a short branch $H > 0$. Proposition 4.2 gives $A_m = 0$ exactly at its nondegenerate minimum when $n \geq 1$; there the kernel is ∂_m and $\partial_m J = -HA_{mm} \neq 0$. This is the corank-one fold criterion, equivalently the parameter-dependent Morse lemma after solving for the transverse output. All remaining short points have $J \neq 0$, including the entire $n = 0$ branch.

For a long branch $A_m > 0$ and $H_m = A_m + sc_\alpha/h^3 > 0$. Moreover H tends to $-\infty$ as $m \downarrow s^2$ and to $+\infty$ as $m \uparrow 1$. This proves existence and uniqueness in (27). At that point the full rank is two, with kernel ∂_a. To evaluate the fold criterion, put $u = a - \theta/2$ at fixed $\theta = 2\alpha$. Reflection $u \mapsto -u$ makes A even and B odd. The components of X in the fixed basis at α are consequently $U = A\cos u - B\sin u$ even and $V = A\sin u + B\cos u$ odd. Their Jacobian $U_u V_m - V_u U_m$ is even, so $\partial_a J = 0$ at $u = 0$. The fold criterion therefore fails, although $J_m = -H_m A_m \neq 0$ and the critical set is smooth there. □

Corollary 4.4 (Long-branch cusps)**.** *Every long-branch singularity in Proposition 4.3 is an ordinary corank-one cusp of the full exponential map, including period $n = 0$.*

Proof. Extend U, V to target coordinates using the basis at $\theta/2$, and write $u = a - \theta/2$. Since $U_m = A_m > 0$ at the singularity, solve locally for $m = m(u, U, \theta)$ and put $W(u, U, \theta) = V(u, m(u, U, \theta), \theta)$. The full map is then $(u, U, \theta) \mapsto (W, U, \theta)$. At the singularity $W_u = 0$, and reflection gives $W_{uu} = 0$, while

$$W_{uU} = \frac{H_m}{A_m} > 0, \qquad \mathcal{C} := W_{uuu} = V_{uuu} - 3\frac{V_{um}U_{uu}}{A_m}.$$

All derivatives here are evaluated at the singular source. Differentiating the local endpoint formulas and using $A = 2sc_\alpha/h$ gives

$$U_{uu} = -\frac{2sc_\alpha m}{h^3}, \qquad V_{uuu} = -\frac{6sc_\alpha^3 m}{h^5}, \qquad V_{um} = A_m + \frac{sc_\alpha}{h^3}.$$

Consequently

$$\mathcal{C} = \frac{6sc_\alpha m}{h^6 A_m}\{sc_\alpha - (1-m)hA_m\}. \tag{29}$$

To determine its sign, define

$$I = \int_0^\alpha \frac{\sin^2 v}{(m - \sin^2 v)^{3/2}}\, dv, \qquad \mathcal{E}_\alpha = \int_0^\alpha \frac{\cos^2 v}{\sqrt{m - \sin^2 v}}\, dv.$$

The long-branch formula gives $A_m = 2(n+1)E(m)/(1-m) + I$. Direct differentiation shows

$$\frac{d}{dv}\frac{\sin v \cos v}{\sqrt{m - \sin^2 v}} = \frac{(1-m)\sin^2 v}{(m - \sin^2 v)^{3/2}} + \frac{\cos^2 v}{\sqrt{m - \sin^2 v}}.$$

Integration therefore yields $sc_\alpha/h = (1-m)I + \mathcal{E}_\alpha$. The substitution $\sin v = \sqrt{m}\sin\phi$ gives $\mathcal{E}_\alpha = E(\arcsin(s/\sqrt{m}) \mid m)$, so $0 < \mathcal{E}_\alpha < E(m)$. Thus

$$\mathcal{C} = \frac{6sc_\alpha m}{h^5 A_m}\{\mathcal{E}_\alpha - 2(n+1)E(m)\} < 0.$$

For $\lambda = W_u$ and the kernel extension ∂_u, one has $\lambda = \lambda_u = 0$, $\lambda_{uu} = \mathcal{C} \neq 0$. Moreover (λ, λ_u) has rank two, since its derivatives in the U, u directions have determinant $W_{uU}\mathcal{C} \neq 0$. The cusp criterion [11, Theorem A.1] applies to this three-dimensional map. □

Proposition 4.5 (Regular joins and local seam counts)**.** *At the common endpoint $m = s^2$ of each pair of branches (26), the full exponential map is regular. In the original coordinates (γ_0, c_0, t) for the positive rotation sign, its Jacobian determinant there is $m^2 > 0$.*

Fix α and vary the target radius R. In a sufficiently small source neighborhood of a short-branch minimum S_n, $n \geq 1$, the numbers of sources are $0, 1, 2$ according as $R < S_n$, $R = S_n$, or $R > S_n$. At the long-branch cusp of period $n \geq 0$, write its radius as R_n^{cusp}. The corresponding local counts are $1, 1, 3$ according as $R < R_n^{\mathrm{cusp}}$, $R = R_n^{\mathrm{cusp}}$, or $R > R_n^{\mathrm{cusp}}$. The extra two sources on the last side have $B \neq 0$. Near a common branch endpoint, the local count remains one. These statements concern radii sufficiently close to the respective threshold.

Proof. Use the initial and terminal pendulum amplitudes ϕ_0, ϕ_1 and m as source coordinates. At the join,

$$\phi_0 = \pi/2, \quad \phi_1 = 3\pi/2 + 2\pi n, \quad k = \sqrt{m} = s, \quad k' = \sqrt{1-m} = c_\alpha.$$

In a neighborhood of this source the defining-integral endpoint is

$$a = \arcsin(k \sin\phi_0), \quad b = \arcsin(k \sin\phi_1), \quad B = k(\cos\phi_0 - \cos\phi_1),$$
$$A = \int_{\phi_0}^{\phi_1} \frac{m \sin^2 v}{\sqrt{1 - m\sin^2 v}}\, dv, \qquad X = A\mathbf{e}(a) + B\mathbf{e}^{\perp}(a), \quad \theta = a - b.$$

These expressions are analytic, including at the join. There, $a_{\phi_0} = b_{\phi_1} = 0$, and differentiation gives

$$X_{\phi_0} = -\frac{m}{k'}\mathbf{e}(\alpha) - k\mathbf{e}^{\perp}(\alpha), \quad X_{\phi_1} = \frac{m}{k'}\mathbf{e}(\alpha) - k\mathbf{e}^{\perp}(\alpha),$$
$$\theta_{\phi_0} = \theta_{\phi_1} = 0, \qquad \theta_m = \frac{1}{kk'}.$$

Hence $\det \partial(X, \theta)/\partial(\phi_0, \phi_1, m) = 2m/(1-m)$. The conversion to the original source coordinates is

$$\gamma_0 = 2\phi_0, \quad c_0 = \frac{2}{k}\sqrt{1 - m\sin^2\phi_0}, \quad t = k\int_{\phi_0}^{\phi_1} \frac{dv}{\sqrt{1 - m\sin^2 v}},$$

with γ_0 taken locally modulo 4π. Its determinant at the join is $2/[m(1-m)]$: indeed $(c_0)_m = -1/(k^3k')$ and $t_{\phi_1} = k/k'$. The ratio is $m^2 > 0$. The inverse function theorem proves the local count one there.

At a short minimum, the transverse derivative $V_u = H > 0$ in the coordinates of Proposition 4.3. Reflection and the implicit function theorem force $u = 0$ for nearby targets on the same ray. The nondegenerate radial minimum then gives the local $0, 1, 2$ rule.

At a long cusp, use $W(u, R, 2\alpha)$ from Corollary 4.4. Analyticity and oddness in u give $W = uG(u^2, R)$ for an analytic G. At $(0, R_n^{\mathrm{cusp}})$,

$$G = 0, \qquad G_z = \mathcal{C}/6 < 0, \qquad G_R = H_m/A_m > 0.$$

The unique local solution of $G(z, R) = 0$ therefore satisfies $z(R_n^{\mathrm{cusp}}) = 0$ and $z'(R_n^{\mathrm{cusp}}) = -6H_m/(A_m\mathcal{C}) > 0$. For $R < R_n^{\mathrm{cusp}}$ this root is negative and contributes no real u; for $R > R_n^{\mathrm{cusp}}$ it gives the two roots $u = \pm\sqrt{z(R)}$, in addition to $u = 0$. At equality all three coincide as one source. Along the fixed-radius reduction, $B_u = -2sc_\alpha/h \neq 0$ at the cusp, so the two new sources have $B \neq 0$. The actual period is locally constant at these sources. □

Lemma 4.6 (Regularity of nonzero transverse sources on the ray)**.** *Every rotating source of $(R\mathbf{e}(\alpha), 2\alpha)$ with $B \neq 0$ is regular, for every $R > 0$ and $0 < \alpha < \pi/2$.*

Proof. Put $p = t/(2k)$, $\tau = \psi + p$, so τ is the midpoint phase, and set

$$\epsilon_p = \int_0^p \mathrm{dn}^2\, v\, dv, \qquad q_p = p - \epsilon_p > 0,$$
$$f_1(p) = -q_p \,\mathrm{cn}\, p - \mathrm{dn}\, p \,\mathrm{sn}\, p, \qquad \Delta = 1 - m\,\mathrm{sn}^2\,\tau\,\mathrm{sn}^2\, p > 0.$$

For the positive rotation sign, the Jacobi addition formulas applied to (3) give the half-heading components

$$\begin{aligned} U = X\cdot \mathbf{e}(\theta/2) &= \frac{2q_p\,\mathrm{dn}\,\tau}{\sqrt{\Delta}} > 0, \qquad V = X\cdot \mathbf{e}_\perp(\theta/2) = \frac{-2k\,\mathrm{sn}\,\tau\, f_1(p)}{\sqrt{\Delta}}, \\ B &= \frac{2k\,\mathrm{sn}\,\tau\,\mathrm{sn}\,p\,\mathrm{dn}\,\tau\,\mathrm{dn}\,p}{\Delta}, \qquad \sin(\theta/2) = \frac{-k\,\mathrm{cn}\,\tau\,\mathrm{sn}\,p}{\sqrt{\Delta}}. \end{aligned} \tag{30}$$

For example, if $\delta = (a+b)/2$, the addition formulas give $\cos\delta = \mathrm{dn}\,\tau/\sqrt{\Delta}$ and $\sin\delta = k\,\mathrm{sn}\,\tau\,\mathrm{cn}\,p/\sqrt{\Delta}$; inserting these into $U = A\cos\delta - B\sin\delta$ and $V = A\sin\delta + B\cos\delta$ proves the first two identities. The negative rotation sign negates U and cannot reach the stated ray.

On the ray $V = 0$, while $B \neq 0$ forces $\mathrm{sn}\,\tau \neq 0$. Thus $f_1(p) = 0$. In particular $\mathrm{cn}\,p \neq 0$, and $q_p = -\,\mathrm{dn}\,p\,\mathrm{sn}\,p/\,\mathrm{cn}\,p$. Substitution into Sachkov's reduced Jacobian [2, (2.10)–(2.14)] gives

$$J_2 = (1-m)q_p\epsilon_p\,\mathrm{sn}^2\,\tau > 0. \tag{31}$$

Indeed the coefficient of $\mathrm{sn}^2\,\tau$ simplifies to $(1-m)q_p\epsilon_p$, and the other coefficient contains f_1. The full Jacobian differs by a nonvanishing factor, proving regularity. □

Theorem 4.7 (Exact rotating count on the ray at every radius)**.** *Fix $0 < \alpha < \pi/2$. For each $n \geq 0$, let $\mu_n \in (s^2, 1)$ be the unique long-branch cusp modulus, and put*

$$R_n^{\mathrm{cusp}} = \frac{2sc_\alpha}{\sqrt{\mu_n - s^2}}.$$

In actual period n, there are exactly two nonzero-B sources of $(R\mathbf{e}(\alpha), 2\alpha)$ when $R > R_n^{\mathrm{cusp}}$, and none when $R \leq R_n^{\mathrm{cusp}}$. Both are regular. Consequently the full rotating-source count is

$$\begin{aligned} N_{\mathrm{rot}}(R,\alpha) = \mathbf{1}_{\{R>2Q\}} &+ 2\#\{n \geq 1 : S_n < R\} + \#\{n \geq 1 : S_n = R\} \\ &+ 2\#\{n \geq 0 : R_n^{\mathrm{cusp}} < R\}, \qquad R > 0. \end{aligned} \tag{32}$$

The cusp thresholds increase strictly and satisfy $R_n^{\mathrm{cusp}} > (n+1/2)L_*$, so all sums are finite. For $R > 2s$, the nonzero-B count also equals $2\max(0, \lceil M_0 \rceil)$ with $M_0 = N_R(0)$ as in Proposition 4.1. This count identity does not assert unimodality of the scalar at every radius.

Proof. By [1, Lemma 5.3], for each $m \in (0,1)$ the positive roots of f_1 consist of one simple root $P_n(m)$ in each interval $((2n+1)K, 2(n+1)K)$, $n \geq 0$. To check this directly,

$$\left(\frac{f_1(p)}{\mathrm{cn}\,p}\right)' = -\frac{\mathrm{dn}^2\,p}{\mathrm{cn}^2\,p} < 0.$$

On the stated interval the quotient decreases from $+\infty$ to $-q_{2(n+1)K} < 0$. On the complementary intervals $(2nK, (2n+1)K)$ it is $-q_p - \mathrm{dn}\,p\tan(\mathrm{am}\,p) < 0$. The endpoints are not positive roots. Simplicity and analyticity give analytic $P_n(m)$; its actual period is $\lfloor P_n/(2K) \rfloor = n$.

Fix n, abbreviate $p = P_n(m)$ and $w = m\,\mathrm{sn}^2 p$, and retain $q_p = p - \epsilon_p$. Equation (30) for heading 2α is equivalent to

$$\mathrm{sn}^2\,\tau = \frac{w - s^2}{w c_\alpha^2}, \qquad \mathrm{sgn}(\mathrm{cn}\,\tau) = -\,\mathrm{sgn}(\mathrm{sn}\,p). \tag{33}$$

It gives exactly two phases modulo $4K$ when $w > s^2$, one when $w = s^2$, and none when $w < s^2$. The two phases have opposite nonzero $\mathrm{sn}\,\tau$ and hence opposite nonzero B. Their common radius is

$$R_n(m)^2 = 4(q_p^2 + s^2 - w) > 4s^2. \tag{34}$$

This follows from $U = 2q_p\,\mathrm{dn}\,\tau/\sqrt{\Delta}$ and $q_p^2 = (1 - m\,\mathrm{sn}^2 p)\,\mathrm{sn}^2 p/\,\mathrm{cn}^2 p$.

The equality $w = s^2$ is precisely the long-branch cusp condition: write $p = 2(n+1)K - v$, where $v = F(\arcsin(s/\sqrt{m}) \mid m)$, and substitute into $f_1 = 0$. It becomes $R_n^{\mathrm{long}}(m) = 2sc_\alpha/\sqrt{m - s^2}$. Its unique solution is μ_n by Proposition 4.3. For $m < s^2$ one has $w < s^2$. As $m \uparrow 1$,

$$q_p > (2n+1)K - 2(n+1)E \longrightarrow \infty.$$

The identity $q_p^2 = (1 - mx)x/(1 - x)$, $x = \mathrm{sn}^2 p$, implies $x \to 1$, and therefore $w \to 1$. Continuity and the unique equality give $w > s^2$ exactly on $(\mu_n, 1)$.

On this interval both phase branches are analytic and regular. Moreover $\theta_\tau = B \neq 0$ and, on $f_1 = 0$, $V_p = -2k\,\mathrm{sn}\,\tau\, f_1'(p)/\sqrt{\Delta} \neq 0$, while $V_\tau = 0$. Thus the constraints $V = 0$, $\theta = 2\alpha$ have independent derivatives in (p, τ) and leave m as a local parameter. Lemma 4.6 then implies $R_n'(m) \neq 0$. The radius tends to R_n^{cusp} as $m \downarrow \mu_n$ and to infinity as $m \uparrow 1$, by (34). Hence it increases strictly and maps $(\mu_n, 1)$ onto $(R_n^{\mathrm{cusp}}, \infty)$. This proves the asserted exhaustive count of nonzero-B sources.

The function $R_n^{\mathrm{long}}(m) - 2sc_\alpha/\sqrt{m - s^2}$ increases strictly in m, and increasing n adds $L(m) > 0$. Consequently $\mu_{n+1} < \mu_n$ and the cusp radii increase strictly. Also $R_n^{\mathrm{cusp}} = R_n^{\mathrm{long}}(\mu_n) > (n + 1/2)L_*$. Adding the complete $B = 0$ count (24) proves (32); the strict inequality at a cusp avoids counting its single $B = 0$ source again.

Finally, for $R > 2s$ put $m_c = s^2 + (2sc_\alpha/R)^2$. The long-branch equation at this selected modulus reads

$$R_n^{\mathrm{long}}(m_c) - R = L(m_c)(n - M_0).$$

Its sign and strict monotonicity in m_c show that $R > R_n^{\mathrm{cusp}}$ is equivalent to $n < M_0$. Counting the nonnegative integers with that strict inequality gives $2\max(0, \lceil M_0 \rceil)$. □

Corollary 4.8 (Minimizing part of the ray count)**.** *Write $C = R_0^{\mathrm{cusp}}$ and let M_{rot} count the minimizing rotating sources of $(R\mathbf{e}(\alpha), 2\alpha)$. Then*

$$M_{\mathrm{rot}} = \begin{cases} 0, & 0 < R \le 2Q, \\ 1, & 2Q < R \le C, \\ 2, & R > C. \end{cases}$$

For $2Q < R < C$ the minimizing source is strictly pre-cut; at $R = C$ it is the period-zero cusp at its cut time. For $R > C$ the two minimizing sources form the period-zero regular Maxwell pair, both at their cut time. Every other rotating source is strictly post-cut, so their number is $N_{\mathrm{rot}} - M_{\mathrm{rot}}$. In the notation of [3, (2.13), (2.20)], $2Q = R_1^2(2\alpha)$ and $C = R_1^1(2\alpha)$.

Proof. The established rotating cut time is $2kP_0(m)$, with $K < P_0 < 2K$, and a geodesic minimizes through that time [3, (2.2), Theorem 3.3]. Every source of actual period $n \ge 1$ is therefore post-cut. The nonzero-B sources have $p = P_n(m)$, so precisely their period-zero pair is at the cut time.

For $B = 0$, period zero has one source exactly when $R > 2Q$. On its short branch $p = u \le K < P_0$, where $u = F(\arcsin(s/\sqrt{m}) \mid m)$. On its long branch $p = 2K - u$ and, with $h = \sqrt{m - s^2}$,

$$f_1(2K - u) = \frac{h}{2k}\left(R_0^{\text{long}}(m) - \frac{2sc_\alpha}{h}\right).$$

The parenthesis increases through zero at $m = \mu_0$. On $(K, 2K)$, f_1 is negative before P_0 and positive after it. Since R_0^{long} increases strictly, this source is pre-cut, at cut, or post-cut according as $R < C$, $R = C$, or $R > C$. The branch join has $p = K$ and is strictly pre-cut as well.

At $m = \mu_0$, put $v = \operatorname{am} P_0 \in (\pi/2, \pi)$. Then $\sin\alpha = k \sin v$ and $C = 2(P_0 - \epsilon_{P_0})$. These are exactly the parametric equations of $R_1^1(2\alpha)$ in [3, (2.18)–(2.20)]; the formula for R_1^2 gives $2Q$ directly. In those rectifying coordinates the present ray has $R_1 = -R$, $R_2 = 0$. This recovers the known minimizing thresholds without identifying the additional post-cut sources with minimizers. □

Proposition 4.9 (Cusp scale near zero heading)**.** *Fix an actual period $n \ge 0$, put $j = n+1$ and $\theta = 2\alpha$, and define $z = (\theta/(j\pi))^{2/3}$. As $\theta \downarrow 0$,*

$$\mu_n = z - \tfrac{1}{4}z^2 + O_j(z^3), \qquad R_n^{\text{cusp}} = j\pi\left(z + \tfrac{1}{8}z^2 + O_j(z^3)\right). \tag{35}$$

In particular $R_n^{\text{cusp}} \sim (j\pi)^{1/3}\theta^{2/3}$. These are fixed-period estimates; no uniform remainder as $n \to \infty$ is asserted.

Proof. Write the Maxwell amplitude as $\operatorname{am} P_n = j\pi - \delta$ and define

$$D = \sqrt{1 - m\sin^2\delta}, \qquad \mathcal{Q} = F(j\pi - \delta \mid m) - E(j\pi - \delta \mid m).$$

The root condition is $\mathcal{Q} = D\tan\delta$. At $(m, \delta) = (0, 0)$ its derivative in δ is -1, so the implicit function theorem gives an analytic $\delta(m)$ with $\delta(0) = 0$. From the defining integral,

$$\mathcal{Q} = m\int_0^{j\pi-\delta} \frac{\sin^2 v}{\sqrt{1 - m\sin^2 v}}\,dv = \frac{j\pi}{2}m + \frac{3j\pi}{16}m^2 + O_j(m^3).$$

Here $\delta = O_j(m)$, and removing the terminal interval contributes $O_j(m\delta^3)$. Since $D = 1 + O_j(m^3)$, the root equation gives $\delta = j\pi m/2 + 3j\pi m^2/16 + O_j(m^3)$. Its positive leading coefficient places this root in the Maxwell interval used in Theorem 4.7. On the cusp,

$$\begin{aligned}\theta &= 2\arcsin(\sqrt{m}\sin\delta) = j\pi m^{3/2}\left(1 + \tfrac{3}{8}m + O_j(m^2)\right),\\ R_n^{\text{cusp}} &= 2\mathcal{Q} = j\pi m\left(1 + \tfrac{3}{8}m + O_j(m^2)\right).\end{aligned}$$

Consequently $z = m + m^2/4 + O_j(m^3)$; inversion and substitution give (35). □

Remark 4.10 (Normalization of the published first threshold). For $n = 0$, this yields $R_1^1(\theta) = \pi^{1/3}\theta^{2/3} + O(\theta^{4/3})$ in the notation of [3]. Lemma 2.5(6) of that paper prints half this leading coefficient. Its exact parametrization (2.19), $R_1 = 2(F - E)$, instead gives $R_1 = \pi k^2 + O(k^4)$, whereas the displayed expansion on p. 304 gives $\pi k^2/2$. We use the exact parametrization. The correction concerns this asymptotic coefficient, not the curve or the optimal synthesis.

Theorem 4.11 (Count blow-up near zero heading)**.** *For $R > \max(2s, 2Q)$, put $m_c = s^2 + 4s^2c_\alpha^2/R^2$. Then the full rotating count on the half-heading ray satisfies*

$$-4 < N_{\text{rot}} - \frac{2R}{L_*} - \frac{2R}{L(m_c)} < 2. \tag{36}$$

Consequently, for each fixed $R > 0$, as $\theta = 2\alpha \downarrow 0$,

$$N_{\mathrm{rot}}(R, \theta/2) = \frac{16R(R^2+2)}{\pi(R^2+4)\theta^2} + O_R(1). \tag{37}$$

The remainder is uniform for R in any compact subinterval of $(0, \infty)$. Exactly two of these sources minimize for sufficiently small θ; the strictly post-cut count has the same asymptotic law.

Proof. Put $x = R/L_*$. From (25), $nL_* < S_n < (n + 1/2)L_*$. If $A = \#\{n \geq 1 : S_n < R\}$ and $E_0 = \#\{n \geq 1 : S_n = R\}$, then

$$A \geq \max(0, \lfloor x - 1/2 \rfloor), \qquad A + E_0 \leq \max(0, \lceil x \rceil - 1).$$

Since $R > 2Q$, formula (24) gives

$$-2 < N_{B=0} - 2x < 1. \tag{38}$$

This includes a fold equality, where $E_0 = 1$.

For the nonzero-B count, Theorem 4.7 gives $2\max(0, \lceil M_0 \rceil)$ and

$$M_0 = \frac{R + 2q_c}{L(m_c)} - 1, \qquad q_c = F(\arctan(R/(2c_\alpha)) \mid m_c) - E(\arctan(R/(2c_\alpha)) \mid m_c).$$

The defining integral is positive and its amplitude is below $\pi/2$, so $0 < q_c < L(m_c)/4$. The ceiling bound, including the case $M_0 \leq 0$, therefore yields

$$-2 < N_{B\neq 0} - 2R/L(m_c) < 1.$$

Adding this to (38) proves (36) without a large-radius assumption.

As $\theta \downarrow 0$, the defining complete integral gives $L(m) = \pi m + O(m^2)$, while

$$s^2 = \theta^2/4 + O(\theta^4), \qquad m_c = \tfrac{1}{4}(1 + 4/R^2)\theta^2 + O_R(\theta^4).$$

It follows that

$$\frac{2R}{L_*} = \frac{8R}{\pi\theta^2} + O_R(1), \qquad \frac{2R}{L(m_c)} = \frac{8R^3}{\pi(R^2+4)\theta^2} + O_R(1).$$

All expansions and the condition $R > \max(2s, 2Q)$ are uniform on the stated compact radius intervals. This proves (37). Finally, $R_0^{\mathrm{cusp}} \to 0$ by Proposition 4.9; Corollary 4.8 then gives exactly two minimizers. □

Remark 4.12 (At most one short-minimum comparison). The exact ray count does not require evaluating every threshold. For $R > 2Q$, let $x = R/L_*$, $\rho = 2Q/L_*$, and $N_{\mathrm{sure}} = \max(0, \lfloor x - 1/2 \rfloor)$. The first N_{sure} short minima are strictly below R. Any undecided index lies in

$$\{n \geq 1 : x - 1/2 < n < x - \rho\}.$$

Since $0 < \rho < 1/2$, this set has at most one integer. If it is empty, $N_{B=0} = 1 + 2N_{\mathrm{sure}}$; otherwise add 0, 1, or 2 according as $R < S_n$, $R = S_n$, or $R > S_n$. Then add the nonzero-B ceiling count when $R > 2s$; for $R \leq 2s$ that count is zero. For $R \leq 2Q$ there are no rotating sources. Thus the number of threshold comparisons does not grow with the period count. Exact numerical decisions near a threshold still require validated enclosures or an analytic equality.

Corollary 4.13 (Finiteness on the target seam)**.** *For every $X \neq 0$ and $0 < |\theta| < \pi$, the full rotating fiber is finite. On the target seam this includes every positive radius, rather than only the far-field regime. Uniform boundedness on compact target families, including those crossing the seam, follows in Proposition 4.14 below.*

Proof. Corollary 2.5 treats targets off the seam. Spatial negation and the reflection

$$(x, y, \theta) \longmapsto (-x, y, -\theta)$$

reduce every remaining target to $(R\mathbf{e}(\alpha), 2\alpha)$ with $R > 0$ and $0 < \alpha < \pi/2$. The exact count in Theorem 4.7 is finite by (25) and the cusp-threshold bound in that theorem. □

Proposition 4.14 (Uniform finiteness through the seam)**.** *The cardinality of the full rotating fiber is uniformly bounded on every compact subset of*

$$\mathcal{D} = \{(X, \theta) : X \neq 0, \quad 0 < |\theta| < \pi\}.$$

In particular, the compact family may cross the target seam. The bound is not effective and does not give an exact count.

Proof. Every such compact family is contained in a band

$$\mathcal{T} = \{r_0 \leq |X| \leq r_1, \quad \theta_0 \leq |\theta| \leq \theta_1\}, \qquad 0 < r_0 \leq r_1, \quad 0 < \theta_0 \leq \theta_1 < \pi.$$

Set $m_* = \sin^2(\theta_0/2) > 0$ and choose an integer $M > r_1/L(m_*)$. The circle equations and longitudinal decomposition give $m \geq m_*$ and $n < r_1/L(m_*)$ for every positive period index; period zero also satisfies $n < M$. For any of these sources,

$$\frac{t}{k} = A + \int_{\psi}^{\psi + t/k} \mathrm{dn}^2\, u\, du < r_1 + 4(n+1)E(m) \leq r_1 + 2\pi M.$$

Here the integration interval consists of n full periods and a strictly shorter residual. Thus

$$r_0 \leq t < T := r_1 + 2\pi M, \qquad |c| \leq C := 2/\sqrt{m_*},$$

using the covector bound from Corollary 2.5 and unit speed. These bounds hold also at seam targets and for both rotation signs.

In the original coordinates let $F(\gamma, c, t) = (x, y, \widetilde{\theta})$ be the endpoint map with continuous heading. The Hamiltonian vector field is analytic and its real solutions exist at all finite times: $|\dot{c}| \leq 1$, while the horizontal controls are bounded by one. Analytic dependence on initial data and time makes F analytic on a neighborhood of the compact box $[0, 4\pi] \times [-C, C] \times [r_0, T]$. After affine rescaling, its restriction is a restricted analytic map. Consider the graph with the additional restrictions

$$0 \leq \gamma < 4\pi, \qquad c^2/2 - \cos\gamma > 1, \qquad F(\gamma, c, t) \in \mathcal{T}.$$

It is definable in the o-minimal structure $\mathbb{R}_{\mathrm{an}}$ of restricted analytic functions [10, Section 2]. The half-open angle interval counts each normalized covector once. For rotating sources the continuous heading lies in $(-\pi, \pi)$, so no additional heading representatives enter this graph. By the bounds above, its fiber over each $q \in \mathcal{T}$ is exactly the full rotating fiber.

Each such fiber is finite by Corollary 4.13. The uniform bound on connected components of fibers in $\mathbb{R}_{\mathrm{an}}$ [10, 4.4] therefore bounds their cardinalities. This theorem applies to definable sets without requiring closedness or properness; removing the energy-one boundary causes no difficulty. The band $\mathcal{T}$ is definable, and the original compact family need not be. Restricting the resulting bound to that family proves the claim. □

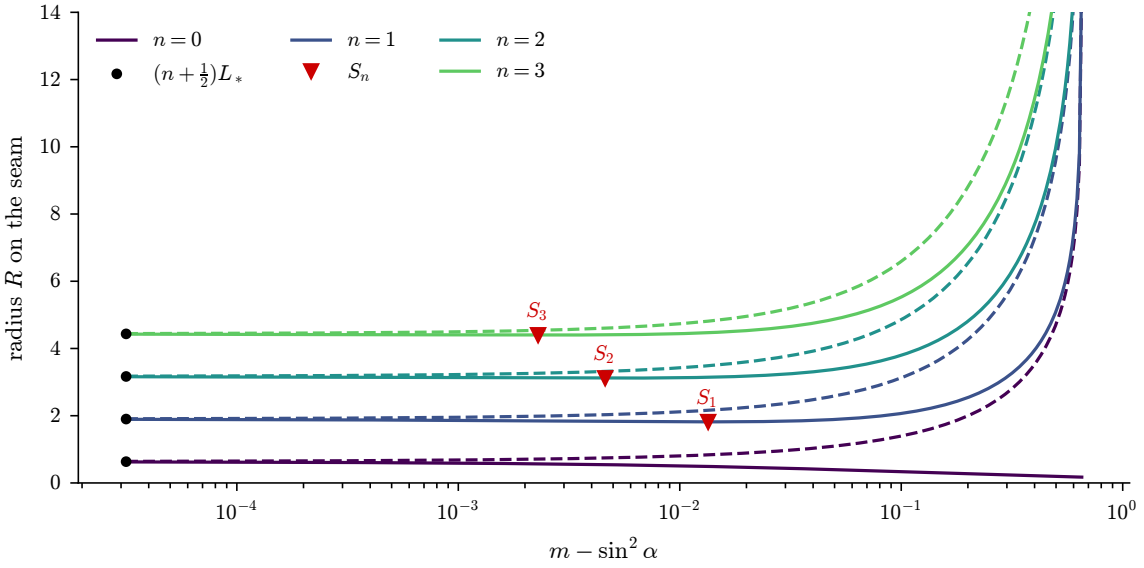


Figure 2: The seam branches R_n^{short} (solid) and R_n^{long} (dashed) of (26), $n = 0, \dots, 3$, at $\alpha = \pi/5$, against $m - \sin^2\alpha$ (logarithmic axis). For $n \geq 1$, the short branch dips to its minimum S_n (triangles); the $n = 0$ short branch decreases. Both branches meet at $H_n = (n + \frac{1}{2})L_*$ as $m \to \sin^2\alpha$ (dots).

Figure 2 shows the branches (26) for $\alpha = \pi/5$, with the minima S_n for $n \geq 1$ and the common endpoints H_n.

Adding Propositions 4.1 and 4.2 proves Theorem 1.1 for positive α. The identity-fixing isometry $(x, y, \theta) \mapsto (-x, y, -\theta)$, which sends controls to $(-u_1, -u_2)$, reduces negative α to this case. Notice that the scalar maximum at $B = 0$ does not imply a full-map fold there: the source chart of Proposition 2.2 is singular on this seam. Proposition 4.3 instead uses a regular source chart: the short-branch minima are full-map folds, whereas each long branch has one cusp (Corollary 4.4). Neither classification is needed for the count.

5 Fixed-cell sources near the separatrix

Return to the generic case $d_0 \neq 0$. Over the four arms, the coefficient c in (20) has values zero, one half, one, and one half.

Proposition 5.1 (Fixed-cell scale)**.** *Fix a nonnegative integer n and an endpoint arm with $n + c > 0$. For all sufficiently large R there is exactly one source of cell n on that arm. Define*

$$C_\sigma = \sigma_0\{\operatorname{atanh}(\sin a_0) - \sin a_0\} - \sigma_1\{\operatorname{atanh}(\sin b_0) - \sin b_0\}.$$

Its modulus satisfies

$$1 - k = 8e^{-2} \exp\left(-\frac{R + C_\sigma}{2(n + c)}\right)\left(1 + O(R^{-1})\right). \tag{39}$$

There are exactly three cell-zero sources eventually, with exponential rates $e^{-R}, e^{-R/2}, e^{-R}$. For each fixed $n \geq 1$, there are eventually four sources. These are fixed-cell statements, not estimates uniform in n growing with R.

Proof. Existence and uniqueness follow from the monotone arm, its limit $-c$, and its diverging maximum. A fixed-level root must tend to its endpoint: otherwise the positive

compact profile in (17) would force N_R to diverge. The short-integral limit is $q(s,1) = \operatorname{atanh}(s) - s$, with bounded derivatives on the strict angle range. Hence along the root $P_R = R + C_\sigma + O(R^{-1}) + O(1-m)$. The standard expansion [7, Section 19.12] reads

$$L(m) = 2\log\frac{16}{1-m} - 4 + O\big((1-m)|\log(1-m)|\big).$$

The equation $P_R = (n+c)L$ first gives exponential decay of $1-m$. Substitution back into the remainders yields $1-m = 16e^{-2}\exp(-(R+C_\sigma)/(2(n+c)))(1+O(R^{-1}))$. Dividing by $1+k$ proves (39). The $c=0$ arm approaches level zero from above, so it contributes no cell-zero root. The other three arms give the stated rates. □

6 The full normal fiber by heading winding

We now retain all pendulum regimes. In the usual vertical coordinates, $\dot\gamma = c$, $\dot c = -\sin\gamma$ and the pendulum energy is $\mathsf{E} = c^2/2 - \cos\gamma$. The oscillating stratum C_1 has $-1 < \mathsf{E} < 1$; C_3 is the separatrix $\mathsf{E} = 1$, $c \neq 0$; C_4 and C_5 are the equilibria $\mathsf{E} = -1$ and $\mathsf{E} = 1$, respectively. These definitions, the endpoint formulas, and the cut times are the standard ones in [1, 2, 3]. For $|\theta| < \pi$, use the smooth target coordinates

$$W = \tan(\theta/2), \qquad U = -\frac{y - xW}{2}, \qquad V = -\frac{x + yW}{2}. \tag{40}$$

The letter U here is a target coordinate and no longer the auxiliary quantity in (5). Write am for the continuously lifted Jacobi amplitude, so $\operatorname{am}(u+2K) = \operatorname{am}(u) + \pi$.

Lemma 6.1 (All positive oscillating winding cells)**.** *Fix a finite-chart target with $(U,V) \neq (0,0)$ and $W \neq 0$. For every integer $n \geq 1$ and sign $s_1 \in \{1,-1\}$ there is exactly one C_1 source in*

$$p = t/2 = 2nK + r, \qquad -K < r < K.$$

It is regular and its heading winding is $\ell = -s_1 n$.

Proof. A C_1 source has phase φ modulo $4K$, modulus $0 < k < 1$, time $t > 0$, and sign s_1. Set $\tau = \varphi + p$, $\eta = \operatorname{am}\tau$ modulo 2π, $q = p - \mathcal{E}(p)$, and $f_2 = k^2 \operatorname{cn} p \operatorname{sn} p + \operatorname{dn} p\, q$. The published endpoint equations give

$$U = \frac{q\cos\eta}{k\operatorname{cn} p}, \quad V = -s_1\frac{f_2\sin\eta}{k\operatorname{cn} p}, \quad W = -s_1\frac{\operatorname{sn} p\sqrt{1-k^2\sin^2\eta}}{\operatorname{cn} p}. \tag{41}$$

There is no finite-chart source with $\operatorname{cn} p = 0$: its principal heading is π. Direct differentiation yields

$$\left(\frac{f_2}{\operatorname{dn} p}\right)' = \frac{k^2\operatorname{cn}^2 p}{\operatorname{dn}^2 p}.$$

Its initial value is zero and its derivative is positive on a set of positive measure in every interval, so $f_2 > 0$ for all $p > 0$.

We first prove the assertion for $UV \neq 0$. Put $Q = \max(2, \sqrt{U^2+4V^2})$ and $z = q/k$. Since $\operatorname{dn} p \geq |\operatorname{cn} p|$, when $q > k^2$ we have $f_2 \geq |\operatorname{cn} p|(q-k^2)$. Recovering sine and cosine from (41) therefore gives

$$1 \leq \frac{U^2}{z^2} + \frac{V^2}{(z-k)^2}.$$

If $z \geq 2k$, this implies $z^2 \leq U^2 + 4V^2$; otherwise $z < 2$. The omitted case $q \leq k^2$ gives $z \leq k < 1$. Thus $q/k \leq Q$. Since $q_p = k^2 \operatorname{sn}^2 p \geq 0$, in cell n this implies

$$(2n-1)\frac{K(k^2) - E(k^2)}{k} \leq Q. \tag{42}$$

The function $H(k) = (K(k^2) - E(k^2))/k$ increases from zero to infinity:

$$H'(k) = \frac{E - (1-k^2)K}{k^2(1-k^2)} > 0,$$

where the numerator equals $k^2 \int_0^{\pi/2} \cos^2 u / \sqrt{1 - k^2 \sin^2 u}\, du$.

Fix n, s_1 and the signs of the nonzero target coordinates. Use $\rho = \operatorname{am} r \in (-\pi/2, \pi/2)$, η, and k as source coordinates. Equations (41) imply

$$\operatorname{sgn} U = (-1)^n \operatorname{sgn}(\cos\eta), \quad \operatorname{sgn} V = -s_1(-1)^n \operatorname{sgn}(\sin\eta), \quad \operatorname{sgn} W = -s_1 \operatorname{sgn}\rho.$$

These signs select one half-interval for ρ, one quadrant for η, and $0 < k < 1$: a nonempty connected source box. At fixed k, $p = 2nK + F(\rho \mid k^2)$ and $\varphi = F(\eta \mid k^2) - p$ have nonsingular Jacobian in (ρ, η). The absence of C_1 conjugate points [2, Theorem 2.1] makes the target map a local diffeomorphism.

This map is proper over the chosen target octant. For targets in a compact subset, (42) bounds k away from one. The equation

$$|W| = |\tan\rho| \sqrt{1 - k^2 \sin^2 \eta}$$

then keeps ρ away from both ends of its half-interval. If $k \to 0$, with n fixed and ρ in this compact interval, uniformly $q, f_2 = O(k^2)$, whence $U, V = O(k)$, contradicting their positive lower absolute bounds. Thus k also stays away from zero. The first two target equations now keep η away from its quadrant edges. All inverse images of compact target sets are therefore compact. A proper local diffeomorphism has nonempty open and closed image here, so it covers the entire connected octant. The octant is simply connected and the source box connected, hence this covering has exactly one sheet.

For $V = 0$, positivity of f_2 forces $\sin\eta = 0$, and $U \neq 0$ gives $\cos\eta = (-1)^n \operatorname{sgn} U$. The remaining equations reduce to

$$\rho = \arctan(-s_1 W), \quad r = F(\rho \mid k^2), \quad \frac{q}{k\cos\rho} = |U|. \tag{43}$$

Define $I_\rho(k) = \int_0^\rho \sin^2 u / \sqrt{1 - k^2 \sin^2 u}\, du$ and $I_* = I_{\pi/2}$. Then $q/k = k(2nI_* + I_\rho)$. For $\rho \geq 0$ the bracket is a positive sum of increasing integrals; for $\rho < 0$ it is

$$(2n-1)I_* + \int_{|\rho|}^{\pi/2} \frac{\sin^2 u}{\sqrt{1 - k^2 \sin^2 u}}\, du.$$

The left side of the last equation in (43) therefore increases strictly from zero to infinity. It gives exactly one modulus, and (43) recovers exactly one source. Its regularity follows from the same published nonconjugacy theorem.

It remains to treat $U = 0$, $V \neq 0$. Put

$$w = |W|, \quad h_k = \sqrt{1 - k^2}, \quad L_k = \sqrt{h_k^2 + w^2}, \quad C_w = \sqrt{1 + w^2}, \quad \epsilon = -s_1 \operatorname{sgn} W.$$

Since $q > 0$, (41) forces $\cos\eta = 0$ and $\sin\eta = -s_1(-1)^n \operatorname{sgn} V$. It also gives $\rho = \epsilon a(k)$, where $a(k) = \arctan(w/h_k)$. The remaining scalar equation is

$$\mathcal{F}_\epsilon(k) = C_w k(2nI_* + \epsilon I_{a(k)}) + \frac{\epsilon k w}{L_k} = |V|. \tag{44}$$

Define $A_a = \int_0^a \sin^2 u/(1-k^2\sin^2 u)^{3/2}\,du$ and $A_* = A_{\pi/2}$. Differentiating, including the moving endpoint, yields

$$\mathcal{F}'_\epsilon = C_w(2nA_* + \epsilon A_{a(k)}) + \frac{\epsilon w}{h_k^2 L_k}. \tag{45}$$

The sign is positive for $\epsilon = 1$. For $\epsilon = -1$, $n \geq 1$, use $2nA_* - A_{a(k)} \geq A_*$ and

$$h_k^2 A_* = J_k, \qquad J_k = \int_0^{\pi/2} \frac{\cos^2 u}{\sqrt{1-k^2\sin^2 u}}\,du.$$

The identity $h_k^2 A_* = J_k$ follows by integrating the derivative of $\sin u\cos u/\sqrt{1-k^2\sin^2 u}$ between its zero endpoint values. Cauchy–Schwarz gives

$$J_k E \geq \left(\int_0^{\pi/2} \cos u\,du\right)^2 = 1.$$

Also $E < 1 + h_k$, by integrating the strict pointwise inequality

$$\sqrt{\cos^2 u + h_k^2 \sin^2 u} < \cos u + h_k \sin u, \qquad 0 < u < \pi/2.$$

Finally

$$C_w^2 L_k^2 - (1+h_k)^2 w^2 = (w^2 - h_k)^2 \geq 0, \qquad J_k > \frac{1}{1+h_k} \geq \frac{w}{C_w L_k}.$$

These inequalities prove strict positivity in (45) for the negative sign too. The scalar tends to zero as $k \to 0$ and to infinity as $k \to 1$, since $q \geq (2n-1)(K-E)$ while kw/L_k stays bounded. Thus it supplies exactly one modulus and one source. Again full-map regularity follows from C_1 nonconjugacy.

Finally the continuous heading is

$$\begin{aligned}\widetilde{\theta} &= s_1\{\operatorname{am}(\tau - p) - \operatorname{am}(\tau + p)\} \\ &= -2\pi s_1 n + s_1\{\operatorname{am}(\tau - r) - \operatorname{am}(\tau + r)\}.\end{aligned}$$

The residual is strictly in $(-\pi, \pi)$ because $|r| < K$, hence equals the principal target heading. This proves $\ell = -s_1 n$. □

Proposition 6.2 (Nonzero winding at every radius)**.** *If $X \neq 0$ and $0 < |\theta| < \pi$, every nonzero integer winding class contains exactly one normal source. It belongs to C_1 and is regular.*

Proof. The target chart (40) is invertible, so $(U, V) \neq (0, 0)$. Lemma 6.1 exhausts all positive C_1 cells; their winding is $-s_1 n$. The remaining C_1 cell is $0 < p < K$ and has winding zero. The continuous C_2 heading is the difference of two strict principal arcsines in (2), so it has winding zero. For C_3, the continuous heading is

$$\widetilde{\theta}(t) = s_1\{\arcsin(\tanh\varphi) - \arcsin(\tanh(\varphi + t))\},$$

also strictly in $(-\pi, \pi)$. Finally C_4 has $X = 0$ and C_5 has principal heading zero, excluding both. Thus no other stratum contributes a nonzero winding source. □

Lemma 6.3 (Effective separatrix exclusion)**.** *For the angles of Theorem 1.1, set* $h = (a_0 - b_0)/2$ *and* $\beta = (a_0 + b_0)/2$*. There is no* C_3 *source of* q_R *if*

$$R > \frac{2\sin^2 h}{\cos(|\beta| + |h|)}. \tag{46}$$

Proof. Here $|\beta| + |h| < \pi/2$ and (40) gives $W = \tan h$, $U = Ru$, $V = Rv$, where

$$u = \frac{\cos\beta}{2\cos h}, \quad v = -\frac{\sin\beta}{2\cos h}, \quad g = u - |W||v| = \frac{\cos(|\beta| + |h|)}{2\cos^2 h} > 0.$$

For C_3, the exact endpoint equations in midpoint variables $p = t/2 > 0$ and τ are

$$U = s_2\frac{(p - \tanh p)\cosh p}{\cosh\tau}, \quad V = -s_1 s_2 p\tanh\tau, \quad W = -s_1\frac{\sinh p}{\cosh\tau}.$$

Eliminating τ and the signs gives the necessary equations

$$\frac{V^2}{p^2} + \frac{W^2}{\sinh^2 p} = 1, \qquad |U| = |W|(p\coth p - 1). \tag{47}$$

The first left side strictly decreases from infinity to zero, so its root is unique. It lies in $(|V|, |V| + |W|)$: for $V \neq 0$ the lower endpoint gives a value above one and $\sinh p > p$ makes the upper value less than one; for $V = 0$ the root is $\operatorname{arsinh}|W| < |W|$. Since $e^{2p} > 1 + 2p$, one has $p\coth p - 1 < p$. Thus $Ru - |W|(p\coth p - 1) > Rg - W^2 > 0$ under (46), excluding the second equation. This is a sufficient bound for C_3 alone. □

Proof of Theorem 1.2. The target has $U > 0$, $W \neq 0$, and $V = 0$ precisely when $\beta = 0$. Lemma 6.1 supplies exactly one source in every nonzero winding class. The bounded continuous heading in (2) shows that every C_2 source has winding zero. It remains to exclude C_1 cell zero and the other regimes.

Suppose first $\beta \neq 0$, equivalently $\sin^2 a_0 \neq \sin^2 b_0$. Proposition 5.1 supplies an actual cell-zero source on each opposite-cosine arm $\sigma_1 = -\sigma_0$. With the source angles a, b and signed short integrals $u_a = F(\arcsin(\sin a/k) \mid m)$ and $u_b = F(\arcsin(\sin b/k) \mid m)$, phase reconstruction gives exactly

$$\delta = 2K - \sigma_0(u_a + u_b).$$

As $R \to \infty$, the sum tends to $\operatorname{atanh}(\sin a_0) + \operatorname{atanh}(\sin b_0)$, which has the nonzero sign of β by oddness and strict monotonicity of $\operatorname{atanh}(\sin a)$. The arm $\sigma_0 = \operatorname{sgn}\beta$ therefore has $t = k\delta < 2kK$ eventually. For $f_1 = \operatorname{cn} p(\mathcal{E}(p) - p) - \operatorname{dn} p\operatorname{sn} p$, we have $f_1 < 0$ on $(0, K]$; thus its first positive root p_1 satisfies $p_1 > K$. The published cut time is $2kp_1$ [3, (2.2)]. This source is strictly pre-cut, and the known synthesis makes it the unique minimizer [3, Theorems 3.1–3.2]. Any C_1 cell-zero source would have $t = 2p < 2K$, its own cut time [3, (2.1)], and would be another strictly pre-cut minimizer. This contradiction excludes cell zero.

For $\beta = 0$, Proposition 4.1 gives two distinct cell-zero sources with $B \neq 0$ eventually. They are exchanged by $B \mapsto -B$, with equal modulus and time. Put $p = t/(2k)$ and $\tau = \psi + p$. The C_2 target formula and Jacobi addition identity give

$$V = \frac{kf_1(p)\operatorname{sn}\tau}{\operatorname{dn} p}, \qquad B = \frac{2k\operatorname{sn}\tau\operatorname{sn} p\operatorname{dn}\tau\operatorname{dn} p}{1 - m\operatorname{sn}^2\tau\operatorname{sn}^2 p}.$$

The denominator and the dn factors are positive. Thus $B \neq 0$ implies $\operatorname{sn}\tau \neq 0$, so $V = 0$ forces $f_1(p) = 0$. Cell zero means $0 < p < 2K$. There is exactly one such root: besides $f_1 < 0$ on $(0, K]$, on $(K, 2K)$ we have

$$(f_1/\operatorname{cn})' = -\operatorname{dn}^2 p/\operatorname{cn}^2 p < 0,$$

and the quotient decreases from $+\infty$ to $-2(K-E) < 0$. Hence both sources are at the first cut time $2kp_1$ and minimize. The published synthesis gives exactly two minimizers at a Maxwell endpoint [3, Theorems 3.2 and 3.4]. A strictly pre-cut C_1 cell-zero source would contradict this pair. Negative h follows either from the same identities or from the isometry used in Section 4.

Lemma 6.3 excludes C_3 eventually. The C_4 image has $x = y = 0$, and the C_5 image has $y = \theta = 0$, so neither reaches the present target. Theorem 1.1 now gives the finite zero-winding count; the remaining classes each contain one regular C_1 source. Spatial negation preserves winding and all counts. No regularity conclusion for every C_2 source is needed. □

6.1 The exact rotating image and its parity

We retain the target coordinates (40), with $X \neq 0$ and $0 < |\theta| < \pi$. The hyperbolic source formulas are classical [1, Section 3.3]; the comparison below applies to every positive-time rotating source, including all post-cut periods.

Lemma 6.4 (The separatrix graph)**.** *For $V \in \mathbb{R}$ and $w > 0$, let $P = P(V, w) > 0$ be the unique root of*

$$\frac{V^2}{P^2} + \frac{w^2}{\sinh^2 P} = 1, \qquad \mathcal{F}(V, w) = w(P \coth P - 1).$$

The complete C_3 image in the stated domain is $\Sigma = \{|U| = \mathcal{F}(V, |W|)\}$, with exactly one C_3 source at each target. The function $\mathcal{F}$ is analytic for $w > 0$, even in V, increasing for $V > 0$, and satisfies

$$0 \leq \partial_V \mathcal{F}(V, w) < w \qquad (V \geq 0). \tag{48}$$

Proof. Existence and uniqueness of P follow from the strictly decreasing left side, as in (47). Its P derivative is nonzero, so the implicit function theorem gives analyticity. Necessity of the graph condition follows from the same endpoint equations. For sufficiency, set $s_1 = -\operatorname{sgn} W$, $s_2 = \operatorname{sgn} U$ and

$$\tau = \operatorname{atanh}(-s_1 s_2 V/P), \qquad \varphi = \tau - P, \qquad t = 2P.$$

The root equation gives $\cosh \tau = \sinh P/|W|$. Substitution recovers all three target coordinates. These signs, time and phase are forced and determine one normalized hyperbolic covector [1]. Also $P \coth P > 1$, so $U \neq 0$ on the graph.

For $V > 0$, put $a = w^2/\sinh^2 P \in (0, 1)$ and $H = P \coth P - 1$. Then $V = P\sqrt{1-a}$ and

$$\frac{dV}{dP} = \frac{1 + aH}{\sqrt{1-a}} > 1, \qquad 0 < H' = \frac{\sinh P \cosh P - P}{\sinh^2 P} < 1.$$

The last inequalities follow from $\sinh(2P) > 2P$ and $1 - e^{-2P} < 2P$. This proves (48) for $V > 0$; evenness and analyticity give derivative zero at $V = 0$. □

Lemma 6.5 (An all-period rotating barrier)**.** *Every positive-time C_2 source in the stated target domain satisfies*

$$|U| > \mathcal{F}(V, |W|). \tag{49}$$

In particular, its full critical-value image is disjoint from Σ.

Proof. Use $p = t/(2k) > 0$, $\tau = \psi + p$ and $\eta = \operatorname{am} \tau$ modulo 2π. Write

$$s = \operatorname{sn} p, \quad c_p = \operatorname{cn} p, \quad d = \operatorname{dn} p, \quad q_p = p - \epsilon_p = \int_0^p k^2 \operatorname{sn}^2 u \, du.$$

The original endpoint formulas [1, Section 3.3] and Jacobi addition give, in the coordinates (40),

$$U = s_2 \frac{q_p \operatorname{dn} \tau}{d}, \qquad V = s_2 \frac{k(-c_p q_p - ds) \sin \eta}{d}, \qquad W = -\frac{ks \cos \eta}{d}. \tag{50}$$

These U, V are the scaled chart coordinates (40), not the unscaled components used temporarily in (30). For example, $\cos(\theta/2) = d/\sqrt{\Delta}$ in that formula, which supplies the scaling and the sign of V.

Put $r = k|s|$, $b = k|c_p|$ and $Q(r) = \operatorname{atanh} r - r$. Since $d^2 - b^2 = 1 - k^2 > 0$, one has $b < d$ and

$$\frac{d}{dp}\{q_p - Q(k|\operatorname{sn} p|)\} = k^2 s^2 \left(1 - \frac{k \operatorname{sgn}(s) c_p}{d}\right) > 0 \quad (s \neq 0).$$

The derivative extends as zero at the zeros of s. The difference starts at zero and increases strictly over every nontrivial interval; thus $q_p > Q(r)$ for every $p > 0$, with no period restriction.

Set $z = |\sin \eta|$, $h = |\cos \eta|$, $D_\tau = \sqrt{1 - k^2 z^2}$ and $q_* = Q(r)$. Strict nonzero heading gives $r, h > 0$. Equation (50) implies

$$|U| = \frac{q_p D_\tau}{d}, \qquad w := |W| = \frac{rh}{d}, \qquad |V| \le z\left(r + \frac{b q_p}{d}\right) =: \overline{V}(q_p).$$

At $P_* := \operatorname{atanh} r$, one has $\overline{V}(q_*) \le z(r + q_*) = zP_*$ and

$$\frac{(zP_*)^2}{P_*^2} + \frac{w^2}{\sinh^2 P_*} = z^2 + h^2 = 1, \qquad \mathcal{F}(zP_*, w) = \frac{hq_*}{d}.$$

Monotonicity in $|V|$ and (48) therefore give

$$\mathcal{F}(V, w) \le \frac{hq_*}{d} + \frac{wzb}{d}(q_p - q_*).$$

Subtracting this from $|U|$, and using $D_\tau \ge h$, $b/d < 1$, yields

$$|U| - \mathcal{F}(V, w) \ge \frac{q_*(D_\tau - h) + (q_p - q_*)(D_\tau - wzb)}{d} \ge \frac{(q_p - q_*)h(1 - rz)}{d} > 0. \tag{51}$$

The non-strict comparison is intentional: equality holds in the last comparison when $z = 0$. Its lower bound is still strictly positive. All phase and rotation signs are retained in this argument. □

Theorem 6.6 (Exact rotating image and regular-fiber parity)**.** *For $X \neq 0$ and $0 < |\theta| < \pi$, a rotating source exists if and only if $|U| > \mathcal{F}(V, |W|)$. Every target in this region has a finite nonempty rotating fiber. If every rotating preimage is regular, the fiber cardinality is odd. At critical values only existence and finiteness are asserted.*

Proof. The preceding lemma proves necessity. For each fixed pair of signs of U, W, the strict region is analytically parametrized by

$$(V, w, z) \in \mathbb{R} \times (0, \infty)^2, \qquad W = \pm w, \quad U = \pm(\mathcal{F}(V, w) + z).$$

It is therefore diffeomorphic to $\mathbb{R}^3$. Restrict the rotating endpoint map to the full inverse of one such target component. The source is an open three-dimensional manifold, with γ modulo 4π; it is not assumed compact or connected.

This map is proper. For a compact target subset, the bounds in the proof of Proposition 4.14 bound all initial covectors and positive times, including every period. A limiting

source has energy at least one. At energy one, C_5 has heading zero and C_3 maps to Σ, both excluded by the compact target subset. Thus every source limit remains in the restricted domain. The angle's half-open representative introduces no physical boundary.

There is a regular target with exactly one source in each component. Indeed any adjacent point of Σ has a unique C_3 source, regular by [2, Theorem 2.4], and no C_2 source by Lemma 6.5. The same source bounds and local injectivity exclude any additional C_2 inverse branches arriving near it. The local energy-greater-than-one side lies in the strict region by the barrier, giving the required single-source base value.

We use degree modulo two [9, Section 4], with the needed proper-map extension made explicit. For two regular target values, the component's $\mathbb{R}^3$ coordinates give an isotopy h_t carrying one to the other. The homotopy $h_t \circ \mathrm{Exp}$ is proper: inverse sets over compact targets lie in the product of $[0,1]$ with a compact inverse set of Exp. A nearby regular value of the homotopy has inverse a compact one-dimensional manifold whose boundary is the two endpoint fibers. Sard's theorem and local constancy at regular values preserve those endpoint counts. The even number of boundary points gives equal parity. Hence every regular value has odd count, since the base count is one. A missing target would be a regular value with empty fiber, contradicting this parity. This proves surjectivity on all four components. Finiteness, also at critical values, follows from Corollary 4.13. □

Corollary 6.7 (The count next to the separatrix)**.** *Every target of Σ has a neighborhood in which the exact rotating count is $\mathbf{1}_{\{|U|>\mathcal{F}(V,|W|)\}}$. The source entering C_2 has actual period zero. A compact subset of Σ admits such a neighborhood, without an effective width bound.*

Proof. The local inverse and source-limit exhaustion in the preceding proof give the count. Time remains bounded while $K \to \infty$ on the entering rotating branch, so its actual period is zero. A finite cover gives the compact-subset assertion. □

Remark 6.8. Oddness cannot be extended to critical values: at the first short minimum $R = S_1$ on the half-heading ray, (32) gives two zero-transverse sources plus an even nonzero-transverse contribution. The image criterion and parity do not determine the exact count at an arbitrary target.

Corollary 6.9 (All-radius zero-winding regimes)**.** *Let $X \neq 0$, $0 < |\theta| < \pi$, and put $\mathcal{G}(q) = |U| - \mathcal{F}(V,|W|)$. The complete zero-winding normal fiber has the following form:*

condition	*stratum*	*cardinality $N_0(q)$*
$\mathcal{G}(q) < 0$	C_1, *cell zero*	1
$\mathcal{G}(q) = 0$	C_3	1
$\mathcal{G}(q) > 0$	C_2	$\nu_2(q) < \infty$.

In the first two rows the single source is regular and is the unique minimizer. In the third row every minimizer belongs to actual C_2 cell zero. Every nonzero winding class contains one regular C_1 source, so the full normal fiber is countably infinite at every target in this domain. No exact arbitrary-target formula for ν_2 is asserted.

Proof. We use the classical cut times $t_{\mathrm{cut}} = 2K$ on C_1, $t_{\mathrm{cut}} = 2kp_1$ on C_2, and $t_{\mathrm{cut}} = \infty$ on C_3 [2, Theorem 3.3][3, (2.1)–(2.3)], together with the published optimal synthesis [3, Theorem 3.2]. In particular, a strictly pre-cut source is the unique minimizer of its target. Every nontrivial target admits a normal minimizer [1, Sections 1–2].

For each fixed sign s_1, the full C_1 cell-zero domain is parametrized by $0 < k < 1$, $0 < p/K < 1$ and $\eta \in \mathbb{R}/(2\pi\mathbb{Z})$; it is connected. The continuous heading has absolute value $\int_{\tau-p}^{\tau+p} \mathrm{dn}\, u\, du \in (0,\pi)$, because $2p < 2K$ and every full $2K$ integral is π. Also $q_p =$

$p - \mathcal{E}(p) > 0$ and $f_2 = k^2 \operatorname{cn} p \operatorname{sn} p + \operatorname{dn} p\, q_p > 0$ on $0 < p < K$. Equation (41) therefore gives $(U, V) \neq (0, 0)$. Thus its entire image lies in the stated target domain.

That image cannot meet Σ. Such a target would have both a strictly pre-cut C_1 source and the C_3 source of Lemma 6.4, also strictly pre-cut, contradicting uniqueness in the synthesis. Consequently $\mathcal{G}$ has a constant nonzero sign on each connected C_1 cell-zero domain. At $\eta = \pi/2$, (41) gives $U = 0$, $V \neq 0$ and $W \neq 0$, hence $\mathcal{G} < 0$. This proves that the entire C_1 cell-zero image lies in $\{\mathcal{G} < 0\}$.

Now choose a normal minimizer of any target in the stated domain. A positive C_1 winding cell has $p > K$, hence lies after its cut time and cannot supply this minimizer. The equilibria C_4, C_5 are excluded by $X \neq 0$ and $\theta \neq 0$, respectively. If $\mathcal{G} < 0$, the rotating barrier excludes C_2 and the separatrix graph excludes C_3, so the minimizer must be in C_1 cell zero; its strict pre-cut uniqueness gives exactly one such source. If $\mathcal{G} = 0$, the unique C_3 source is the only zero-winding source. If $\mathcal{G} > 0$, neither C_1 cell zero nor C_3 can occur, so every minimizer is rotating. Since $p_1 < 2K$, its time satisfies $t \leq 2kp_1 < 4kK$, giving actual period zero.

The previous winding catalogue and rotating finiteness now give all three rows and the full-fiber assertion. Regularity in the first two rows is the known nonconjugacy of C_1, C_3 [2, Theorems 2.1, 2.4]. The minimizing existence argument also gives an alternative proof of surjectivity onto $\{\mathcal{G} > 0\}$ using only the rotating barrier, without the degree argument used for parity. □

6.2 Convexity and sharp radial existence intervals

The graph in Lemma 6.4 comes from the classical hyperbolic formulas [1, Section 3.3]. The following geometry, combined with the all-period barrier, determines exactly where a fixed target ray admits rotating sources. It does not determine their number.

Lemma 6.10 (Strict convexity at fixed heading)**.** *For each fixed $w > 0$, $\mathcal{F}(V, w)$ is strictly convex in V. The two spatial components of the rotating image at a fixed strict heading are open convex sets. There is a unique $V_*(w) > 0$ such that*

$$\mathcal{F}(V_*, w) = V_* \partial_V \mathcal{F}(V_*, w).$$

Writing $\sigma(w) = \mathcal{F}(V_*, w)/V_*$, one has $0 < \sigma(w) < w$. The quotient $\mathcal{F}(V, w)/V$ decreases strictly from infinity to $\sigma(w)$ and then increases strictly to w. Both V_* and σ are analytic for $w > 0$.

Proof. Put $P_0 = \operatorname{arsinh} w$, $a = w^2/\sinh^2 P$ and $H = P \coth P - 1$. For $P > P_0$, parametrize the graph by $V = P\sqrt{1-a}$ and $\mathcal{F} = wH$. Differentiation in P gives

$$V' = \frac{1 + aH}{\sqrt{1-a}} > 0, \qquad H' = \coth P - P \operatorname{csch}^2 P > 0, \qquad H'' = 2H \operatorname{csch}^2 P > 0,$$

$$V'' = \frac{a}{(1-a)^{3/2}} \big\{ (1-a)(H' - 2H \coth P) - (1 + aH) \coth P \big\} < 0.$$

For the last sign, with $x = 2P$, observe that

$$\sinh^2 P \, (H' - 2H \coth P) = -\tfrac{1}{2}(x \cosh x + 2x - 3 \sinh x),$$

$$x \cosh x + 2x - 3 \sinh x = \sum_{j \geq 2} \frac{(2j-2)x^{2j+1}}{(2j+1)!} > 0.$$

Thus $\mathcal{F}_{VV} = w(H''V' - H'V'')/(V')^3 > 0$ for $V > 0$. At $V = 0$, implicit differentiation gives

$$P_V = 0, \qquad P_{VV} = \frac{1}{P_0^2 \coth P_0}, \qquad \mathcal{F}_{VV}(0, w) = \frac{wH'(P_0)}{P_0^2 \coth P_0} > 0.$$

Evenness proves the assertion on the whole line. The regions $U > \mathcal{F}(V, w)$ and $U < -\mathcal{F}(V, w)$ are convex, and the fixed-heading change from (x, y) to (U, V) is invertible and linear.

As $V \to \infty$, the identities $P - V = Pa/(1 + \sqrt{1-a})$, $V < P < V + w$ and $H = P - 1 + 2P/(e^{2P} - 1)$ imply

$$\mathcal{F} = wV - w + O_w(Ve^{-2V}), \qquad \mathcal{F}_V = w + O_w(Ve^{-2V}). \tag{52}$$

The derivative estimate follows directly from $\mathcal{F}_V = wH'/V'$. Consequently $J(V) = \mathcal{F} - V\mathcal{F}_V$ starts at $J(0) = \mathcal{F}(0, w) > 0$, has derivative $-V\mathcal{F}_{VV} < 0$ for $V > 0$, and tends to $-w$. Its unique zero gives the claims about $\mathcal{F}/V$, since $(\mathcal{F}/V)' = -J/V^2$. The slope bound (48) gives $0 < \sigma < w$, and the implicit function theorem at $J_V(V_*) < 0$ gives analytic dependence. □

Proposition 6.11 (Exact radial existence classification)**.** *Fix $0 < |\theta| < \pi$ and a unit spatial direction $\mathbf{e} = (e_x, e_y)$. For $q_R = (R\mathbf{e}, \theta)$, $R > 0$, put*

$$W = \tan(\theta/2), \quad w = |W|, \quad u = \tfrac{1}{2}|e_y - We_x|, \quad v = \tfrac{1}{2}|e_x + We_y|.$$

If $v = 0$, rotating sources exist precisely for $R > \mathcal{F}(0, w)/u$. If $v > 0$, put $b = u/v$. With $\sigma = \sigma(w)$ from Lemma 6.10, the complete classification is

direction	*radii on Σ*	*radii with C_2 sources*
$b \geq w$	*one transverse* R_-	(R_-, ∞)
$\sigma < b < w$	*two transverse* $R_- < R_+$	(R_-, R_+)
$b = \sigma$	*one tangency* $R_* = V_*/v$	$\varnothing$
$0 \leq b < \sigma$	$\varnothing$	$\varnothing$.

At every radius on Σ, the zero-winding source is the unique regular C_3 source. In particular, the target-ray tangency is not a conjugate point or a singularity of the exponential map.

Proof. The exact image criterion is $g(R) = Ru - \mathcal{F}(Rv, w) > 0$. Since $u^2 + v^2 = (1 + w^2)/4$, the case $v = 0$ has $u > 0$ and is immediate. For $v > 0$, $g'' = -v^2\mathcal{F}_{VV} < 0$ and $\lim_{R \downarrow 0} g(R) = -\mathcal{F}(0, w) < 0$; the latter is only a scalar limit, since the spatial origin is excluded. If $b \geq w$, $g' = v(b - \mathcal{F}_V) > 0$. Equation (52) gives $g \to \infty$ for $b > w$ and $g \to w > 0$ for $b = w$. Hence there is one transverse crossing in either case. For $b < w$, the last three rows follow from the unique strict minimum and the two monotone branches of $\mathcal{F}(V, w)/V$. At $b = \sigma$, g touches zero at its unique maximum; it is negative elsewhere. Corollary 6.9 identifies the sources at each boundary and in the complementary regions. □

The tangent parameter can be found without differentiating an inverse: $P_* > \operatorname{arsinh} w$ is the unique root of

$$\sinh^2 P_* = (1 + w^2)P_*^2 - w^2(P_* \coth P_* - 1).$$

Indeed, $H - PH' = P^2/\sinh^2 P - 1$ and $H^2 + PH' = P^2 - H$ reduce $J = 0$ to this equation. Uniqueness follows from the lemma; no monotonicity of the multiplied residual is needed.

In the directional notation of Theorem 1.3,

$$u^2 - w^2v^2 = \tfrac{1}{4}(1 + w^2)^2\Delta, \qquad \Delta = e_y(e_y \cos\theta - e_x \sin\theta).$$

Thus $\Delta \geq 0$ gives one onset of rotating existence. For $\Delta < 0$, a finite interval, one C_3 tangency, or neither can occur. Finite intervals occur at every strict heading, since $\sigma(w) < w$ and every nonnegative ratio u/v is attained by a direction. These existence thresholds do not identify where the eventual count formulas become valid. At $v = 0$ the threshold reduces to $2(\operatorname{atanh} s - s)$, $s = w/\sqrt{1+w^2}$, the classical function in [3, (2.13), Lemma 2.3]. No novelty of that threshold or of the minimizing synthesis is asserted.

A certified finite interval. Take $\theta = 2\arctan(3/4)$ and $\mathbf{e} = (-1,-2)/\sqrt{5}$. Then $\Delta = -4/25$, $U = V/2 > 0$ and $R = 4V/\sqrt{5}$. Strict interval gap signs at $V = 1/10, 1, 4$ are negative, positive, negative. Proposition 6.11 therefore gives exactly two boundary crossings. Outward interval bisection at 256 bits gives

$$R_- \simeq 0.46581453181801560948, \qquad R_+ \simeq 5.22604376092815354863.$$

The exact rational brackets and strict endpoint signs are recorded in `data/e105_radial_window.json`; their V widths are below 10^{-26}. A separate 90-digit inversion in P recovers both brackets, and original Hamiltonian integrations corroborate the boundary targets. Corollary 6.13 proves that exactly one rotating source exists for $R_- < R < R_+$, with none outside or on its boundaries.

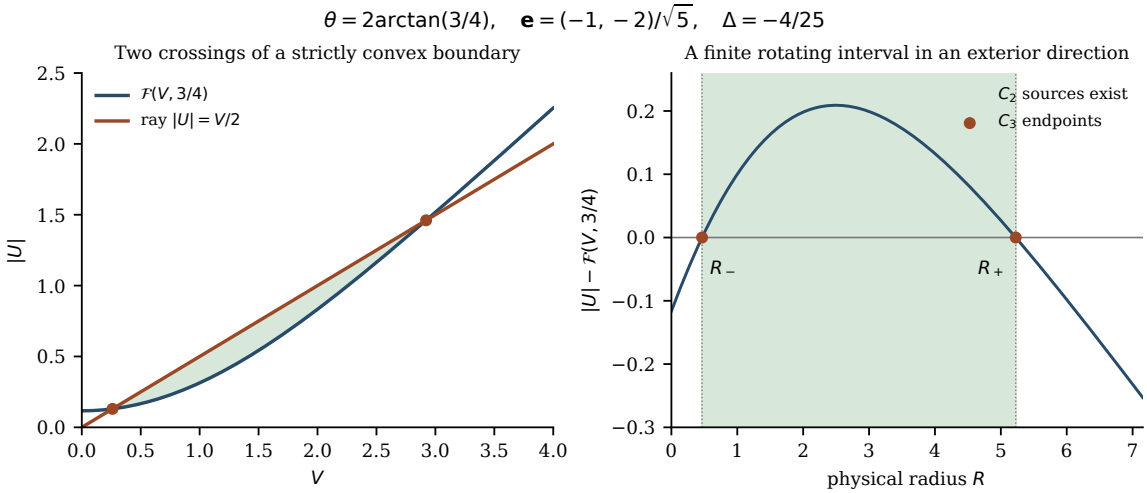


Figure 3: A finite rotating interval in an exterior direction of SE(2). The boundary graph and its gap from the target ray are sampled illustrations; the two crossing locations have separate interval certificates. Corollary 6.13 proves count one in the shaded interval and zero elsewhere. Both boundary targets have a regular C_3 source.

6.3 Exact counts in an exterior sector

Exteriority in Theorem 1.3 is equivalent to $|U| < |WV|$. The following restrictions apply at every radius and retain the actual rotating period index $n = \lfloor t/(4kK) \rfloor$.

Lemma 6.12 (Exterior source restrictions)**.** *Let a rotating source reach an exterior target, and put $w = |W|$, $b = |U|/|V|$ and $m_h = w^2/(1+w^2)$. Then $0 < b < w$ and, with $m = k^2$, $p = t/(2k)$ and $\mathcal{Q}(m) = K(m) - E(m)$,*

$$p - \mathcal{E}(p) < \frac{b}{w-b}, \qquad n < \frac{b}{2(w-b)\mathcal{Q}(m_h)}, \qquad 2nK < p < (2n+1)K. \tag{53}$$

If any rotating source exists at an exterior target, exactly one has actual period zero. That source is regular and strictly pre-cut; all additional rotating sources have positive actual period.

Proof. Write $\kappa = \sqrt{1-m}$, $s = \operatorname{sn} p$, $c = \operatorname{cn} p$, $d = \operatorname{dn} p$, $q_p = p - \mathcal{E}(p)$, and $z = |\sin\eta|$, $h = |\cos\eta|$, $D = \sqrt{1-mz^2}$ in the midpoint chart. Its endpoint equations give

$$|U| = \frac{q_p D}{d}, \quad |V| = \frac{kz|cq_p + ds|}{d}, \quad w = \frac{k|s|h}{d}.$$

Here s, h, z are nonzero. Since $k|c| < d$ and $D \geq h$, $wk|c|z < D$. Also $D^2 = \kappa^2 + w^2 d^2/s^2$, so $D > wd/|s|$. The identity $q_p D = bkz|cq_p + ds|$ and $b < w$ therefore yield

$$q_p \leq \frac{bkzd|s|}{D - bkz|c|} < \frac{b}{1-b/w}\frac{kzd|s|}{D} < \frac{b}{w-b}kzs^2 < \frac{b}{w-b}.$$

The heading constraint gives $ms^2 \geq m_h$. Moreover $q_p \geq 2n\mathcal{Q}(m)$ and $\mathcal{Q}$ is increasing, which proves the index bound.

We next exclude the second half of each period. The complete-integral derivatives [7, Section 19.4], converted to the parameter m, give

$$\mathcal{Q}'(m) = \frac{E(m)}{2(1-m)}, \qquad \frac{d}{dm}\big[\mathcal{Q}(m) - (1-\kappa)\big] = \frac{E(m)-\kappa}{2\kappa^2} > 0.$$

Since $\mathcal{Q}(0) = 0$ and $E(m) > \pi\kappa/2 > \kappa$, $\mathcal{Q}(m) > 1 - \kappa =: \lambda$. The square identity

$$1 - mz^2 - (1+\kappa)^2 z^2(1-z^2) = [1-(1+\kappa)z^2]^2$$

implies $mhz/D \leq \lambda$. If $(2n+1)K \leq p \leq (2n+2)K$, then $sc \leq 0$ and $q_p \geq \mathcal{Q} > \lambda$. Put $S = |s|$, $C = |c|$. When $dS \geq Cq_p$,

$$q_p d - \lambda S|dS - Cq_p| = d(q_p - \lambda S^2) + \lambda SCq_p > 0;$$

when $dS < Cq_p$, the same difference is $q_p(d - \lambda SC) + \lambda dS^2 > 0$, because $d \geq C$ and $\lambda S < 1$. These inequalities imply $|U| > w|V|$, a contradiction. The endpoints $p = 2jK$ have zero heading and are excluded.

For $n = 0$ we now have $p < K < p_1$, hence $t < 2kK < 2kp_1 = t_{\text{cut}}$ [1, (5.17)][3, (2.2)]. Strict pre-cut uniqueness and regularity follow from the known synthesis and conjugate-time bound [3, Theorem 3.2][2]. If the target admits a rotating source, Corollary 6.9 supplies a rotating cell-zero minimizer. It is the unique cell-zero source. □

Corollary 6.13 (An exact uniqueness sector)**.** *For $X \neq 0$, $W = \pm 3/4$ and $|U| \leq |V|/2$, the zero-winding normal fiber contains exactly one source. It is regular, strictly pre-cut and uniquely minimizing, and*

$$\nu_2(q) = \mathbf{1}_{\{|U| > \mathcal{F}(V,3/4)\}}, \qquad N_0(q) = 1.$$

In particular, this fixed-heading sector contains no rotating conjugate value. The stated angular boundary is sufficient; optimality is not asserted.

Proof. There is no rotating source when $U = 0$. Otherwise $0 < b \leq 1/2 < w = 3/4$, and Lemma 6.12 gives $q_p < 2$. We show that $n \geq 1$ is impossible. With the notation of its proof, put $r = ms^2 \geq 9/25$. Then

$$D \geq \tfrac{5}{4}\sqrt{1-m}, \qquad bk|c|z \leq \tfrac{1}{2}\sqrt{m - 9/25}, \qquad (bkzd|s|)^2 = b^2\left(\tfrac{25r-9}{16}\right)(1-r) \leq \tfrac{1}{25}.$$

The last quadratic has maximum $4/25$ at $r = 17/25$ before multiplication by $b^2 \le 1/4$. Whenever its denominator is positive,

$$q_p \le \frac{1/5}{(5/4)\sqrt{1-m} - (1/2)\sqrt{m-9/25}}. \tag{54}$$

For $9/25 \le m \le 3/5$, that denominator is greater than $1/2$: use $\sqrt{2/5} > 3/5$ and $\sqrt{6/25} < 1/2$ at $m = 3/5$. Thus $q_p < 2/5$, while $n \ge 1$ requires $q_p \ge 2\mathcal{Q}(m) > 3m/2 \ge 27/50$. Here $\mathcal{Q}(m) > \pi m/4 > 3m/4$ follows from its positive integral. For $3/5 \le m \le 4/5$, the denominator is at least $\sqrt{5}/4 - \sqrt{11}/10 > 2/9$, using $\sqrt{5} > 20/9$ and $\sqrt{11} < 10/3$. Hence $q_p < 9/10$, whereas $2\mathcal{Q}(m) > 3m/2 \ge 9/10$.

Finally $\mathcal{Q}(4/5) > 1$. Indeed, retaining five positive terms in the binomial expansion of its defining integral gives

$$\mathcal{Q}(m) > \frac{\pi m}{4}\left(1 + \frac{3m}{8} + \frac{15m^2}{64} + \frac{175m^3}{1024} + \frac{2205m^4}{16384}\right).$$

At $m = 4/5$, $\pi > 157/50$ makes the right side greater than $2000337/2000000 > 1$. Thus $m \ge 4/5$ would give $q_p \ge 2\mathcal{Q}(m) > 2$, contradicting $q_p < 2$. These three ranges exhaust all admissible moduli.

Every rotating source consequently has $n = 0$ and is strictly pre-cut by Lemma 6.12. The exact image theorem supplies it precisely when $|U| > \mathcal{F}(V, 3/4)$; the other cases and their regularity follow from Corollary 6.9. □

For the ray of Figure 3, the corollary gives exactly one rotating source for $R_- < R < R_+$ and none elsewhere. The zero-winding source is unique and regular at every positive radius. In original covector and time coordinates it varies analytically with R, by the inverse function theorem and uniqueness, changing from C_1 through C_3 to C_2 and back through C_3 to C_1.

The midpoint identity also bounds every positive period away from the separatrix at a fixed strict-heading target. With $w = |W|$ and the notation of Lemma 6.12, $D^2 = 1 - m + w^2 d^2/s^2$ gives

$$q_p = \frac{|U|d}{D} < \frac{|U|\,|s|}{w} \le \frac{|U|}{w}, \qquad 2n\mathcal{Q}(m) \le q_p. \tag{55}$$

This inequality does not require exteriority. For $n \ge 1$, it implies $m < \mu_n$, where $\mathcal{Q}(\mu_n) = |U|/(2nw)$. The solution $\mu_n \in (0,1)$ exists uniquely because $\mathcal{Q}$ is strictly increasing from zero to infinity [7, Sections 19.4, 19.12]. Together with $m \ge m_h$, this excludes the period if $\mu_n \le m_h$. The following pointwise certificate combines this bound with the unique exterior cell-zero source; it uses interval computation to count the remaining period-one sources.

Proposition 6.14 (A certified exterior three-source fiber)**.** *Choose the exact source data*

$$m_0 = \frac{9999}{10000}, \qquad \alpha_0 = \frac{29}{20}, \qquad p_0 = 2K(m_0) + F(\alpha_0 \mid m_0),$$

$$\cos\eta_0 = \frac{3\sqrt{1 - m_0 \sin^2\alpha_0}}{4\sqrt{m_0}\sin\alpha_0}, \qquad 0 < \eta_0 < \frac{\pi}{2}, \qquad \psi_0 = F(\eta_0 \mid m_0) - p_0.$$

Let q_* be the endpoint of $(\psi_0, \sqrt{m_0}, 2\sqrt{m_0}p_0, +1)$. Its zero-winding fiber consists of exactly three rotating sources, all regular: one strictly pre-cut cell-zero minimizer and two post-cut sources of actual period one. There are no higher-period rotating sources. In particular, exterior uniqueness does not hold in every direction.

Computer-assisted proof. Direct outward evaluation at 256 bits gives $U, V > 0$, $W = 3/4$ and

$$\frac{7}{10} < \frac{U}{V} < \frac{18}{25} < \frac{3}{4}, \qquad 4\sqrt{m_0}K(m_0) < 2\sqrt{m_0}p_0 < 6\sqrt{m_0}K(m_0).$$

Thus q_* is exterior and has a period-one source. Lemma 6.12 supplies exactly one regular, strictly pre-cut cell-zero source. All other sources have positive actual period. The exact target is defined by the source, not by its approximation

$$(-7.6417489246, -23.6582865360, 2\arctan(3/4)).$$

For the scalar inverse of Proposition 2.1, $\sin\theta = 24/25$ and $\cos\theta = 7/25$. The two roots of $P - 2C$, with $P = C^2 + (7z/25 - 24/25)^2 - z^2$, lie below $-7/24$, the heading lower bound. The roots of $P + 2C$ lie inside

$$I_{\mathrm{out}} = [-0.281164, -0.280274].$$

The quadratic factorization (8) therefore gives one admissible component, between the latter two roots. On I_{out}, $C < 0$, $xz - y > 0$ and $\cos\theta + z\sin\theta > 0$. There is no component at infinity, and $m > 0$ since $z \neq 0$. The negative rotation sign is excluded by $U > 0$ and $U = \varepsilon q_p D/d$. The $B = 0$ seam is excluded by $V \neq 0$. Equivalently, $C < 0$ on the entire required heading half-plane, since $y < 0$ and $C(-7/24) < 0$.

Thirty-two adjacent rational intervals prove $m > 0.9998$ on all I_{out}. Since the target satisfies $U/W < 12$ and

$$4\mathcal{Q}(0.9998) > 12, \qquad 2\mathcal{Q}(0.99999) > 12,$$

(55) excludes $n \geq 2$ everywhere, and excludes $n \geq 1$ wherever $m \geq 0.99999$. Sixteen rational intervals at each end prove that latter inequality on

$$[-0.281164, -0.28115] \quad \text{and} \quad [-0.28029, -0.280274].$$

It remains to count $N(z) = 1$ on $I = [-0.28115, -0.28029]$, whose endpoints lie strictly inside the admissible component.

An adjacent 256-bit interval cover of I has 1282 leaves. Each either excludes zero from $N - 1$, or excludes zero from N_z and proves strict endpoint signs. The latter test counts one root exactly when the endpoint signs differ. The only two root intervals are

$$\left[-\frac{14378669}{51200000}, -\frac{7189313}{25600000}\right], \qquad \left[-\frac{898519}{3200000}, -\frac{3594033}{12800000}\right].$$

The derivative is strictly positive on the first and strictly negative on the second. Every shared endpoint is excluded as a root by the same strict tests; no interval remains unresolved. Proposition 2.2 proves both sources regular. Their actual period is one, so $t > 4kK > 2kp_1 = t_{\mathrm{cut}}$. Adding the unique cell-zero source gives three. Corollary 6.9 excludes other zero-winding strata.

The exact rational cover, target enclosures and signs are in `data/e109_exterior_count.json`, produced by `research/e109_exterior_count.py`. A full 384-bit replay using unfactored geometry with mean-value enclosures reproduces the count in 1328 nodes and rechecks all 64 modulus intervals. It shares the Arb library and elliptic derivative primitives. A separate exact rational Bernstein calculation checks the domain and modulus polynomial inequalities without that library. The replay and rational audit are recorded in `data/e109_replay.json` and `data/e110_rational_audit.json`. □

6.4 Sharp angular thresholds for positive periods

The previous explicit sector can be sharpened by a variational characterization. The thresholds below concern existence at some radius on a fixed ray, rather than the count at a prescribed radius. Fix $w > 0$, put $m_h = w^2/(1+w^2)$, and define

$$\mathcal{D}_w = \{(m,\alpha) : m_h < m < 1, \quad \arcsin\sqrt{m_h/m} < \alpha < \pi/2\}.$$

For $(m,\alpha) \in \mathcal{D}_w$, set

$$S = \sin\alpha, \quad C = \cos\alpha, \quad d = \sqrt{1-mS^2}, \quad k = \sqrt{m},$$
$$h = \frac{wd}{kS}, \qquad z = \sqrt{1-h^2}, \qquad D = \sqrt{1-mz^2},$$
$$q_n = 2n\mathcal{Q}(m) + F(\alpha \mid m) - E(\alpha \mid m), \qquad B_n(m,\alpha;w) = \frac{Dq_n}{kz(Cq_n + dS)}, \quad n \geq 1.$$

All factors are positive. The letter B_n here denotes a slope function, not the seam coefficient B of the scalar inverse.

Theorem 6.15 (Sharp exterior period thresholds)**.** *For each $w > 0$ and integer $n \geq 1$, the minimum*

$$\beta_n(w) = \min_{(m,\alpha)\in\mathcal{D}_w} B_n(m,\alpha;w) \tag{56}$$

is attained in $\mathcal{D}_w$. With $\sigma(w)$ from Proposition 6.11,

$$\sigma(w) < \beta_1(w) < \beta_2(w) < \cdots < w, \qquad 0 < w - \beta_n(w) < \frac{w}{1+2n\mathcal{Q}(m_h)}. \tag{57}$$

In particular, $\beta_n(w) \to w$.

Fix a strict heading with $|W| = w$ and an exterior spatial ray with $b = |U|/|V| < w$. A rotating source of actual period n reaches some positive radius on that ray if and only if $b \geq \beta_n(w)$. At $b = \beta_n(w)$ every such period-n source is post-cut and conjugate. The zero-winding fiber consists of exactly one source, which is regular and strictly pre-cut, at every positive radius on the ray if and only if $b < \beta_1(w)$. Different periods need not occur at the same radius.

Proof. Lemma 6.12 puts every exterior period-n source at $p = 2nK + F(\alpha \mid m)$ in the stated domain. Its absolute endpoint coordinates are

$$U_n = \frac{q_n D}{d}, \qquad V_n = \frac{kz(Cq_n + dS)}{d}, \qquad |W| = w,$$

so its slope is B_n. Conversely every domain point gives a source. For prescribed signs $s_U, s_V, s_W \in \{-1, 1\}$, choose

$$\varepsilon = s_U, \quad \cos\eta = (-1)^{n+1} s_W h, \quad \sin\eta = (-1)^{n+1}\varepsilon s_V z.$$

With a consistent real amplitude lift, $\psi = F(\eta \mid m) - p$ and $t = 2kp$ then give $(U,V,W) = (s_U U_n, s_V V_n, s_W w)$. The physical radius is $2\sqrt{U_n^2 + V_n^2}/\sqrt{1+w^2} > 0$. For $j > i \geq 1$, direct subtraction gives

$$B_j - B_i = \frac{DdS(q_j - q_i)}{kz(Cq_i + dS)(Cq_j + dS)} > 0, \qquad q_j - q_i = 2(j-i)\mathcal{Q}. \tag{58}$$

We first show that every sublevel $B_n \le B_* < w$ is compact inside $\mathcal{D}_w$. The exterior drift bound implies

$$2n\mathcal{Q}(m) \le q_n < \frac{B_*}{w - B_*}.$$

Since $\mathcal{Q}(m) \to \infty$ as $m \to 1$, this bounds m above by $m_{\max} < 1$. The numerator Dq_n is bounded below by $2n\sqrt{1 - m_{\max}}\,\mathcal{Q}(m_h) > 0$, whereas the denominator is at most $z(q_n + 1)$. Thus the sublevel stays away from $z = 0$, equivalently $mS^2 = m_h$, and also from $m = m_h$. At the remaining boundary $\alpha = \pi/2$,

$$B_n(m, \pi/2; w) = \frac{(2n+1)\mathcal{Q}(m)}{\sqrt{m - m_h}} > w.$$

Indeed, $\mathcal{Q} > 1 - \kappa$, $\kappa = \sqrt{1-m}$, and

$$(1-\kappa)^2 - w^2(m - m_h) = \left(\sqrt{1+w^2}\,\kappa - \frac{1}{\sqrt{1+w^2}}\right)^2 \ge 0.$$

Continuity on the remaining compact modulus interval excludes this boundary too, including joint approaches to the original corners.

Such an exterior sublevel is nonempty for every fixed n, w. Let $e \downarrow 0$ and choose $m = 1 - e^2$, $C = \sqrt{e}$, $S = \sqrt{1-e}$. These points eventually lie in $\mathcal{D}_w$. Writing $a = dS/C$ and $G = D/(kzC)$, rational algebra yields

$$a^2 = 1 - 2e^2 + e^3, \qquad \frac{G^2}{w^2} = 1 + (w + w^{-1})^2 e + O_w(e^2).$$

Moreover $2n\mathcal{Q} < q_n < (2n+1)\mathcal{Q}$ and $\mathcal{Q}(1 - e^2) = \log(4/e) - 1 + o(1)$ [7, Section 19.12]. Thus $q_n \to \infty$, $eq_n \to 0$, and

$$q_n\left(\frac{B_n}{w} - 1\right) = \frac{q_n(G/w - 1) - a}{1 + a/q_n} \longrightarrow -1.$$

Hence $B_n < w$ eventually. Compactness of its sublevel proves that the positive global minimum is attained in the interior and is less than w. An attained minimum is an actual exterior rotating target, so Proposition 6.11 gives $\beta_n > \sigma(w)$. Evaluate (58) at a minimizer of B_{n+1} to get $\beta_n < \beta_{n+1}$. At a minimizer of B_n, $2n\mathcal{Q}(m_h) \le q_n < \beta_n/(w - \beta_n)$ gives the final bound in (57).

The domain is connected, and B_n is unbounded above as $z \to 0$ at fixed m. Its image is therefore $[\beta_n, \infty)$. Source reconstruction with arbitrary prescribed signs proves the claimed support on each exterior ray.

At an interior minimum, both derivatives of B_n at fixed W vanish. The source coordinates (m, α, η) are nonsingular since $k_m > 0$, $p_\alpha = 1/d > 0$ and $\psi_\eta = 1/D > 0$. Replacing η by W is legitimate because $W_\eta = k\,\mathrm{sn}\,p \sin\eta/d \ne 0$. At fixed W, the nonzero row $(1/V, -U/V^2)$ annihilates the derivative of (U, V), so the exponential map is singular. Every source on the threshold ray with actual period n is a minimizer of B_n, hence conjugate. It is post-cut because $p > 2nK \ge 2K > p_1$.

For $b < \beta_1$ there is no positive-period rotating source. Lemma 6.12 and Corollary 6.9 give exactly one regular pre-cut zero-winding source at every radius, including $b = 0$. For $\beta_1 \le b < w$, some radius has a positive-period source as well as the distinct cell-zero minimizer. This proves the sharp strict inequality for all-radius uniqueness. □

The theorem does not classify the corank or fold/cusp type at a threshold, nor assert uniqueness of the minimizing parameters in (56). It concerns the zero-winding fiber; the

full normal fiber still contains its nonzero-winding sources. Corollary 6.13, together with attainment, already gives $\beta_1(3/4) > 1/2$. A complete interval cover gives the following much narrower enclosure.

Corollary 6.16 (A certified angular threshold)**.** *The first exterior threshold at $w = 3/4$ satisfies*

$$0.70555041348314 < \beta_1(3/4) < 0.705550413483143. \tag{59}$$

Computer-assisted proof. An upper witness is given by the exact source parameters

$$m_* = \frac{999956771591962}{10^{15}}, \qquad \alpha_* = \frac{1469648769750615}{10^{15}}, \qquad w = \frac{3}{4}.$$

Direct outward evaluation at 384 bits proves $(m_*, \alpha_*) \in \mathcal{D}_{3/4}$ and $B_1(m_*, \alpha_*; 3/4) < 0.705550413483143$. These rational parameters define a source, not an asserted exact minimizer. The resulting upper bound is also less than $b_* = 0.705551$.

To prove the lower bound, parametrize the full domain by

$$m = 1 - \tfrac{16}{25}e^{-\xi}, \quad \Lambda = \sqrt{m - \tfrac{9}{25}}, \quad v = \tfrac{\pi}{2}\zeta,$$
$$r = \tfrac{9}{25} + \Lambda^2 \sin^2 v, \quad \alpha = \operatorname{atan2}(\sqrt{r}, \Lambda \cos v), \qquad \xi > 0, \quad 0 < \zeta < 1.$$

Here $r = m \sin^2 \alpha$; ξ, ζ are source coordinates, unrelated to the target coordinates x, y. Let $m_c = 1 - (16/25)e^{-16}$. Outward complete-integral evaluation gives

$$2\mathcal{Q}(m_c) > \frac{b_*}{3/4 - b_*}.$$

Lemma 6.12 therefore excludes $\xi \geq 16$ for every source with $B_1 \leq b_*$, in particular for every minimizer.

On $[0, 16] \times [0, 1]$, put $q = q_1$ and

$$D_c = \sqrt{1 - m + \tfrac{9}{16} m(1/r - 1)}, \qquad A_c = \frac{5\Lambda^2 \sin v \cos v}{4\sqrt{r}},$$
$$C_c = \frac{5}{4}\Lambda \sin v \sqrt{1 - r}, \qquad \mathcal{R} = q(D_c - \underline{b} A_c) - \underline{b} C_c, \quad \underline{b} = 0.70555041348314.$$

In the open source domain, $B_1 = qD_c/(A_c q + C_c)$ with positive denominator, so $\mathcal{R} > 0$ proves $B_1 > \underline{b}$. The residual has continuous boundary values on the closed rectangle. A complete 192-bit cover has 7825 rational boxes, each with a strictly positive residual enclosure. Each box uses either direct interval evaluation or a centered mean-value bound with derivatives enclosed on the whole box. Nonfinite derivatives disable that tightening; such a box must pass its direct bound or be subdivided.

The incomplete-integral corner bounds use only strictly positive amplitudes. Their signs follow from the defining integrals: q increases with m and α, whereas $E(\alpha \mid m)$ decreases with m and increases with α. A separate implementation using explicit, simplified derivatives rechecks all 7825 boxes at 256 bits in 7849 nodes. Exact rational partition reconstruction and an independent sweep both prove full coverage without gaps or overlapping interiors. No box remains unresolved. The cover was assembled in two runs, 7176 leaves inherited from a first run and 649 nodes added, and the replay re-evaluates every leaf. The two numerical implementations share Arb special functions; their agreement is not an independent-library review.

The full cover and alternate replay are `data/e113_threshold_cover.json` and `data/e113_threshold_replay.json`. The separate sweep, symbolic derivative audit and rational upper witness are recorded in `data/e114_certificate_audit.json`. Together with attainment from Theorem 6.15, the strict residual bounds prove the strict lower endpoint in (59). □

This encloses the global minimum to an interval of width $3 \cdot 10^{-15}$. A separate localization and second-derivative certificate identifies its unique attaining source and local type in this particular case.

Theorem 6.17 (A unique threshold fold and its radial count window)**.** *Put $w = 3/4$ and $\beta = \beta_1(w)$. The function B_1 has exactly one global minimizer in $\mathcal{D}_w$. In the source coordinates of Corollary 6.16, it belongs to*

$$\mathcal{C} = [9.6027, 9.60275] \times [0.91943, 0.919431]. \tag{60}$$

For each prescribed set of signs of U, V, W, its reconstructed source is an ordinary fold of the full exponential map. Its target radius R_* satisfies

$$26.8980 < R_* < 26.8989. \tag{61}$$

The certificate encloses R_* in $(26.89806, 26.89887)$; the separate audit of the same quantity gives only $(26.8979, 26.8991)$, so the displayed digits rest on one implementation. On that signed ray at slope $b = \beta$, the zero-winding count N_0 is two at R_* and one at every other positive radius.

There exists $\delta > 0$ such that, for each $\beta < b < \beta + \delta$, two radii $R_-(b) < R_+(b)$ give the complete count

$$N_0(R) = \begin{cases} 1, & R \notin [R_-(b), R_+(b)], \\ 2, & R = R_-(b) \text{ or } R = R_+(b), \\ 3, & R_-(b) < R < R_+(b). \end{cases} \tag{62}$$

The additional sources have actual period one and are post-cut: one fold at either endpoint, two regular sources inside. The remaining source is regular and strictly pre-cut. Both radii tend to R_* as $b \downarrow \beta$. A certified choice of δ is supplied in Corollary 6.18.

Computer-assisted proof. Let $b_U = 0.705550413483143$. At this upper comparison level, repeat the residual test in Corollary 6.16. Cut its verified rectangular partition at the four rational boundaries of $\mathcal{C}$. Retain pieces inside $\mathcal{C}$ and prove strict positivity of $q(D_c - b_U A_c) - b_U C_c$ on every other piece, subdividing when necessary. The resulting cover has 7465 excluded boxes and 1343 retained boxes, with none unresolved. A 384-bit alternate implementation rechecks all 7465 exclusions. Exact rational area and pairwise overlap checks within each of the 7825 original boxes independently verify the partition; missing and duplicated pieces fail these checks. The analytic tail exclusion is unchanged and its strict margin is reevaluated. Since $\beta < b_U$, every global minimizer lies in $\mathcal{C}$. Retained pieces carry no positive-residual claim.

Write $H = D^2_{\xi,\zeta} B_1$ and $\mathbf{k} = (V_\zeta, -V_\xi)$. On the whole convex rectangle $\mathcal{C}$, two interval implementations give

$$H_{11} > 0.0033, \qquad \det H > 0.0089, \qquad V_\xi > 1.0231, \qquad \mathbf{k}^T H \mathbf{k} > 9.5.$$

One implementation differentiates the simplified drift gradients in the transformed chart. The separate audit differentiates the original (m, α) formulas symbolically and pulls back their Hessian, including the second derivatives of the coordinate change. It uses

$$q_m = \frac{2E + E(\alpha \mid m) - \sin\alpha\cos\alpha/d}{2(1-m)}, \qquad q_\alpha = \frac{m\sin^2\alpha}{d}, \qquad d = \sqrt{1 - m\sin^2\alpha},$$

with the standard elliptic derivative identities [7, Section 19.4]. The implementations share Arb special functions; this is not an independent-library review. The positive principal

minors make B_1 strictly convex on $\mathcal{C}$. Attainment from Theorem 6.15 therefore gives exactly one global minimizer. It is interior to the original source domain, so its full (ξ, ζ) gradient vanishes, whether or not it lies on the artificial boundary of $\mathcal{C}$. Outward evaluation of $R = 2\sqrt{U^2 + V^2}/\sqrt{1 + w^2}$ on $\mathcal{C}$ gives (61).

The source chart (ξ, ζ, W) is nonsingular by the chart argument in Theorem 6.15; explicitly, $m_\xi = 1 - m > 0$ and $\alpha_\zeta = \pi\Lambda \sin v/(2\sqrt{r}) > 0$. Target coordinates $(B = U/V, V, W)$ are nonsingular since $V \neq 0$. At the minimum, $\nabla B = 0$ and $V_\xi \neq 0$, so the full map has rank two and kernel $(\mathbf{k}, 0)$. Its transverse quadratic derivative is $\mathbf{k}^T H \mathbf{k} > 0$. To see the full-map fold directly, choose source coordinates (s, V, W). The map is $(f(s, V, W), V, W)$, with $f_s = 0$, $f_{ss} > 0$ at the point. The implicit function theorem gives a critical section $s_c(V, W)$. Taylor's integral formula writes $f - f(s_c, V, W)$ as $(s - s_c)^2$ times a smooth positive factor. Rescaling the first source coordinate and subtracting $f(s_c, V, W)$ in the target gives (z^2, V, W). This proves the ordinary fold type in three dimensions. The signed reconstruction transfers the result to every sign choice.

The strict inequality $\beta_2 > \beta$ excludes all higher periods on the threshold ray. Every period-one source there minimizes B_1, so uniqueness gives one such source at one radius R_*. Lemma 6.12 supplies the distinct pre-cut cell-zero source, proving the count at $b = \beta$.

Choose δ with $\beta + \delta < \min(\beta_2, b_U)$. Outside any neighborhood of the unique minimum, the compact complement inside $\mathcal{C}$ has minimum strictly greater than β. Decreasing δ thus places every period-one source in one fold neighborhood. Choose a smaller product neighborhood in (z, V) so that all solutions considered below remain in its interior. At fixed W, the fold representation retaining the original B, V coordinates is

$$B = c(V) + z^2, \qquad c(V_*) = \beta, \qquad c'(V_*) = 0, \qquad c''(V_*) = \frac{1}{\nabla V^T H^{-1} \nabla V} = \frac{\det H}{\mathbf{k}^T H \mathbf{k}} > 0.$$

The curvature formula follows by differentiating the constrained minimum of B at fixed V. Hence $c(V) = b$ has exactly two nearby solutions $V_-(b) < V_+(b)$: between them there are two period-one sources, at either endpoint one fold, and outside none. Compactness excludes remote period-one sources, and $b < \beta_2$ excludes higher periods. Corollary 6.9 and Lemma 6.12 supply exactly one remaining pre-cut source at every radius, in its appropriate stratum. Finally $R = 2\sqrt{1 + b^2}|V|/\sqrt{1 + w^2}$ proves (62) and convergence of the endpoints.

The localization and local bounds are recorded in `data/e115_threshold_localize.json` and `data/e115_threshold_local.json`; the separate Hessian and full exclusion audits are `data/e116_threshold_audit.json` and `data/e116_localization_replay.json`. □

The same proof gives a square-root opening law. With $\kappa = c''(V_*) > 0$,

$$R_+(b) - R_-(b) = 2C_R\sqrt{b - \beta} + O(b - \beta), \qquad C_R = \frac{2\sqrt{1 + \beta^2}}{\sqrt{1 + w^2}}\sqrt{\frac{2}{\kappa}}.$$

These statements concern distinct zero-winding sources; a fold counts once. The full normal fiber remains countably infinite. Theorem 6.19 below separately orders the post-cut travel times over the explicit slope range. The following certificate quantifies a valid slope range for the radial count window.

Corollary 6.18 (An effective first fold window)**.** *Put $w = 3/4$, $\beta = \beta_1(w)$ and $b_c = 0.705551$. For every prescribed signed exterior ray with*

$$\beta < b \leq b_c, \tag{63}$$

the complete count (62) *and the source types in Theorem 6.17 hold. In particular, one may take* $\delta = 0.000000586516857$ *in that theorem. At the exact slope* $b = b_c$*, the fold radii satisfy*

$$\begin{aligned} 26.8301460 < R_-(b_c) < 26.8301463, \\ 26.9671811 < R_+(b_c) < 26.9671814. \end{aligned} \tag{64}$$

This is an interval in the spatial slope $b = |U|/|V|$ *at fixed heading parameter* w*, not an interval of heading angles.*

Computer-assisted proof. Work first with $U, V, W > 0$. In the same source coordinates as above, set

$$\mathcal{C}_b = [9.55, 9.66] \times [0.918, 0.921].$$

The residual test at b_c gives a complete partition with 3111 excluded boxes and 5167 retained boxes in $\mathcal{C}_b$; the $\xi \geq 16$ tail is excluded by the drift inequality. An alternate 384-bit replay checks every excluded box in 3203 nodes. Exact parentwise area and overlap checks verify the partition. Separate edge subdivisions verify that the excluded closed boxes cover all four sides of $\mathcal{C}_b$. Thus $B_1 > b_c$ on its boundary and every source with $B_1 \leq b_c$ lies in its interior. This last boundary assertion is needed below, not just localization to a closed rectangle.

Write $B = B_1$, $H = D^2_{\xi,\zeta} B$, and

$$\Delta = B_\xi V_\zeta - B_\zeta V_\xi, \qquad \mathbf{k} = (V_\zeta, -V_\xi).$$

A 2048-leaf interval cover of $\mathcal{C}_b$ proves

$$H \text{ positive definite}, \qquad V_\xi > 1, \quad V_\zeta > 12, \qquad \mathbf{k}(\Delta) > 6, \qquad B_2 > b_c + 0.02.$$

An independent symbolic calculation in (m, α), followed by the full Hessian pullback for both B and V, replays every local leaf at 384 bits in 2080 nodes. The evaluated kernel derivative is

$$\begin{aligned} \mathbf{k}(\Delta) = \mathbf{k}^T H \mathbf{k} + B_\xi(V_{\xi\zeta} k_\xi + V_{\zeta\zeta} k_\zeta) \\ - B_\zeta(V_{\xi\xi} k_\xi + V_{\xi\zeta} k_\zeta). \end{aligned}$$

No stationary-point assumption is used in these bounds. For B_2 the evaluator uses $q_2 = q_1 + 2\mathcal{Q}(m)$ in the original endpoint formulas. If any $n \geq 2$ had $B_n \leq b_c$, the strict period comparison would give $B_1 < B_n \leq b_c$, placing its parameters in $\mathcal{C}_b$, where $B_n \geq B_2 > b_c$ is a contradiction. Hence no higher period contributes in (63).

The positive coordinate derivatives of V make every nontrivial intersection of $V = v$ with $\mathcal{C}_b$ a connected graph $\xi = \xi_v(\zeta)$. Indeed, the allowed ζ values are determined by $V(\xi_{\text{left}}, \zeta) \leq v \leq V(\xi_{\text{right}}, \zeta)$, and both boundary functions increase strictly. Put $g_v(\zeta) = B(\xi_v(\zeta), \zeta)$. Then

$$g_v' = -\frac{\Delta}{V_\xi}, \qquad g_v'' = \frac{\mathbf{k}(\Delta)}{V_\xi^2} > 0 \quad \text{at every stationary point.}$$

There is at most one stationary point: two negative-to-positive crossings of g_v' would require an intervening zero that is not such a crossing. The graph endpoints lie on the rectangle boundary, where $B > b_c$. Consequently $g_v = b \leq b_c$ has zero, one double, or two simple roots according as its minimum is above, equal to, or below b.

To exclude several disconnected radial windows, use the positive Hessian. For $\beta < b \leq b_c$, the sublevel $S_b = \{B \leq b\}$ is compact and convex with nonempty interior, strictly inside $\mathcal{C}_b$.

Its image under V is an interval $[v_-, v_+]$. Since V is a submersion, $V(\operatorname{int} S_b)$ is open and connected; density of the interior and compactness imply that it equals (v_-, v_+). For every v in this open interval, g_v has minimum below b, hence two roots. At $v_\pm$ the minimum equals b, giving one fold. Outside the closed interval there is no root. The positivity of $\mathbf{k}(\Delta)$ gives ordinary folds of the full map, while the simple roots are regular. The signed reconstruction, radius conversion and unique remaining pre-cut source then prove (62) for the whole range (63).

For (64), interval arithmetic isolates two solutions of $(B - b_c, \Delta) = 0$ in disjoint rational source boxes. For each box X with center c, a nonsingular rational matrix A satisfies

$$c - AF(c) + (I - A\,DF(X))(X - c) \subset \operatorname{int} X, \qquad F = (B - b_c, \Delta).$$

The weighted infinity norm of $I - A\,DF(X)$ is less than 0.001. Thus $z \mapsto z - AF(z)$ is a contraction of X into itself, proving a unique root. An original-coordinate derivative implementation independently repeats both inclusions. Evaluating $R = 2\sqrt{1 + b_c^2}\,V/\sqrt{1 + w^2}$ on their enclosures gives (64). The preceding global argument proves that these roots are the two endpoints, not merely two selected conjugate sources. Finally, $b_c - 0.705550413483143 = 0.000000586516857$ and the strict threshold upper bound give the claimed choice of δ.

The original covers and root certificates are `data/e117_localize.json`, `data/e117_local_window.json` and `data/e117_window_geometry.json`. The replays and boundary audit are `data/e118_global_replay.json`, `data/e118_local_replay.json` and `data/e118_geometry_audit.json`. All interval implementations share Arb special functions. □

The cap b_c is a certified extent, not a maximal one. The wider target of Proposition 6.14 has $b > b_c$ and retains its separate pointwise certificate. Ordering the post-cut travel times requires the additional momentum condition proved next.

Theorem 6.19 (Time order in the first fold window)**.** *In the range of Corollary 6.18, fix a radius $R_-(b) < R < R_+(b)$. Label the two period-one sources by their coordinates $\zeta_L < \zeta_R$ in the chart of Corollary 6.16, and let T_0 be the travel time of the remaining pre-cut source. Then*

$$T_0 < T_L < T_R. \tag{65}$$

The gap $T_R - T_L$ tends to zero at both radial endpoints. At fixed $|V|$, it is strictly increasing with b wherever the pair exists and $b \le b_c$. No radial monotonicity of the gap is asserted.

Proof. It suffices first to take $U, V, W > 0$, with $W = w = 3/4$. Write $S = \sin\alpha$, $C = \cos\alpha$, $d = \sqrt{1 - mS^2}$, $k = \sqrt{m}$, $h = wd/(kS)$, $z = \sqrt{1 - h^2}$ and $D = \sqrt{1 - mz^2}$. For period one,

$$q = 2(K - E) + F(\alpha \mid m) - E(\alpha \mid m), \qquad T = 2k(2K + F(\alpha \mid m)),$$
$$U = qD/d, \qquad V = kz(Cq + dS)/d.$$

Here $z = \sin\eta$ is distinct from the chart coordinate ζ.

The cotangent Gauss lemma and Hamiltonian homogeneity give $dT = \lambda_f \cdot d(x, y, \theta)$ for a smooth fixed-start family of unit-speed normal extremals, including nonminimizing and conjugate sources [8, Section 8.6]. Indeed, scaling the initial covector to $T\lambda_0$ in the time-one exponential gives $T\lambda_f \cdot d(x, y, \theta) = dH(T\lambda_0) = T\,dT$. In the retained convention the conserved spatial momenta are $p_x = \operatorname{sn}\psi$ and $p_y = -\operatorname{dn}\psi/k$. Jacobi subtraction at $\psi = F(\eta \mid m) - 2K - F(\alpha \mid m)$ gives

$$p_x = \frac{-zCd + ShD}{H_0}, \qquad p_y = -\frac{Dd + mzShC}{kH_0}, \qquad H_0 = d^2(1 + w^2).$$

Substitute $x = 2(wU - V)/(1 + w^2)$ and $y = -2(U + wV)/(1 + w^2)$. At fixed W this yields the smooth source identity

$$dT = \ell_U \, dU + \ell_V \, dV, \qquad \ell_U = \frac{2D}{kd(1 + w^2)}, \qquad \ell_V = \frac{2zC}{d(1 + w^2)}. \tag{66}$$

No inverse-map determinant is divided out. Direct differentiation of the displayed clock and endpoint formulas gives the same identity.

The additional interval condition is

$$\left.\frac{d\ell_U}{d\zeta}\right|_V > 0 \quad \text{on } \{B \le b_c\} \cap \mathcal{C}_b, \qquad B = U/V, \tag{67}$$

where $\mathcal{C}_b$ is the rectangle in the preceding proof. To make its certificate explicit, put $g = 1 - m$ and $r = m \sin^2 \alpha$. Then

$$P = \frac{g}{m(1 - r)} + \frac{w^2}{r}, \qquad \ell_U = \frac{2\sqrt{P}}{1 + w^2}, \qquad \left.\frac{d\ell_U}{d\zeta}\right|_V = \frac{V_\xi P_\zeta - V_\zeta P_\xi}{(1 + w^2)\sqrt{P}\, V_\xi}.$$

A complete mixed cover has 117 boxes with positive V_ξ and positive momentum derivative, and 38 boxes with $B > b_c$. An exact area and disjointness audit verifies coverage; a separate sweep rejects missing and duplicated-box controls. The latter boxes cannot contain a point of $\{B \le b_c\}$. An alternate 384-bit calculation in the original (m, α) coordinates replays all 155 boxes in 287 nodes. It uses

$$V_\xi(\ell_U)_\zeta - V_\zeta(\ell_U)_\xi = g\alpha_\zeta\big(V_m(\ell_U)_\alpha - V_\alpha(\ell_U)_m\big)$$

and independently simplified derivatives. Both implementations share Arb special functions, and the first reuses the residual evaluator of the threshold-cover replay. Positivity on the entire rectangle is neither used nor true: at $(\xi, \zeta) = (9.66, .918)$ the derivative is negative, but $B > b_c$. The certificates are `data/e119_time_order.json` and `data/e120_time_order_replay.json`; exact algebra and signed reconstruction checks accompany them.

Fix $V = v > 0$ and follow the connected graph $\xi = \xi_v(\zeta)$ from Corollary 6.18. Put $f(\zeta) = B(\xi_v(\zeta), \zeta)$. Between its two roots $f(\zeta_L) = f(\zeta_R) = b$, that corollary gives $f < b$. The intervening interval therefore lies in $\{B \le b_c\}$. Equation (66) and integration by parts give

$$T_R - T_L = v \int_{\zeta_L}^{\zeta_R} (b - f(\zeta)) \left.\frac{d\ell_U}{d\zeta}\right|_V d\zeta > 0. \tag{68}$$

The unique pre-cut source is minimizing, while both period-one sources are post-cut, so $T_0 < T_L$. Signed reconstruction preserves (m, α, T) and the ζ labels, proving the assertion for every prescribed sign choice. At each radial endpoint the two sources coalesce, and smoothness of T gives a zero limiting gap. Finally, on the regular branches at fixed v,

$$\frac{d(T_R - T_L)}{db} = v(\ell_{U,R} - \ell_{U,L}) > 0$$

by (67). Fixing $|V|$ as b varies changes $R = 2|V|\sqrt{1 + b^2}/\sqrt{1 + w^2}$; this is not radial monotonicity. □

The integral also displays the usual local fold scaling. At a fixed v with constrained minimum $f(\zeta_0) = c(v) < b_c$, put $a = f''(\zeta_0) > 0$ and $\ell = (d\ell_U/d\zeta)|_V(\zeta_0) > 0$. Taylor expansion in (68) gives

$$T_R - T_L = \frac{4\sqrt{2}}{3} \frac{v\ell}{\sqrt{a}} (b - c(v))^{3/2} + O\big((b - c(v))^2\big).$$

This is a local asymptotic at fixed v, with no uniform remainder claimed. First variation and the fold mechanism are classical; the certificate establishes their time-order consequence throughout the specified SE(2) window. No time comparison outside that window or between arbitrary periods follows.

6.5 Time comparison along a continuous period label

A different comparison applies when sources in two actual periods can be joined through the following formal family. Fix $w > 0$ and use S, C, d, k, h, z, D as in the preceding proof, on the open domain $0 < m < 1$, $0 < \alpha < \pi/2$, $h = wd/(kS) < 1$. For a real label $\nu > 0$, put

$$\begin{gathered} q_\nu = 2\nu(K - E) + F(\alpha \mid m) - E(\alpha \mid m), \qquad p_\nu = 2\nu K + F(\alpha \mid m), \\ U_\nu = q_\nu D/d, \qquad V_\nu = kz(Cq_\nu + dS)/d, \qquad T_\nu = 2kp_\nu. \end{gathered} \tag{69}$$

At an integer $\nu = n \geq 1$, these are the positive-signed endpoint and clock formulas in the first quarter of actual period n. At noninteger labels they are formal analytic interpolants, not asserted geodesics. In particular, physical first variation alone does not determine the ν derivative.

Proposition 6.20 (Period action on a connected branch)**.** *On this formal source domain, with w fixed,*

$$\begin{gathered} dT_\nu = \ell_U \, dU_\nu + \ell_V \, dV_\nu + \mathcal{A}(m) \, d\nu, \\ \mathcal{A}(m) = \frac{4[E(m) - (1-m)K(m)]}{\sqrt{m}}, \qquad 0 < \mathcal{A}(m) < 4, \qquad \mathcal{A}'(m) = \frac{2(K-E)}{m^{3/2}} > 0, \end{gathered} \tag{70}$$

where ℓ_U, ℓ_V are as in (66). Consequently, on any C^1 branch $\nu \mapsto (m(\nu), \alpha(\nu))$ with U_ν, V_ν constant,

$$0 < T_{\nu_2} - T_{\nu_1} = \int_{\nu_1}^{\nu_2} \mathcal{A}(m(\nu)) \, d\nu < 4(\nu_2 - \nu_1) \qquad (\nu_1 < \nu_2). \tag{71}$$

Here both labels must lie on that same branch, parametrized by ν.

Proof. The standard elliptic derivatives [7, Sections 19.2 and 19.4], with parameter $m = k^2$, give, on writing $J = 2\nu E + E(\alpha \mid m)$,

$$(q_\nu)_m = \frac{J - SC/d}{2(1-m)}, \quad (q_\nu)_\alpha = \frac{mS^2}{d}, \quad (T_\nu)_m = \frac{J - mSC/d}{k(1-m)}, \quad (T_\nu)_\alpha = \frac{2k}{d}.$$

Substitution in U_ν, V_ν verifies the m and α components of (70). In the remaining direction, $(q_\nu)_\nu = 2(K - E)$ and $(T_\nu)_\nu = 4kK$. The identity $D^2 + mz^2C^2 = d^2(1 + w^2)$ gives

$$\ell_U (U_\nu)_\nu + \ell_V (V_\nu)_\nu = 4(K - E)/k.$$

The difference is $\mathcal{A}(m)$, which proves the full differential identity without treating a noninteger label as a physical extremal. The defining integrals give

$$\mathcal{A}(m) = 4 \int_0^{\pi/2} \cos t \, \frac{\sqrt{m} \cos t}{\sqrt{1 - m \sin^2 t}} \, dt.$$

The fraction lies strictly between zero and one in the open integration interval. Thus $0 < \mathcal{A} < 4$. Differentiating the complete integrals gives $(E - (1-m)K)' = K/2$ and the displayed formula for $\mathcal{A}'$. Finally, restrict the differential identity to the assumed branch and integrate. All source denominators are strictly positive on the domain. □

Dominated convergence gives $\mathcal{A}(m) \to 4$ as $m \to 1$. Therefore a family of such branches uniformly approaching the separatrix has clock increment asymptotic to four times its label increment. This requires a connected continuation and uniform approach to $m = 1$; it is not a uniform comparison between arbitrary sources in adjacent cells.

Proposition 6.21 (Nine rotating sources with ordered periods)**.** *The target*

$$(U, V, W) = (6, 3, 3/4), \qquad (x, y, \theta) = (48/25, -264/25, 2\arctan(3/4)) \tag{72}$$

has exactly nine zero-winding normal sources, all rotating and regular. Their actual-period counts are three in period zero, four in period one and two in period two; no other rotating periods occur. All nine travel times are distinct. Every source in period zero arrives before every source in period one, and every source in period one before every source in period two, with a gap greater than 2 between adjacent groups. Exactly one source is pre-cut; the other eight are post-cut.

The two positive-arc sources in periods one and two having the smaller auxiliary parameter a in the proof below are also the integer endpoints of one branch in Proposition 6.20 over $1 \le \nu \le 2$. Their clocks satisfy

$$3.68729216 < T_2 - T_1 < 3.68729218. \tag{73}$$

These time comparisons concern this target's rotating sources. Its nonzero-winding sources are the countably infinite family of Proposition 6.2.

Computer-assisted proof. We first exhaust the possible rotating sources analytically. Put $p = t/(2k)$, $n = \lfloor p/(2K) \rfloor$ and $\alpha = \mathrm{am}(p - 2nK) \in (0, \pi)$. A boundary $p = 2nK$ would have $W = 0$, so it is excluded. Write $S = \sin\alpha$, $C = \cos\alpha$, $\sigma = S^2$, $d^2 = 1 - m\sigma$, $\eta = \mathrm{am}(\psi + p)$ and $D^2 = 1 - m\sin^2\eta$. The sign s_2 in (50) equals $+1$, because $U > 0$ and $q = p - \epsilon(p) > 0$. Introduce the signed auxiliary scalar

$$a = -(-1)^n kS\sin\eta;$$

it is distinct from the endpoint angle in (2). Then $0 < |a| < 1$: $a = 0$ would give $V = 0$. The midpoint equations become

$$m\sigma = \frac{a^2 + w^2}{1 + w^2}, \qquad V = a\left(1 + \frac{UC}{\sqrt{\sigma - a^2}}\right), \qquad w = 3/4.$$

Since $V = 3 > |a|$, $\operatorname{sgn} C = \operatorname{sgn} a$. In particular, $C = 0$ cannot reach this target. Solving the second identity and restoring this cosine sign gives

$$\begin{aligned} \sigma(a) &= \frac{a^2[36 + (3-a)^2]}{36a^2 + (3-a)^2}, \qquad m(a) = \frac{(16a^2+9)(37a^2 - 6a + 9)}{25a^2(a^2 - 6a + 45)}, \\ q_*(a) &= \frac{4}{5}\sqrt{36 + (3-a)^2}, \qquad a \in (1/3, 1) \text{ or } a \in (-1, -3/7). \end{aligned} \tag{74}$$

The two intervals are exact. Indeed,

$$1 - m = \frac{(1-a^2)[36a^2 - w^2(3-a)^2]}{(1+w^2)a^2[36 + (3-a)^2]}, \qquad 36a^2 - w^2(3-a)^2 = \frac{27}{16}(3a-1)(7a+3).$$

The displayed domains ensure $0 < \sigma < 1$ and $\sigma > a^2$. Conversely these formulas reconstruct all endpoint equations with positive radicals and the forced cosine sign. Thus squaring has introduced no extra arm or lost sign choice.

Set $\alpha_0 = \arcsin\sqrt{\sigma(a)} \in (0, \pi/2)$, $Q = K - E$ and $B = F(\alpha_0 \mid m) - E(\alpha_0 \mid m)$. On the positive arc $\alpha = \alpha_0$; on the negative arc $\alpha = \pi - \alpha_0$. All sources are therefore exactly the nonnegative integer intersections of

$$\begin{aligned} f_n^+(a) &= q_*(a) - 2nQ - B = 0 \qquad (1/3 < a < 1), \\ f_n^-(a) &= q_*(a) - 2(n+1)Q + B = 0 \quad (-1 < a < -3/7). \end{aligned} \tag{75}$$

Each intersection reconstructs one source modulo the $4K$ phase circle, with actual period n. The omitted quarter boundaries have already been excluded, so these are exhaustive scalar charts for this target.

For the positive arc, exact Bernstein coefficients prove $m > .73$ throughout $[1/3, 1]$, and $m > .995$ on $[1/3, .334]$ and $[.999, 1]$. There $q_* \le 8\sqrt{97}/15$, and outward complete-integral evaluation gives

$$6Q(.73) > 8\sqrt{97}/15, \qquad 2Q(.995) > 8\sqrt{97}/15.$$

Since $B > 0$ and Q increases with m, the first inequality excludes $n \ge 3$, and the second excludes every $n \ge 1$ on both tails. On $[.334, .999]$, the two problems $n = 1, 2$ have a complete cover of 1062 sign-excluded intervals and four retained root intervals. Each retained interval has opposite endpoint signs and $0 \notin (f_n^+)'$, proving one root. A separate 448-bit implementation, using rational source derivatives and explicit elliptic partials, replays all exclusions in 1274 nodes and recalculates the root derivative and clock enclosures. A separately included $n = 0$ root is the only one on this arc: every such root has $p = F(\alpha_0 \mid m) < K < p_1$, hence is strictly pre-cut, and classical optimal synthesis implies uniqueness at a fixed target [3, Theorems 3.1–3.2].

For the negative arc, exact polynomial bounds give $m > .86$ globally and $m > .99999$ on $[-1, -.99999]$ and $[-.428572, -3/7]$. Here $q_* \le 8\sqrt{13}/5$ and $0 < B < Q$, so the inequalities

$$5Q(.86) > 8\sqrt{13}/5, \qquad Q(.99999) > 8\sqrt{13}/5$$

exclude every $n \ge 2$ globally and every $n \ge 0$ on the two tails. The compact remainder has 1953 excluded intervals and four unique-root intervals across $n = 0, 1$. An alternate implementation uses generic forward derivatives and the original residual $U_n(a) - 6$, rather than f_n^-, to replay all exclusions in 11841 nodes and verify all four root inclusions. Exact partition audits cover every interval boundary and reject missing and duplicated intervals. Both implementations share Arb special functions; these are separate internal replays, not external review. The certificates are `data/e123_first_quarter.json`, `data/e124_all_quarters.json` and `data/e124_remaining_replay.json`.

Simple scalar roots give full exponential-map regularity. For each sign of a, the source change $(a, \sigma) \mapsto (m, \alpha)$ has determinant

$$\frac{|a|}{(1 + w^2)\sigma^{3/2}\sqrt{1 - \sigma}} > 0.$$

Replace the endpoint residuals by $G_1 = q_n - Ud/D$ and $G_2 = a + UaC/\sqrt{\sigma - a^2} - V$; the residual transformation has determinant $d/D > 0$. On the chosen cosine branch,

$$(G_2)_\sigma = -\frac{U|a|(1 - a^2)}{2\sqrt{1 - \sigma}(\sigma - a^2)^{3/2}} < 0.$$

Along $G_2 = 0$, one has $G_1 = -f_n^\pm$, so its source determinant is $-(G_2)_\sigma (f_n^\pm)' \ne 0$. The full source change $(m, \alpha, w) \mapsto (m, p, \eta)$ has determinant $-1/a \ne 0$; the standard rotating and target charts are also nonsingular. These facts give full rank. Reconstructing all nine root-box centers and integrating the original Hamiltonian system independently corroborates

the target, actual periods and zero lifted heading winding; these finite integrations do not replace the interval count.

The outward clock enclosures are listed below. Within each arc and period, "left" and "right" mean smaller and larger a.

period	arc	branch	lower clock bound	upper clock bound
0	positive	sole	12.119269	12.119271
0	negative	left	13.537705	13.537707
0	negative	right	13.667421	13.667423
1	positive	left	15.805793	15.805796
1	positive	right	16.037329	16.037331
1	negative	left	17.416405	17.416408
1	negative	right	17.436377	17.436379
2	positive	left	19.493085	19.493088
2	positive	right	19.551252	19.551255

These disjoint intervals prove all stated time comparisons. The positive $n = 0$ source is pre-cut, and all $n \geq 1$ sources have $p > 2K > p_1$. For a negative-arc $n = 0$ source, $K < p < 2K$ and $V = 3 > 0$ force $-\operatorname{cn} p\, q - \operatorname{dn} p \operatorname{sn} p > 0$, placing it after the unique Maxwell root $p_1 \in (K, 2K)$. Hence the other eight sources are post-cut. Finally, Corollary 6.9 identifies this complete rotating fiber with the zero-winding fiber.

For the connected comparison in (73), the earlier parameterized certificate remains applicable to the two smaller-a positive-arc sources. In source coordinates $s = -\log(1 - m), \alpha$, its 256 label intervals cover $[1, 2]$. Uniform Krawczyk contraction proves one root in each source rectangle for every label. At all 257 subdivision values, separate root boxes lie inside the adjacent rectangles; uniqueness identifies the roots, and the implicit function theorem joins one analytic branch. An independent 448-bit replay checks all 513 inclusions with maximum contraction below .377, preserving the exact affine label dependence before interval multiplication. Its data are `data/e121_period_action.json` and `data/e122_period_replay.json`. Endpoint clock evaluation gives (73). □

This example has $|U|/|V| = 2 > w$, outside the exterior fold window. It gives a complete target-specific rotating count and time order, as well as the selected connected continuation. Comparisons between arbitrary targets or arbitrary period branches remain open.

6.6 A complete nine-source interval

Proposition 6.22 (Two consecutive folds on a fixed target line)**.** *Let* $(U, V, W) = (u, 3, 3/4)$ *with* $11/2 \leq u \leq 13/2$. *There are exactly two critical values of the rotating endpoint map on this segment, denoted by* $u_- < u_+$, *with*

$$\begin{aligned} 5.5172084304 < u_- < 5.5172084325, \\ 6.4218626764 < u_+ < 6.4218626785. \end{aligned} \tag{76}$$

Each has exactly one critical source, an ordinary post-cut fold: its actual period is one, on the negative scalar arc, at u_- *and three, on the positive arc, at* u_+*: the pair created at* u_- *is the family* $\mathcal{T}_1^-$ *of Corollary 6.24, the pair created at* u_+ *is* $\mathcal{T}_3^+$. *The complete zero-winding counts, including critical sources once, are*

target coordinate	*total*	*period* 0	*period* 1	*period* 2	*period* 3
$11/2 \le u < u_-$	7	3	2	2	0
$u = u_-$	8	3	3	2	0
$u_- < u < u_+$	9	3	4	2	0
$u = u_+$	10	3	4	2	1
$u_+ < u \le 13/2$	11	3	4	2	2

All these sources are rotating. At each target exactly one is pre-cut; all others are post-cut. Away from $u_\pm$ every source is regular.

Computer-assisted proof. The sign argument in Proposition 6.21 applies throughout $u > 3$. With $w = 3/4$ it gives the exhaustive scalar charts

$$\begin{gathered} a \in \left(\frac{3w}{u+w}, 1\right) \quad \text{or} \quad a \in \left(-1, -\frac{3w}{u-w}\right), \\ \sigma = \frac{a^2[u^2 + (3-a)^2]}{u^2a^2 + (3-a)^2}, \qquad m = \frac{a^2+w^2}{(1+w^2)\sigma}, \qquad q_* = \frac{\sqrt{u^2+(3-a)^2}}{\sqrt{1+w^2}}. \end{gathered} \tag{77}$$

Let $\alpha_0 = \arcsin\sqrt{\sigma}$, $Q = K - E$ and $B = F(\alpha_0 \mid m) - E(\alpha_0 \mid m)$. The residuals are $f_n^+ = q_* - 2nQ - B$ and $f_n^- = q_* - 2(n+1)Q + B$. As before, n is the actual period in both expressions.

An analytic monotonicity removes the need to search freely in both a and u. Fix $a \neq 0$, put $r = (a^2+w^2)/(1+w^2)$, and use $r < m < 1$ as the second coordinate. Then

$$u = \frac{3-a}{|a|}\sqrt{\frac{r-ma^2}{m-r}}, \qquad q_*\sqrt{m-r} = \frac{3-a}{|a|}\sqrt{r(1-r)} =: \Lambda(a), \qquad u_m < 0.$$

Changing variable $t = \sqrt{m}\sin\alpha$ in the defining integral gives

$$B_r(m) = \int_0^{\sqrt{r}} \frac{t^2\,dt}{\sqrt{m-t^2}\sqrt{1-t^2}}, \qquad B_r'(m) < 0.$$

The derivative here holds r fixed, not α_0. Moreover,

$$\frac{d}{dm}\left[\sqrt{m-r}\,B_r(m)\right] = \frac{1}{2\sqrt{m-r}}\int_0^{\sqrt{r}} \frac{t^2(r-t^2)\,dt}{(m-t^2)^{3/2}\sqrt{1-t^2}} > 0. \tag{78}$$

Thus both $H_n^+ = \sqrt{m-r}(2nQ + B_r)$ and $H_n^- = \sqrt{m-r}(2(n+1)Q - B_r)$ increase strictly in m. For the second expression, use $2(n+1)Q - B_r > 0$ and its positive m derivative. For positive $n \ge 1$ and negative $n \ge 0$, each H_n increases from zero to infinity. Indeed, $B_r \to Q(r)$ as $m \downarrow r$, whereas B_r has a finite limit and $Q \to \infty$ as $m \uparrow 1$. There is therefore exactly one solution of $H_n = \Lambda$ for each signed a. In particular $G_n = \sqrt{m-r}\,f_n = \Lambda - H_n$ has $\partial_u G_n > 0$, and $\partial_u f_n > 0$ at every root. Each post-cut family is a single analytic graph $u_n(a)$. This does not assert that it has only one critical point.

The following finite certificate excludes all additional critical points on the stated segment. Since $m_u < 0$ and $(q_*)_u > 0$, bounds at $u = 13/2$ apply throughout the band. Four exact polynomial intervals prove $m > .71$ on the positive arc, and $8Q(.71) > 2\sqrt{233}/5 \ge q_*$ excludes $n \ge 4$. On the negative arc, 346 polynomial intervals retain the common a dependence and prove $5Q(m) > q_*$, excluding $n \ge 2$. All 350 intervals have a separate exact-coverage and elliptic replay; their data are `data/e126_period_bounds.json` and `data/e126_period_replay.json`.

Only positive periods $1, 2, 3$ and negative periods $0, 1$ remain post-cut. Four short endpoint tails are excluded analytically by polynomial bounds $m > .999$ on the positive arc

and $m > .999999$ on the negative arc, together with $2Q(.999) > q_*$ and $Q(.999999) > q_*$. The remaining five compact parameter intervals have an exact partition into 285 intervals. On each one, signs of f_n at fixed u, interpreted using the monotonicity of G_n, either exclude the whole target band or enclose every possible root in a smaller u interval. On the resulting rectangle, either $0 \notin \partial_a f_n$, or the rectangle is contained in one of two independently certified local uniqueness boxes. The criterion $f_n = \partial_a f_n = 0$ is equivalent to full conjugacy by the nonsingular source and residual changes in the preceding proposition. The positive $n = 0$ source has $p < K$ and is regular.

The two retained boxes contain the points enclosed in (76). Both have $\partial_u f_n > 0$ and $\partial_{aa} f_n < 0$. To check the full fold condition, write $b = a\cos\alpha/\sqrt{\sigma - a^2} > 0$ and $\widehat{u} = (3-a)/b$. The source change $(a, \sigma) \mapsto (a, \widehat{u})$ has positive second-coordinate derivative. With $R = U_{\text{actual}} - \widehat{u} = -(D/d)f_n$, the two endpoint residuals are $(R + \widehat{u} - u_*, bR)$. At $R = R_a = 0$ their derivative has rank one, and a cokernel applied to the second kernel derivative gives $-bR_{aa} = b(D/d)(f_n)_{aa} \neq 0$. Restoring W proves an ordinary fold of the full endpoint map. Each pair is born with increasing u.

An alternate implementation replays all 285 intervals, all four tails and both larger uniqueness boxes, using the original residual $ud/D - q_n$ and generic directional derivatives. It uses 1768 interval evaluation nodes and rejects ten missing/duplicated cover controls. The two implementations share Arb special functions; these are internal independent replays, not external review. The complete data are `data/e127_critical_cover.json` and `data/e127_critical_replay.json`.

Finally this entire target segment lies strictly inside the rotating image. If P defines $\mathcal{F}(3, 3/4)$, elementary bounds give $3 < P < 4$ and $\coth P < 5/4$, hence $\mathcal{F}(3, 3/4) < 3$. Properness over the strict image and Corollary 6.9 exclude source arrivals from other regimes. Starting with the complete $u = 6$ count in Proposition 6.21, the two simple pair births give the table. The period and arc labels cannot change on a regular continuation, since the excluded quarter joins, period boundaries and $a = 0$ are the only such label boundaries. The same cut inequalities as in that proposition give one pre-cut source and all remaining sources strictly post-cut. □

6.7 Analytical comparison within a signed period family

The within-family comparison extends beyond the preceding target segment. Here $u = U > 0$, $V > 1$ and $w = W > 0$; all three are the target coordinates already used above, with heading $2\arctan w$. The scalar a remains an auxiliary source parameter, not a heading angle.

Theorem 6.23 (Component time order at general parameters)**.** *Fix $V > 1$ and $w > 0$. For each positive arc with actual period $n \geq 1$, and each negative arc with actual period $n \geq 0$, the rotating sources form a proper real-analytic graph $u = g_n^{\pm}(a)$ on $(0, 1)$ or $(-1, 0)$, respectively. Both ends of each graph tend to $+\infty$. Write $T(a)$ for its travel time and $\lambda(a, u)$ for the coefficient of du in first variation at fixed V, w.*

If $a_L < a_R$ are any two distinct sources in the same signed period family at the same target u, then their moduli and momenta satisfy

$$m_R < m_L, \qquad \lambda(a_R, u) > \lambda(a_L, u) > 0. \tag{79}$$

If, in addition, $[a_L, a_R]$ is a nondegenerate connected component of $\{a : g_n^{\pm}(a) \leq u\}$, then

$$T(a_R) > T(a_L). \tag{80}$$

Where its endpoints vary as regular source branches, $d(T_R - T_L)/du > 0$. In particular, these time conclusions apply whenever the family has exactly two regular sources at the target. No uniqueness of a minimum of $g_n^{\pm}$ is asserted.

Proof. The signed reconstruction in Proposition 6.21 uses only $u > 0$ and $V > |a|$, so it applies for every $V > 1, w > 0$. The identity $V = a + uaC/\sqrt{\sigma - a^2}$ forces $\operatorname{sgn} C = \operatorname{sgn} a$ and excludes $a = 0$ and $C = 0$. Solving it with this sign restored gives

$$\sigma = \frac{a^2[u^2 + (V-a)^2]}{u^2a^2 + (V-a)^2}, \qquad m = \frac{a^2 + w^2}{(1+w^2)\sigma}, \qquad q_* = \frac{\sqrt{u^2 + (V-a)^2}}{\sqrt{1+w^2}},$$
$$a \in \left(\frac{wV}{u+w}, 1\right) \quad \text{or} \quad a \in \left(-1, -\frac{wV}{u-w}\right). \tag{81}$$

The positive arc is nonempty for $u > w(V-1)$; the negative one requires $u > w(V+1)$. Put $\alpha_0 = \arcsin\sqrt{\sigma} \in (0, \pi/2)$, $Q(m) = K(m) - E(m)$ and $B(\alpha_0, m) = F(\alpha_0 \mid m) - E(\alpha_0 \mid m)$. The source equations are

$$q_* = 2nQ + B \quad (a > 0), \qquad q_* = 2(n+1)Q - B \quad (a < 0). \tag{82}$$

The negative arc uses the full amplitude $\pi - \alpha_0$; n remains the actual period in both equations.

For completeness, the monotone-modulus argument of (78) applies without fixing V, w. For fixed a put $r = (a^2 + w^2)/(1 + w^2)$. Then $r < m < 1$ and

$$u = \frac{V-a}{|a|}\sqrt{\frac{r - ma^2}{m - r}}, \qquad u_m < 0, \qquad q_*\sqrt{m-r} = \frac{V-a}{|a|}\sqrt{r(1-r)}.$$

The rescaled drifts $\sqrt{m-r}(2nQ + B_r)$ and $\sqrt{m-r}(2(n+1)Q - B_r)$ increase strictly from zero to infinity. Here $B_r = B(\arcsin\sqrt{r/m}, m)$; the positive derivative of its weighted form is (78). On the negative arc use $B_r < Q$ and $B_r' < 0$. Thus each equation has exactly one modulus root for each signed a, with nonzero modulus derivative. The implicit function theorem gives the stated analytic graph and $f_u > 0$ at every root of $f = q_* - (2nQ + B)$ or $q_* - (2(n+1)Q - B)$. The physical inequality $u > w(V-a)/|a|$ gives $u \to \infty$ as $a \to 0$. As $|a| \to 1$, $r \to 1$ and the drift is at least $2nQ(r)$ on the positive arc, or $Q(r)$ on the negative arc. It diverges, forcing $u \to \infty$. This proves properness, without a critical-point count.

At fixed target, direct differentiation gives

$$\sigma_a = \frac{2a(V-a)[u^2(V-a^3) + (V-a)^3]}{[u^2a^2 + (V-a)^2]^2}, \qquad (q_*)_a = -\frac{V-a}{(1+w^2)q_*} < 0. \tag{83}$$

The bracket is positive because $V > 1$ and $|a| < 1$. Thus α_0 increases with a on the positive arc and decreases on the negative arc. From the defining integrals [7, Equations 19.2.4–19.2.8],

$$B(\alpha, m) = \int_0^{\alpha} \frac{m\sin^2 t}{\sqrt{1 - m\sin^2 t}}\, dt, \qquad \partial_m \frac{m\sin^2 t}{\sqrt{1 - m\sin^2 t}} = \frac{\sin^2 t(2 - m\sin^2 t)}{2(1 - m\sin^2 t)^{3/2}} \geq 0.$$

The last derivative is strictly positive for $0 < t < \pi/2$. With $0 < \alpha < \pi/2$ fixed, $0 < B_m < Q_m$ and $B_\alpha > 0$. Consequently $2nQ + B$ increases in both m and α, while $2(n+1)Q - B$ increases in m and decreases in α. If $m_R \geq m_L$, the signed amplitude change in either case would make the right drift strictly larger. This contradicts $q_*(a_R) < q_*(a_L)$ in (82). Hence $m_R < m_L$.

First variation supplies the momentum comparison. Write $S = \sin\alpha > 0$, $C = \cos\alpha$, $z = a/(kS)$, $h = wd/(kS)$, $d^2 = 1 - mS^2$ and $D^2 = 1 - mz^2$, where α is the full amplitude. Jacobi subtraction [7, Equations 22.8.1 and 22.8.3] gives

$$p_x = \frac{-zCd + ShD}{1 - a^2}, \qquad p_y = -\frac{Dd + k^2 zShC}{k(1 - a^2)}.$$

The actual-period parity factors cancel. Since $x = 2(wu - V)/(1 + w^2)$ and $y = -2(u + wV)/(1 + w^2)$,

$$\lambda(a, u) = \frac{2(wp_x - p_y)}{1 + w^2} = \frac{2D}{kd(1 + w^2)} = \frac{2u|a|}{\sqrt{(a^2 + w^2)[u^2a^2 + (V - a)^2]}} > 0. \tag{84}$$

Here $1 - a^2 = (1 + w^2)d^2$. The endpoint identity $u = q_* D/d$ implies

$$\sqrt{m}\, q_* \lambda = \frac{2u}{1 + w^2}. \tag{85}$$

Both m and q_* strictly decrease from a_L to a_R at roots. This proves (79).

It remains to fix the sign of the clock difference, rather than just its derivative. Set $g = g_n^{\pm}$ and let $[A, B]$ be a nondegenerate connected component of $\{g \le u\}$. Properness makes it compact inside one signed arc, with $g(A) = g(B) = u$ and $v_0 := \min_{[A,B]} g < u$. The clock is analytic along the graph, and first variation gives $T'(a) = \lambda(a, g(a))g'(a)$, including at its critical points. Define on $[A, B]$

$$H(a) = T(a) + \int_{g(a)}^{u} \lambda(a, v)\, dv.$$

Then $H'(a) = \int_{g(a)}^{u} \partial_a \lambda(a, v)\, dv$ and $H(A) = T(A)$, $H(B) = T(B)$. For almost every $v \in (v_0, u)$, the section $\{a \in [A, B] : g(a) \le v\}$ is a finite union of intervals $[l_j(v), r_j(v)]$. Indeed, the nonconstant analytic graph has only finitely many critical points on this compact interval. Fubini's theorem therefore yields

$$T(B) - T(A) = \int_{v_0}^{u} \sum_j \big[\lambda(r_j(v), v) - \lambda(l_j(v), v)\big]\, dv > 0. \tag{86}$$

Every bracket is positive by (79); nonempty regular sections occur on a set of positive measure. The integrand $\partial_a \lambda$ is continuous on the compact integration region; no pointwise sign of this derivative away from the source graph is needed. This proves (80). For regular endpoints, first variation directly gives $d(T_R - T_L)/du = \lambda_R - \lambda_L > 0$. Finally, if exactly two roots exist and both are regular, the graph's infinite endpoint limits force them to bound its unique sublevel component. □

This theorem compares momenta within one signed period family and times across one connected sublevel component. It does not order times between separate components, different signs or different periods, or determine the number of sources at an arbitrary target.

6.8 Complete time order on the certified target segment

Corollary 6.24 (Time ordering throughout the fixed target segment)**.** *On the segment of Proposition 6.22, let $\mathcal{T}_n^+(u)$ and $\mathcal{T}_n^-(u)$ be the sets of travel times in the positive and*

negative scalar arcs, with actual period n. For nonempty families, with $A < B$ meaning $\max A < \min B$,

$$\mathcal{T}_0^+ < \mathcal{T}_0^- < \mathcal{T}_1^+ < \mathcal{T}_1^- < \mathcal{T}_2^+ < \mathcal{T}_3^+, \tag{87}$$

where absent families are omitted. Within each two-source family, label the sources by $a_L < a_R$. Then $T_L < T_R$ and

$$\frac{d}{du}(T_R - T_L) > 0$$

on its regular existence interval. At either fold the merging source is counted once. Thus all distinct zero-winding sources of any target on the segment have different travel times.

Computer-assisted proof. Proposition 6.22 gives exactly two regular sources in each present post-cut signed family away from its fold. Theorem 6.23 makes them the endpoints of one sublevel component and proves their strict time order and increasing gap. At a fold the two sources coincide and are counted once.

For comparisons between families, (84) also holds on the positive period-zero branch. With $w = 3/4$,

$$0 < \lambda(a, u) < \frac{2}{w} = \frac{8}{3}, \tag{88}$$

because the exact reciprocal-square identity is

$$\frac{1}{\lambda^2} - \frac{w^2}{4} = \frac{a^2}{4} + \frac{(a^2 + w^2)(V - a)^2}{4u^2a^2} > 0.$$

Continuous clock and momentum limits extend the integrated bounds to fold endpoints. The earlier 14-interval paired-momentum certificate, its 38 root enclosures, two fold corridors and three initial clock-gap anchors remain separately replayed corroboration in `data/e129_pair_cover.json` and `data/e129_pair_replay.json`; the general analytical theorem supplies the within-family proof here.

Finally, any inter-family clock gap has derivative of absolute value less than $8/3$, since both individual derivatives belong to $(0, 8/3)$. The certified $u = 6$ clocks in Proposition 6.21 give 32 comparisons between its distinct signed period families. After subtracting $4/3$, every gap still exceeds .0457; since $|u-6| \le 1/2$, their order persists wherever both branches exist. This includes the direct comparison of $\mathcal{T}_1^+$ with $\mathcal{T}_2^+$ when $\mathcal{T}_1^-$ is absent. Both $\mathcal{T}_3^+$ clocks increase from their common upper-fold clock. Subtracting the largest earlier clock at $u = 6$ and $4/3$ from its certified lower bound leaves a margin greater than 2.524. Thus $\mathcal{T}_3^+$ follows every earlier family. The exact rational comparison audit is `data/e129_analytic_audit.json`. Equation (87) follows, including the fold targets by continuity and counting their single critical sources once. □

This is a time-order result for the complete zero-winding fiber on the specified segment. It makes no comparison with nonzero winding classes or with branches at arbitrary targets.

6.9 Simultaneous scaling of the endpoint and period

Fix $U_0 > 0$, $V_0 > 1$, $w > 0$ and $\nu_0 \ge 1$. We now follow selected period families while both the endpoint and their effective label grow:

$$(U, V, W) = (sU_0, sV_0, w), \qquad P_{s\nu_0}, \quad N_{s\nu_0 - 1}, \qquad s \ge 1. \tag{89}$$

Here P_n and N_n denote the positive and negative scalar arcs with *actual* period n. At non-integer labels these are the formal analytic families of Section 6.5; they represent geodesics

only when the actual labels are integers. The results below count these two families, not all periods or the full endpoint fiber.

Put $c=\sqrt{1+w^2}$, $Q=K-E$, $a=\epsilon t$ with $\epsilon\in\{1,-1\}$ and $0<t<1$, and use the short amplitude $\alpha\in(0,\pi/2)$. The reconstruction (81)–(82) reads

$$\begin{aligned} R=\sqrt{U^2+(V-\epsilon t)^2}, \qquad q=R/c, \qquad B=F(\alpha\mid m)-E(\alpha\mid m),\\ q=2s\nu_0 Q+\epsilon B, \qquad T=4s\nu_0\sqrt{m}K+\epsilon\,2\sqrt{m}F(\alpha\mid m). \end{aligned} \tag{90}$$

For $\epsilon=-1$, the full amplitude is $\pi-\alpha$. Its incomplete integrals are $2K-F$ and $2E-E(\alpha\mid m)$; this explains the minus-one shift in (89).

Lemma 6.25 (Signed action and a positive intercept)**.** *On either formal source family, with w fixed and effective label ν,*

$$dT=\ell_U\,dU+\ell_V\,dV+\mathcal{A}(m)\,d\nu, \quad (\ell_U,\ell_V)=\frac{2(U,V-\epsilon t)}{c\sqrt{m}R}, \tag{91}$$

where $\mathcal{A}$ is given in (70). Define, with $\sin^2\alpha=(t^2+w^2)/(c^2m)$,

$$j=c[E(\alpha\mid m)-(1-m)F(\alpha\mid m)], \qquad p=\frac{t\cos\alpha}{c\sin\alpha\sqrt{1-m\sin^2\alpha}}, \qquad I=j-tp.$$

Then $j_t=p$ at fixed m, $p=(V-\epsilon t)/R$ at a source, and

$$\begin{aligned} I=cm(1-m)\int_0^\alpha\frac{\sin^2 z}{(1-m\sin^2 z)^{3/2}}\,dz\\ +\frac{w^2\cos\alpha}{c\sin\alpha\sqrt{1-m\sin^2\alpha}}>0. \end{aligned} \tag{92}$$

Consequently every regular C^1 source branch parametrized by s in (89) satisfies

$$\frac{d(T/s)}{ds}=-\frac{\epsilon\,2I}{c\sqrt{m}\,s^2}. \tag{93}$$

Proof. The defining elliptic derivatives [7, Sections 19.2 and 19.4] give $j_t=p$ and $j_m=cF(\alpha\mid m)/2$, with the other variable fixed. The amplitude derivative is included in the latter identity. The geometric reconstruction is

$$U=cq\sqrt{1-p^2}, \qquad V=\epsilon t+cqp.$$

It gives $\ell_U dU+\ell_V dV=2\,dq/\sqrt{m}+\epsilon\,2p\,dt/(c\sqrt{m})$. On using $q=2\nu Q+\epsilon B$, the clock decomposes as

$$T=2q/\sqrt{m}+\nu\mathcal{A}(m)+\epsilon\,2j/(c\sqrt{m}).$$

Differentiate this expression. The dt terms use $j_t=p$, and its remaining dm coefficient vanishes because $\mathcal{A}'=2Q/m^{3/2}$ and $2mj_m-j=cB$. This proves (91) for both signs, without assigning physical first variation to a noninteger period.

Write $S=\sin\alpha$, $C=\cos\alpha$, $d=\sqrt{1-mS^2}$. Since $t^2=c^2mS^2-w^2$,

$$I=c[E(\alpha\mid m)-(1-m)F(\alpha\mid m)-mSC/d]+w^2C/(cSd).$$

The expression in brackets vanishes at $\alpha=0$ and its derivative is $m(1-m)\sin^2\alpha/(1-m\sin^2\alpha)^{3/2}$. This proves (92). Finally, apply (91) along the scaling and subtract the clock decomposition from sT'. The complete-period action cancels and $sT'-T=-\epsilon\,2(j-tp)/(c\sqrt{m})$, proving (93). □

Lemma 6.26 (Uniform normalized limit)**.** *Let $R_0 = \sqrt{U_0^2 + V_0^2}$ and $q_0 = R_0/c$. There is a unique $m_0 \in (0,1)$ satisfying $2\nu_0 Q(m_0) = q_0$. Every source in (89) that exists at scale s satisfies, uniformly in the sign and source choice as $s \to \infty$,*

$$m = m_0 + O(s^{-1}), \qquad T/s = C_0 + O(s^{-1}), \qquad C_0 = 4\nu_0\sqrt{m_0}K(m_0). \tag{94}$$

This assertion does not assume or imply existence at every scale.

Proof. The reverse triangle inequality and $0 < B < Q$ give

$$|q/s - q_0| < \frac{1}{cs}, \qquad |2\nu_0 Q - q_0| < \frac{Q + 1/c}{s}.$$

Since $q_0 > 1/c$ and $\nu_0 \geq 1$, all such moduli belong to the fixed compact subinterval determined by

$$0 < \frac{q_0 - 1/c}{2\nu_0 + 1} < Q(m) < \frac{q_0 + 1/c}{2\nu_0 - 1} =: Q_{\mathrm{hi}} < \infty.$$

Moreover $|Q(m) - q_0/(2\nu_0)| \leq (Q_{\mathrm{hi}} + 1/c)/(2\nu_0 s)$. Use $Q' = E/[2(1-m)] > 0$ to invert this bound on the compact interval. The clock assertion follows from (90), since $0 < F(\alpha \mid m) < K(m)$ there. □

If a positive source continues regularly for every $s \geq 1$ along this scaling, Lemmas 6.25–6.26 give $T_P(1) > C_0$. Such a negative continuation gives $T_N(1) < C_0$. These are conditional comparisons: continuation is an essential hypothesis. The obstruction below shows that even joint initial family existence does not guarantee persistence to large scales.

Theorem 6.27 (Eventual counts and clocks away from the critical surface)**.** *Suppose $U_0 > wV_0$ and put*

$$r = U_0/R_0, \quad \beta = V_0/R_0, \quad t_b = wV_0/U_0, \qquad m_\star = \frac{(\beta + wr)^2}{c^2}.$$

If $m_0 > m_\star$, then for all sufficiently large s each selected sign family has exactly two regular sources. If $m_0 < m_\star$, both families are eventually empty. They are also eventually empty when $U_0 \leq wV_0$.

In the four-source case, their limiting absolute source coordinates $t_L < t_R$ satisfy $t_L t_R = t_b$. Define I_L, I_R by (92) at (t_i, m_0) and put $D_i = 2I_i/(c\sqrt{m_0})$. Then $0 < D_L < D_R$ and

$$T_{\epsilon,i}(s) = sC_0 + \epsilon D_i + O(s^{-1}), \qquad T_{+,i}(s) - T_{-,i}(s) = 2D_i + O(s^{-2}). \tag{95}$$

Thus eventually $T_{-,R} < T_{-,L} < T_{+,L} < T_{+,R}$. The scale threshold depends on the fixed base parameters and is not quantified here. Equality $m_0 = m_\star$ is treated separately below.

Proof. Put $\tau = 1/s$ and, for $0 < x < 1$, define

$$\begin{aligned} z_\epsilon &= V_0 - \epsilon\tau x, \qquad \sigma_\epsilon = \frac{x^2(U_0^2 + z_\epsilon^2)}{U_0^2 x^2 + z_\epsilon^2}, \quad \mu_\epsilon = \frac{x^2 + w^2}{c^2\sigma_\epsilon}, \\ H_\epsilon(x,\tau) &= \frac{\sqrt{U_0^2 + z_\epsilon^2}}{c} - 2\nu_0 Q(\mu_\epsilon) - \epsilon\tau B(\arcsin\sqrt{\sigma_\epsilon}, \mu_\epsilon). \end{aligned} \tag{96}$$

These are analytic near every interior limiting source, with the exact reflection $H_-(x,\tau) = H_+(x,-\tau)$. At $\tau = 0$ the geometric modulus obeys

$$\mu_\infty(x) = \frac{r^2x^2 + r^2w^2 + \beta^2 + \beta^2w^2/x^2}{c^2}, \qquad \mu_\infty(x) - m_\star = \frac{(rx - \beta w/x)^2}{c^2}. \tag{97}$$

The limiting physical arc is $(t_b, 1)$, with modulus one at either endpoint and a unique minimum at $x = \sqrt{t_b}$. For $m_0 > m_\star$ the equation $\mu_\infty(x) = m_0$ has two simple roots in that arc. They solve

$$r^2 y^2 + (r^2 w^2 + \beta^2 - c^2 m_0) y + \beta^2 w^2 = 0, \qquad y = x^2,$$

so $t_L t_R = t_b$. The analytic implicit-function theorem applied to H_ϵ gives one source near each root for sufficiently small $|\tau|$. All other sources are excluded by compactness: a sequence of extra sources at unbounded scales would have a limiting absolute coordinate in $[t_b, 1]$, while Lemma 6.26 forces its modulus to m_0. The endpoints, with modulus one, cannot occur; an interior limit must be t_L or t_R, contradicting local uniqueness. The same argument proves eventual emptiness when $m_0 < m_\star$. If $U_0 < wV_0$, the positive physical arc eventually disappears and the negative arc is empty. If $U_0 = wV_0$, any possible positive source has $x \to 1$ and hence $m \to 1$, contradicting the uniform compact modulus bound. Simplicity transfers to full-map regularity in the source chart.

For either limiting root, $p(t_i, m_0) = \beta$. Expanding the source condition gives, with $B_i = B(\alpha_i, m_0)$,

$$m_{\epsilon,i} = m_0 - \frac{\epsilon}{s} \frac{B_i + \beta t_i / c}{2\nu_0 Q'(m_0)} + O(s^{-2}).$$

Using $(\sqrt{m}K)' = E/[2\sqrt{m}(1-m)]$ in the clock formula yields the first expansion in (95). The reflection of H_ϵ extends to the locally unique analytic roots and their normalized clocks. Their even powers of τ cancel in the difference, giving its stated $O(s^{-2})$ remainder. At fixed m_0, $p(t) > \beta$ for $t_L < t < t_R$: the sign of $p(t)^2 - \beta^2$ is the opposite of the quadratic above, with a positive denominator. Since $j_t = p$,

$$I_R - I_L = \int_{t_L}^{t_R} [p(t) - \beta]\, dt > 0.$$

Together with Lemma 6.25, this proves the order. □

Theorem 6.28 (Critical scaling classification)**.** *Assume $U_0 > wV_0$ and $m_0 = m_\star$, and put $t = \sqrt{wV_0/U_0}$. Write $H = H_+$ in (96), and evaluate the following partial derivatives at $(x, \tau) = (t, 0)$, keeping the base target fixed:*

$$A = -H_{xx}/2 > 0, \qquad L = H_\tau, \qquad b = H_{x\tau}, \qquad c_2 = H_{\tau\tau}/2, \qquad \kappa = c_2 + b^2/(4A). \quad (98)$$

For all sufficiently large s, the complete selected-family counts are

	$P_{s\nu_0}$	$N_{s\nu_0 - 1}$
$L > 0$	2	0
$L < 0$	0	2
$L = 0,\ \kappa > 0$	2	2
$L = 0,\ \kappa < 0$	0	0

All counted sources are regular. The case $L = \kappa = 0$ is excluded. If $\epsilon L > 0$, their absolute coordinates are $t \pm \sqrt{\epsilon L / A}\, s^{-1/2} + O(s^{-1})$. If $L = 0, \kappa > 0$, they are $t + [\epsilon b/(2A) \pm \sqrt{\kappa/A}] s^{-1} + O(s^{-2})$. No explicit starting scale or critical clock-gap exponent is asserted.

Proof. The square identity (97) gives $H(t, 0) = H_x(t, 0) = 0$ and $A = 8\nu_0 Q'(m_0) r^2 / c^2 > 0$. The analytic implicit-function theorem gives a unique nearby critical point $x_c(\tau)$, a strict maximum on a fixed small interval about t. Its endpoints retain negative residual values. Taylor expansion gives

$$x_c(\tau) = t + \frac{b}{2A}\tau + O(\tau^2), \qquad g(\tau) := H(x_c(\tau), \tau) = L\tau + \kappa\tau^2 + O(\tau^3).$$

Strict concavity gives exactly two simple roots when $g > 0$ and none when $g < 0$. The negative family has maximum value $g(-\tau)$. The compact exclusion in the preceding proof applies also at equality: every possible source at unbounded scales must converge to the unique limiting root t. Hence these local counts are complete for the selected families. Finally, when $L \neq 0$, substitute $\tau = h^2$, $x = t + hy$ and divide the residual by h^2. The limiting polynomial is $-Ay^2 + \epsilon L$, with two simple roots for the winning sign. When $L = 0, \kappa > 0$, use $x = t + \tau y$ and divide by τ^2; the limiting polynomial is $-Ay^2 + \epsilon by + c_2$. Analytic implicit inversion of these two polynomials proves the expansions. □

The coefficients are explicit in the critical target parameters. For $0 < t < 1$, $w > 0$, set

$$\begin{aligned} D = t^4 + w^2, \quad r = w/\sqrt{D}, \quad \beta = t^2/\sqrt{D}, \quad m = \frac{(t^2+w^2)^2}{c^2 D}, \\ R_0 = 2c\nu_0 Q(m), \quad (U_0, V_0) = R_0(r, \beta), \quad \alpha = \arcsin\sqrt{D/(t^2+w^2)}. \end{aligned} \tag{99}$$

Require $\nu_0 \geq 1$ and $V_0 > 1$. These parameters describe exactly the critical targets under consideration. In particular,

$$L = \frac{t}{c\sqrt{D}}\left[\frac{E(m)(t^2+w^2)}{Q(m)(1-t^2)} - t^2\right] - B(\alpha, m), \tag{100}$$

which is independent of ν_0. There is no single-transition assumption.

Proposition 6.29 (Certified critical examples)**.** *At $w = 3/10$, the parameter choices $t = 3/5$ and $t = 4/5$, both with $\nu_0 = 10$, have $L > 0$ and $L < 0$, respectively. There are at least three distinct zeros of L at this same w. Each interval below contains exactly one zero; with its listed label it satisfies $V_0 > 1$ and the listed sign of κ:*

interval for t	ν_0	$\operatorname{sgn}\kappa$
$(9756330442696770162 2, 9756330442696770162 3)/10^{22}$ (≈ 0.00976)	50000	$-$
$(74226934183767748 39, 74226934183767748 40)/10^{20}$	10	$+$
$(87334017356145181518, 87334017356145181519)/10^{20}$	10	$-$

Thus all four outcomes of Theorem 6.28 occur. The three intervals do not establish an exhaustive zero count.

Computer-assisted proof. The 384-bit certificate and its producer are

`data/e136_critical_certificate.json`
`research/e136_critical_certificate.py`.

They check the real domain and $V_0 > 1$ throughout each rational input interval. It bounds L with opposite signs at each pair of endpoints and bounds L' away from zero throughout the interval, proving existence and uniqueness of its zero. The derivative L' here varies the base critical target in (99); it is not a partial derivative at a fixed endpoint. It also encloses κ with the stated strict sign throughout each interval. The two rational point inputs have the asserted strict L signs. Full coefficient formulas and their chain-rule reductions are recorded in the accompanying proof audit `references/e136/proof_audit.md`. An independent implementation at 140 decimal digits differentiates the full-amplitude residual and places all 45 coefficient and target values inside the raw outward intervals. That replay corroborates the certificate; the interval signs and the analytical classification establish the stated outcomes. □

Proposition 6.30 (Joint existence without persistence)**.** *There is an exact target with at least two regular P_2 sources and two regular N_1 sources, but no source in either scaled family P_{2s}, N_{2s-1} for any real $s \geq 1024$.*

Computer-assisted proof with a uniform tail exclusion. Use a positive source of actual period two, with

$$m = 41/50, \qquad \alpha = 3/10, \qquad w = 3/100.$$

Set $c = \sqrt{1+w^2}$, $S = \sin\alpha$, $C = \cos\alpha$, $d = \sqrt{1-mS^2}$, and put

$$a = \sqrt{c^2 mS^2 - w^2}, \quad p = \frac{aC}{cSd}, \quad q = 4Q(m) + B(\alpha, m), \quad U_0 = cq\sqrt{1-p^2}, \quad V_0 = a + cqp.$$

These exact expressions give

$$(U_0, V_0, w) \approx (2.06467730131716, 4.34439453433560, .03).$$

The 320-bit certificate `data/e135_joint_obstruction.json` encloses four distinct roots in the signed a intervals

$$\begin{array}{r|l} N_1 & (-2729390928, -2729390924)/10^{10} \\ N_1 & (-2548570550, -2548570545)/10^{10} \\ P_2 & \quad (2250499601, 2250499604)/10^{10} \\ P_2 & \quad (2660391869, 2660391872)/10^{10} \end{array}$$

Opposite endpoint signs and a nonzero derivative enclosure prove one regular root in each box. No exclusion of further initial roots is used. The same certificate proves $27/10^5 < m_\star - m_0 < 29/10^5$.

For a hypothetical scaled source write $z = V_0 - \epsilon t/s$. The square identity gives

$$m \geq M(z) := \frac{(z + wU_0)^2}{c^2(U_0^2 + z^2)}.$$

For $S_0 = 1024$, the certificate checks $U_0 > w(V_0 + 1/S_0)$, so M increases on $[V_0 - 1/S_0, V_0 + 1/S_0]$. Conversely, the source equation and $B < Q$ imply, for every $s \geq S_0$,

$$Q(m) \leq \frac{q_0 + 1/(cs)}{2\nu_0 - 1/s} \leq \frac{q_0 + 1/(cS_0)}{2\nu_0 - 1/S_0}, \qquad \nu_0 = 2.$$

Outward evaluation gives

$$Q(M(V_0 - 1/S_0)) - \frac{q_0 + 1/(cS_0)}{2\nu_0 - 1/S_0} > 0.000226.$$

This contradicts both bounds for every real $s \geq S_0$, proving the entire tail empty. The separate full-amplitude replay `data/e135_joint_replay.json` corroborates the four initial roots and interval quantities at 110 digits. It is not a second outward certificate. No exact disappearance scale, initial exhaustive count or clock-order reversal is asserted. □

The last example prevents an argument for arbitrary finite-target time order based solely on continuation to infinity. Bounded source components require a separate comparison. Likewise, the critical case $L = \kappa = 0$ and a complete global description of the zeros in (100) remain open.

7 Boundary directions: a logarithmic drift scale

Fix $\alpha \in (-\pi/2, \pi/2)$ and first take $b_0 = -\pi/2$, $\theta = \alpha + \pi/2$. Write $s = \sin\alpha$, $c = \cos\alpha > 0$ and $q_R = (R\mathbf{e}(\alpha), \theta)$. The identity-fixing isometry $(x, y, \theta) \mapsto (-x, y, -\theta)$ gives the other boundary with α replaced by $-\alpha$.

Lemma 7.1 (Complete boundary scalar chart)**.** *For sufficiently large R, the entire rotating inverse has one component, parametrized by $z \in \mathbb{R}$. Put*

$$w = \tanh z, \quad J = \sqrt{R^2 + 2Rw + 1}, \quad D = \sqrt{c^2 + (s+J)^2}, \quad d = \frac{c^2}{D^2}. \tag{101}$$

The exact endpoint quantities are

$$\begin{aligned} B &= -\frac{Rc}{D}, \quad A = \frac{R(s+J)}{D}, \quad \sin a = \frac{1+sJ}{D}, \quad \cos a = \frac{cJ}{D}, \\ \sin b &= -\frac{s+J}{D}, \quad \cos b = \frac{c}{D}, \quad f = -\frac{c(R+w)}{D}, \quad g = -\frac{cw}{D}, \quad m = 1 - d\,\operatorname{sech}^2 z. \end{aligned} \tag{102}$$

Here z is a new coordinate, not the tangent coordinate of Proposition 2.1. With $q_a = q(\sin a, m)$ from (19), define

$$G(z) = \int_0^z \sqrt{1 - d(z)\frac{\cosh^2 v}{\cosh^2 z}}\, dv, \qquad P_R = A - q_a + G, \qquad N_R = \frac{P_R}{L(m)} - \frac{1}{4}. \tag{103}$$

The intersections $N_R(z) = n \in \mathbb{Z}_{\geq 0}$ are in bijection with the rotating sources, with actual period index n.

Proof. The negative spatial sign is excluded when $Rc > 2$. For the positive sign, $a = \alpha - \arcsin(B/R)$ and $b = -\pi/2 - \arcsin(B/R)$. Principal b forces $B < 0$ and gives $\cos b = -B/R$. Since $m = \sin^2 b + g^2$, the condition $m < 1$ is equivalent to $|Rg/B| < 1$. Set $w = Rg/B$. Solving the algebraic endpoint equations with $A/(-B) > 0$ gives (102). Conversely every $-1 < w < 1$ gives strict principal a, b, $A > 0$, $0 < m < 1$, and $\sin(a-b) = c$, $\cos(a-b) = -s$. It is therefore admissible. There is no $B = 0$ source: that would require $\alpha = \theta/2$, contrary to this boundary condition. Moreover

$$\frac{dB}{dw} = \frac{R^2 c(s+J)}{JD^3} > 0.$$

Both ends have $m = 1$ and are excluded. The point $g = 0$ at $w = 0$ is included.

For completeness, set $u_a = F(\arcsin(\sin a/\sqrt{m}) \mid m)$ and $j = F(\arcsin(w/\sqrt{m}) \mid m)$. The latter is well defined because $m - w^2 = (1-w^2)(1-d) > 0$. The full phase lifts are $\psi = 2K - u_a$ and $v = 3K - j$. Their residual difference is $\delta = K + u_a - j$. For $w < 0$, $3K < v < 4K$ and $K < \psi < 3K$, so $\delta > 0$. For $w \geq 0$, set $u_b = j - K$; then $\delta = u_a - u_b > 0$ because $a > b$. Both phase lifts are in $(0, 4K)$, so $\delta < 4K$. Jacobi addition for the elliptic epsilon function gives

$$Q_0 = (K - E) + q_a - G, \qquad G = j - E(\arcsin(w/\sqrt{m}) \mid m) + \frac{w\sqrt{m - w^2}}{\sqrt{1 - w^2}}.$$

Differentiating the last expression at fixed m and checking its value at zero gives $G = \int_0^w \sqrt{m - u^2}/(1-u^2)^{3/2}\, du$. Substitute $u = \tanh v$ to obtain (103); $d(z)$ is held fixed during this integral. The longitudinal equation is exactly $N_R = n$, with reconstructed time $t = \sqrt{m}(\delta + 4nK)$. Proposition 2.1 then proves the claimed bijection. □

Lemma 7.2 (Uniform estimates on the whole scalar line)**.** *Put $Q_\alpha = \operatorname{atanh}(s) - s$, $C_R = \log(4R/c) - 1$, and $t_z = \operatorname{sech}^2 z$. There are constants κ_1, κ_2 depending only on the fixed angle such that, for all real z and sufficiently large R,*

$$\begin{aligned} P_R &= R - Q_\alpha + z + p_R, & |p_R| &\leq \kappa_1/R, \\ |p_R'| + |p_R''| &\leq \kappa_1(1 + |z|)t_z/R^2, & & \qquad (104) \\ L_R := L(m) &= 4(C_R + \log\cosh z) + e_R, & |e_R| &\leq \kappa_2/R, \\ |e_R'| + |e_R''| &\leq \kappa_2 t_z\{R^{-1} + R^{-2}(\log R + |z|)\}. & & \qquad (105) \end{aligned}$$

Proof. Uniformly for $-1 \le w \le 1$, $J = R + w + O(R^{-1})$ and $D = R + s + w + O(R^{-1})$, also after two w derivatives of the remainders. Hence $d = O(R^{-2})$ and $d', d'' = O(t_z R^{-3})$. The angle $a = \alpha + \arcsin(c/D)$ stays in a fixed strict principal interval; $m - \sin^2 a$ stays bounded away from zero. The required partial derivatives of the short integral are therefore bounded. It follows that $A - R, q_a - Q_\alpha = O(R^{-1})$, with their first two z derivatives $O(t_z R^{-2})$.

Expand $\sqrt{1-x} = 1 - \sum_{j\ge1} a_j x^j$, where $0 < a_j \le 1$. For $d \le 1/2$, (103) becomes

$$G - z = -\sum_{j\ge1} a_j d(z)^j T_j(z), \qquad T_j(z) = \operatorname{sech}^{2j} z \int_0^z \cosh^{2j} v \, dv.$$

Integration by parts gives

$$T_0 = z, \qquad T_j = \frac{\tanh z}{2j} + \frac{2j-1}{2j} t_z T_{j-1}.$$

In particular $|T_j| \le |T_1| \le 1$. Iteration expresses T_j as at most $j+1$ terms of the forms $w t_z^l$ and $z t_z^j$, with coefficients bounded by one. Differentiating these expressions yields $|T_j'| + |T_j''| \le C j^3 (1 + |z|) t_z$. The convergent sum $\sum j^3 d^j$, together with the bounds on d', d'', proves $G - z = O(R^{-2})$ and the derivative bounds in (104). This treats both infinite tails.

Set $\epsilon = 1 - m = d t_z$. The convergent complementary-parameter series in [7, Section 19.12], differentiated termwise for $0 < \epsilon \le \epsilon_0 < 1$, give

$$L = 4\log(4/\sqrt{\epsilon}) - 4 + \Lambda(\epsilon),$$

$$|\Lambda| \le C\epsilon(1 + |\log\epsilon|), \quad |\Lambda'| \le C(1 + |\log\epsilon|), \quad |\Lambda''| \le C/\epsilon.$$

Here $\epsilon', \epsilon'' = O(\epsilon)$ and $|\log\epsilon| = O(\log R + |z|)$. Composition gives the stated remainder bounds. The other error is $4\log(D/R) = O(R^{-1})$; its first two derivatives are $O(t_z/R)$. Since every fixed power of $|z|$ times t_z is bounded, these function and derivative estimates are uniform on $\mathbb{R}$. □

Theorem 7.3 (Boundary count and folds)**.** *For sufficiently large R, N_R has exactly one critical point, a nondegenerate maximum $M_R = N_R(z_R)$. Its endpoint limits are $-1/2$ and 0, both approached from above, and*

$$\nu_2(q_R) = 2\lfloor M_R \rfloor + 1 - \mathbf{1}_{\{M_R \in \mathbb{Z}\}} = \frac{R}{2(\log(4R/c) - 1)} + O(1). \tag{106}$$

All these sources are regular except an integer-maximum source, which is a full-map fold.

Proof. Set $H_R = P_R' L_R - P_R L_R'$, so $N_R' = H_R / L_R^2$. Lemma 7.2 gives, uniformly on the whole line,

$$\frac{H_R}{4} = C_R + I(z) - (R - Q_\alpha)\tanh z + O(1), \qquad I(z) = \log\cosh z - z\tanh z \in [-\log 2, 0]. \tag{107}$$

All error products contain a decaying t_z when multiplying an unbounded factor, so they are uniformly $O(1)$. For $z \le 0$ this is positive eventually; for $z \ge \operatorname{atanh}(1/2)$ it is negative eventually. On the remaining fixed interval, $P_R \asymp R$, $L_R'' = 4t_z + O(R^{-1}) > 0$, $P_R'' = O(R^{-2})$, and $L_R = O(\log R)$. Thus $H_R' = P_R'' L_R - P_R L_R'' < 0$ there. This proves one nondegenerate maximum and excludes critical points escaping toward either end.

Equations (104)–(105) give $N_R(-\infty) = -1/2$ and $N_R(+\infty) = 0$. The derivative signs show approach from above. Level zero has one root on the left; each positive integer below

M_R has two, and an integer maximum has one. This proves the floor formula. At the maximum (107) gives $R\tanh z_R = C_R + O(1)$ and $z_R = O(\log R/R)$. Substitution yields the useful refinement

$$M_R = \frac{R}{4C_R} - \frac{1}{4} - \frac{Q_\alpha}{4C_R} + O(C_R^{-2}),$$

which proves the count asymptotic with bounded error. The changes $z \mapsto w \mapsto B \mapsto a$ have nonzero derivative at every finite point, and B never vanishes. Proposition 2.2 therefore gives the regularity and fold statements for the full map. □

Proposition 7.4 (Full fiber on the boundary)**.** *Eventually the zero-winding fiber on either boundary consists exactly of the sources in Theorem 7.3. Its unique cell-zero source is strictly pre-cut and is the unique minimizer. There are no C_3, C_4, C_5 sources.*

Proof. On the lower boundary, (40) becomes $W = \tan((\alpha + \pi/2)/2) > 0$, $U = RW/2$, $V = R/2$. The C_3 scalar equation (47) has $p > V$ and gives

$$p - V = \frac{W^2 p^2}{(p+V)\sinh^2 p} \le \frac{W^2 V}{\sinh^2 V}.$$

Here $p/\sinh^2 p$ decreases on $p > 0$. Likewise $2p/(e^{2p}-1)$ decreases, so

$$p\coth p - 1 - V \le T_R - 1, \qquad T_R = \frac{W^2 V}{\sinh^2 V} + \frac{2V}{e^{2V}-1} \longrightarrow 0.$$

For $T_R < 1$, the required C_3 value $W(p\coth p - 1)$ is strictly less than the actual $U = WV$. This excludes C_3. The equilibrium images again exclude C_4, C_5.

Since $N_R(0) \to \infty$, the unique cell-zero rotating root has $z < 0$ eventually. Thus $f < 0$, $g > 0$ and $\delta = 2K + u_a + u_b$, where $u_b = F(\arcsin(\sin b/\sqrt{m}) \mid m)$. The initial short integral u_a stays bounded, whereas $u_b \to -\infty$: the terminal angle tends to $-\pi/2$ and $m \to 1$. The joint divergence follows by lower bounding the positive integral on each fixed shorter interval and then letting that interval approach $\pi/2$. Since $m \to 1$, also $K \to \infty$, and $|u_b| < K$ throughout, so the divergence of u_b is compatible with the phase bounds; as u_a stays bounded, $u_a + u_b < 0$ eventually. Hence $\delta < 2K$ eventually and $t = k\delta < 2kK < 2kp_1 = t_{\text{cut}}$. The published synthesis makes this source the unique minimizer. Any C_1 cell-zero source would also be strictly pre-cut and minimizing, a contradiction. Proposition 6.2 supplies the remaining classes. Reflection gives the upper boundary. □

8 Exterior directions and the exact oscillating face

We first complete the $U = 0$ face at all radii.

Proposition 8.1 (The $U = 0$ normal fiber)**.** *For $U = 0$, $V \neq 0$, $W \neq 0$, every normal source belongs to C_1. There is exactly one source in every integer winding class. The cell-zero source is strictly pre-cut and uniquely minimizing; all sources are regular.*

Proof. The positive cells were proved in Lemma 6.1. For cell zero, positive $p = r$ forces $\epsilon = 1$ in (44), hence $s_1 = -\operatorname{sgn} W$. The derivative (45) remains positive when $n = 0$. The scalar tends to zero at $k = 0$. As $k \to 1$, $a(k) \to \pi/2$ and $r = F(a(k) \mid k^2) \to \infty$, whereas $E(a(k) \mid k^2)$ remains bounded. Thus $q = r - E(a(k) \mid k^2) \to \infty$, proving that the scalar tends to infinity. It yields one cell-zero source for each $|V| > 0$. Its time satisfies $t < 2K = t_{\text{cut}}$.

For C_2, midpoint coordinates give

$$U = s_2 \frac{(p - \mathcal{E}(p))\sqrt{1 - k^2 \sin^2 \eta}}{\operatorname{dn} p}, \qquad p = t/(2k) > 0.$$

Every factor except the sign is positive, excluding $U = 0$. The C_3 formula preceding (47) also has nonzero U for $p > 0$. The equilibrium images exclude C_4, C_5. The cut-time and nonconjugacy inputs prove the remaining assertions. □

Theorem 8.2 (An explicit exterior threshold)**.** *Fix $\alpha \in (-\pi/2, \pi/2)$ and $0 < |\theta| < \pi$ with $|b_0| = |\alpha - \theta| > \pi/2$, and let $q_R = (R\mathbf{e}(\alpha), \theta)$. Put*

$$h = \theta/2, \quad \beta = \alpha - h, \quad W = \tan h, \quad u = \frac{\cos \beta}{2 \cos h}, \quad v = -\frac{\sin \beta}{2 \cos h}, \quad g = |Wv| - |u|.$$

Then $g > 0$. With $\eta_0 = \min((|b_0| - \pi/2)/2, \pi/4)$, define

$$R_{\mathrm{ext}} = \max\left\{ \frac{2}{\cos \alpha}, \frac{2}{\sin \eta_0}, \frac{|W|}{g} \right\}. \tag{108}$$

For $R > R_{\mathrm{ext}}$ all normal sources of q_R belong to C_1, with exactly one in every integer winding class. All are regular; the unique zero-winding source is the unique minimizer. Its sign is $s_1 = -\operatorname{sgn} \theta$.

Proof. The negative C_2 spatial sign is excluded by $R \cos \alpha > 2$. The positive sign would require $b = b_0 - \arcsin(B/R)$ with $|B| < 2$. Since $|\arcsin(B/R)| < \eta_0 < |b_0| - \pi/2$, this b cannot be principal. Thus there are no C_2 sources, including $B = 0$. The fixed-angle assumptions imply $|b_0| < 3\pi/2$ and $\cos b_0 < 0$. Direct calculation gives

$$u^2 - W^2 v^2 = \frac{\cos \alpha \cos b_0}{4 \cos^4 h} < 0,$$

so $v \neq 0$ and $g > 0$. A C_3 source would satisfy (47). Its first equation forces $p > |V| = R|v|$, and $p \coth p - 1 > p - 1$ then gives

$$|U| - |W|(p \coth p - 1) < |W| - Rg < 0.$$

The second equation is impossible. The equilibrium images exclude C_4, C_5.

Proposition 6.2 already supplies all nonzero classes. A minimizing trajectory exists by the known synthesis and is normal in this contact problem [8]. It must now belong to C_1. No positive cell can minimize: $p > (2n - 1)K \geq K$ there, so $t > 2K = t_{\mathrm{cut}}$. Hence a cell-zero source exists. It has $0 < p < K$, is strictly pre-cut, and is the unique minimizer. Any further cell-zero source would also be a strictly pre-cut minimizer, contradicting the synthesis. Its sign follows from $\operatorname{sgn} W = -s_1 \operatorname{sgn} r$ and $r = p > 0$. All C_1 sources are regular. □

The nonhorizontal part of Theorem 1.3 follows by combining Theorems 1.2, 7.3, 8.2, and Propositions 7.4, 6.2. The boundary law does not follow by putting $m_* = 1$ in the interior formula: its modulus complement is of order $R^{-2} \operatorname{sech}^2 z$, producing the logarithmic drift in (105).

9 Source reversal and horizontal directions

Proposition 9.1 (Reversal on the full source space)**.** *There is a smooth involution $\mathcal{S}$ of the positive-time unit-energy source space such that*

$$\operatorname{Exp}\circ\mathcal{S} = \iota\circ\operatorname{Exp}, \qquad \iota(X,\theta) = (-\operatorname{Rot}(-\theta)X, -\theta).$$

It preserves the pendulum stratum, modulus, travel time, regularity, full-map local singularity type and minimizing status. For a strict principal heading, it sends winding ℓ to $-\ell$.

Proof. This is the established reflection ε^5 of [1], in the following full-source convention. If $g(s)$ is the source trajectory on $[0,T]$, put $g_{\mathrm{r}}(s) = g(T)^{-1}g(T-s)$. Its controls are $-u(T-s)$. With $u_1 = \sin(\gamma/2)$, $u_2 = -\cos(\gamma/2)$, the vertical coordinates are

$$\gamma_{\mathrm{r}}(s) = \gamma(T-s) + 2\pi \pmod{4\pi}, \qquad c_{\mathrm{r}}(s) = -c(T-s).$$

They satisfy $\dot\gamma_{\mathrm{r}} = c_{\mathrm{r}}$ and $\dot c_{\mathrm{r}} = -\sin\gamma_{\mathrm{r}}$, with unchanged pendulum energy. If Φ_T is the vertical flow and $r(\gamma,c) = (\gamma+2\pi,-c)$, then $r\Phi_T = \Phi_{-T}r$ and $\mathcal{S}(\lambda,T) = (r\Phi_T(\lambda),T)$ is a smooth involution. The endpoint identity follows from the path construction. Both outer maps are diffeomorphisms, including their time dependence, so they preserve the full-map local type. Reversing and translating competitors gives a length-preserving bijection of the two endpoint problems, proving the minimizing assertion. The lifted heading ends at $-\widetilde\theta(T)$, which gives the strict-heading winding rule. □

Proposition 9.2 (Horizontal boundary transport)**.** *For $q_R = (\sigma R, 0, \theta)$, $\sigma = \pm 1$ and $0 < |\theta| < \pi$, the eventual zero-winding fiber has the boundary count*

$$N_0(q_R) = \frac{R}{2(\log(4R/|\sin\theta|) - 1)} + O(1).$$

Its exact eventual floor formula, regularity and fold statements are those of Theorem 7.3, transported by $\mathcal{S}$. The unique minimizer is a strictly pre-cut C_2 source. The threshold remains non-effective.

Proof. The inverse target is $(-\sigma R\cos\theta, \sigma R\sin\theta, -\theta)$. Its spatial direction is nonhorizontal, with absolute vertical component $|\sin\theta|$, and its terminal-body transverse component is zero. It is therefore a boundary target of Section 7. Proposition 9.1 gives the required source bijection, including winding zero, full-map type and minimizing status.

To complete Theorem 1.3, put $\delta_0 = e_y$, $\delta_1 = e_y\cos\theta - e_x\sin\theta$. After spatial sign normalization in the nonhorizontal case, $\mathbf{e} = \pm\mathbf{e}(\alpha)$ and $\delta_0\delta_1 = \cos\alpha\cos(\alpha-\theta)$. Thus the sign of this product is exactly the interior/boundary/exterior condition already proved. In the interior, $m_* = 1 - \min(\delta_0^2, \delta_1^2)$; on either boundary the nonzero transverse component has magnitude $|\sin\theta|$. The horizontal case is the transport just established. □

10 Pure orientation and heading π

Lemma 10.1 (The fiber over pure orientation)**.** *For $(X,\theta) = (0,\theta)$, with $\theta \in (-\pi,\pi]$, the only positive-time sources are in C_4. There is exactly one for every ℓ with $\Theta_\ell = \theta + 2\pi\ell \neq 0$, at time $T = |\Theta_\ell|$ and rotation sign $s_1 = -\operatorname{sgn}\Theta_\ell$. All are regular. A nonzero heading has one minimizer, except at π, where the two shortest lifts minimize. At the identity there is no positive-time winding-zero source and no positive-time minimizing source.*

Proof. Let $a = p_x$ and $b = p_y$ be the conserved canonical spatial momenta. The standard Hamiltonian [6, Section 2.2] and its equations are

$$H = \tfrac{1}{2}[p_\theta^2 + (a\cos\theta + b\sin\theta)^2] = \tfrac{1}{2}, \quad \dot{X} = u(\cos\theta, \sin\theta), \quad u = a\cos\theta + b\sin\theta, \quad \dot{\theta} = p_\theta.$$

Hence $a\dot{x} + b\dot{y} = u^2$. If $X(T) = X(0)$ with $T > 0$, integration forces $u \equiv 0$. The energy condition gives $p_\theta^2 = 1$, and differentiating $u = 0$ gives $-a\sin\theta + b\cos\theta = 0$. These two equations imply $a = b = 0$. Thus the source is a pure C_4 rotation, with $X = 0$ and $\widetilde{\theta} = -s_1 T$, proving the source list. This arbitrary-time argument uses the same conserved-momentum positivity as the period-drift argument in [6, Section 2.3]; here the identity applies on an arbitrary time interval. Regularity follows from [2, Theorem 2.4]; equivalently its full determinant in (γ, c, T) is $(T^2 - \sin^2 T)/16 > 0$. The cut time is π [3, (2.4)]. The shortest lifted rotations give the stated minimizers. At the identity the distance is attained at zero time, which is outside the source domain being counted. □

Theorem 10.2 (Exact heading-π catalogue)**.** *At every target (X, π) there is exactly one source in each winding class. For $R = |X| > 0$ all sources belong to C_1; for $R = 0$ they are the C_4 sources of Lemma 10.1. All are regular. For $R > 0$ set*

$$H(k) = \frac{K(k^2) - E(k^2)}{k}.$$

For $n \geq 0$, let $a = 2n + 1$ and solve $R = 2aH(k)$ for $k \in (0, 1)$. The pair with windings $n, -n - 1$ has common time $T_n(R) = 2aK(k^2)$ and phases determined by

$$\operatorname{sn}\phi = s_1 x/R, \quad \operatorname{cn}\phi = -y/R, \qquad \ell = \begin{cases} n, & s_1 = -1, \\ -n-1, & s_1 = 1. \end{cases}$$

The phase is unique modulo $4K$. Exactly the first pair minimizes. Different pairs have different travel times.

Proof. The C_1 lifted heading is $s_1(\operatorname{am}\phi - \operatorname{am}(\phi + T))$. Strict increase of the real amplitude forces $T = 2(2n+1)K$ at group heading π. The half-period identities in the original spatial equations then give

$$X = 2aH(k)(s_1 \operatorname{sn}\phi, -\operatorname{cn}\phi).$$

Every C_2 or C_3 lifted heading lies strictly in $(-\pi, \pi)$; C_5 has heading zero. Thus these are all sources when $R > 0$. Moreover

$$H(k) = k\int_0^{\pi/2} \frac{\sin^2 u}{\sqrt{1 - k^2\sin^2 u}}\,du, \qquad H'(k) = \int_0^{\pi/2} \frac{\sin^2 u}{(1 - k^2\sin^2 u)^{3/2}}\,du > 0.$$

The endpoint limits are zero and infinity, so the modulus and the full phase-circle point are unique. Opposite s_1 give distinct covectors, since $\cos(\gamma/2) = s_1 \operatorname{dn}\phi$ never vanishes. The lifted heading is $-s_1 a\pi$, giving the displayed labels. Nonconjugacy follows from [2, Theorem 2.1], including these half-period boundaries. The cut time $2K$ [3, (2.1)] selects exactly $a = 1$. Time ordering is proved next. □

Proposition 10.3 (Heading-pi time spectrum)**.** *For $R > 0$ the pair times in Theorem 10.2 satisfy*

$$R + 2a < T_n(R) < R + a\pi, \quad a = 2n + 1, \qquad 4 < T_{n+1}(R) - T_n(R) < 2\pi.$$

At fixed R the gaps increase strictly with n and tend to 2π. At fixed n they decrease strictly with R and tend to 4. For fixed R, as $a = 2n + 1 \to \infty$,

$$T_n(R) = a\pi + \frac{R^2}{a\pi} - \frac{3R^4}{4a^3\pi^3} + O_R(a^{-5}).$$

Proof. Allow $a > 0$ to be real in $R = 2aH(k)$, $T = 2aK$. The complete-integral derivatives give $K_k = kH'$ and $E_k = -H$, hence

$$T_R = k, \qquad T_a = 2E, \qquad T_{aa} = \frac{2H^2}{aH'} > 0.$$

Integrating $2 < 2E < \pi$ over an a interval of length two proves the gap bounds and convexity proves their strict increase. Since k decreases with a, the radial derivative of a gap is negative. The endpoint limits follow by dominated convergence in that same integral for T_a. The quantity $K - H$ decreases strictly from $\pi/2$ to 1, since $(K - H)' = (k-1)H' < 0$; this proves the time-offset bounds. Finally, with $z = 2R/(a\pi)$, the convergent small-modulus series [7, Section 19.5] give $k = z - 3z^3/8 + 3z^5/16 + O(z^7)$ and $T/(a\pi) = 1 + z^2/4 - 3z^4/64 + O(z^6)$. □

11 Zero heading: exact fibers and exceptional axes

In this section write $\mathcal{D}(k) = K(k^2) - E(k^2)$; unadorned K, E at a source mean $K(k^2), E(k^2)$. Source phases still have period $4K$.

Theorem 11.1 (Zero-heading winding and axes)**.** *For every $X \neq 0$ at heading zero, each nonzero winding class has exactly one regular C_1 source. For $n = |\ell| \geq 1$ it is determined by*

$$R = 4nH(k), \quad T = 4nK, \quad s_1 = -\operatorname{sgn}\ell, \quad \operatorname{sn}\phi = s_1 x/R, \quad \operatorname{cn}\phi = -y/R.$$

All these sources are post-cut. There are no C_3 sources. For $(x, 0, 0)$, $x \neq 0$, winding zero consists of the unique straight C_5 source, which is regular and uniquely minimizing.

For $(0, y, 0)$, $y \neq 0$, winding zero consists of four regular C_2 sources for each positive root label n: two with $p = 2nK$ and two with $p = p_n^1 \in ((2n-1)K, 2nK)$, where $f_1(p_n^1) = 0$. The central pair has actual period index n, whereas this f_1 pair has index $n-1$. Thus actual cell zero contains two sources; each positive actual cell j contains the central pair labelled j and the f_1 pair labelled $j+1$. Exactly the first f_1 pair minimizes. For this fixed vertical target, the total number of sources with travel time at most t satisfies

$$N_{\leq t}(0, y, 0) = \frac{t^2}{\pi|y|} + \frac{t}{\pi} + O_y(1). \tag{109}$$

Proof. The continuous C_1 amplitude now forces $T = 4nK$, and substitution in the original map gives the displayed radial equation. The proof of Theorem 10.2 applies with $a = 2n$. The C_3 lifted heading is $s_1[\arcsin(\tanh\phi) - \arcsin(\tanh(\phi + T))]$; it is nonzero and strictly between $-\pi$ and π. Thus it cannot have group heading zero. Every C_2 source here has winding zero. In the chart (40), at this heading $U = -y/2$, $V = -x/2$, $W = 0$. Writing $\eta = \operatorname{am}(\psi + p)$ and $q = p - \mathcal{E}(p) > 0$, the original map gives

$$U = \frac{s_2 q\sqrt{1 - k^2\sin^2\eta}}{\operatorname{dn} p}, \quad V = \frac{s_2 k f_1(p)\sin\eta}{\operatorname{dn} p}, \quad W = -\frac{k\operatorname{sn} p\cos\eta}{\operatorname{dn} p}. \tag{110}$$

In particular $U \neq 0$, excluding C_2 on the horizontal axis. The remaining straight C_5 source minimizes since $T \geq |x|$, with equality forcing that straight control. Its regularity follows from [2, Theorem 2.4].

For the vertical axis put $u = |y|/2$ and $s_2 = -\operatorname{sgn} y$. The two zero factors in (110) give precisely $p = 2nK$, $\eta = 0, \pi$, or $f_1(p) = 0$, $\eta = \pi/2, 3\pi/2$. The function $f_1/\operatorname{cn} p = -q - \operatorname{dn} p\operatorname{sn} p/\operatorname{cn} p$ has derivative $-\operatorname{dn}^2 p/\operatorname{cn}^2 p < 0$ between poles. Thus it has exactly one

zero in each interval $((2n-1)K, 2nK)$ and none in the preceding half-cell. These are the root intervals of [1, Lemma 5.3]. We take this occasion to correct equation (5.15) there: the continuous $k \to 0$ limit of the n-th root is $n\pi$, not the displayed $2\pi n$, as $f_1(p,0) = -\sin p$ and the adjacent first-root limit show. The central modulus solves $2n\mathcal{D}(k) = u$ uniquely. For the f_1 pair put $h = \sqrt{1-k^2}$, $\chi = \arctan(u/h)$. Its equations are

$$p = 2nK - F(\chi \mid k^2), \quad 2n\mathcal{D}(k) = I(u,h), \quad I(u,h) = \int_0^u \sqrt{\frac{1+z^2}{h^2+z^2}}\, dz.$$

The last integral equals $F(\chi) - E(\chi) + \sqrt{1-k^2\sin^2\chi}\tan\chi$, by differentiation in u. At fixed u, substituting $z = h\tan v$ in its modulus derivative gives

$$\frac{d}{dk}[2n\mathcal{D}(k) - I(u,h)] = \frac{k}{h^2}[2nE - E(\chi \mid k^2)] > 0.$$

The endpoint values are $-u$ at $k=0$ and infinity at $k=1$: $I(u,h) = \log(1/h) + O_u(1)$ and $\mathcal{D}(k) = \log(4/h) - 1 + o(1)$. This proves unique reconstruction of both phase choices for every n. These four sources are exhaustive. Sachkov's reduced Jacobian [2, (2.10)–(2.14)] (see also [12]), whose full-map prefactor is nonzero here, is $J_2 = \alpha\,\mathrm{sn}^2\tau + \beta\,\mathrm{cn}^2\tau$ with $\tau = \psi + p$,

$$\begin{aligned}\alpha &= (1-k^2)\,\mathrm{sn}\,p\,\{\mathrm{cn}\,p\,\mathrm{dn}\,p\,(p - 2\mathcal{E}(p)) + \mathrm{sn}\,p\,(\mathrm{dn}^2 p + \mathcal{E}(p)q)\},\\ \beta &= f_1(p)(\mathrm{cn}\,p\,\mathcal{E}(p) - \mathrm{dn}\,p\,\mathrm{sn}\,p),\end{aligned}$$

the brace being the α_1 of Lemma 11.4 below. At the central pair $\mathrm{sn}\,p = \mathrm{sn}\,\tau = 0$, $\mathrm{cn}\,p = (-1)^n$ and $f_1(p) = (-1)^{n+1}q$, so $J_2 = \beta = -2nEq < 0$. At the f_1 pair $\beta = 0$ and $\mathrm{cn}\,\tau = 0$; substituting $\mathrm{sn}\,p = -q\,\mathrm{cn}\,p/\,\mathrm{dn}\,p$ from $f_1(p) = 0$ turns the brace into $-\mathrm{cn}\,p\,\mathcal{E}(p)(\mathrm{dn}^2 p + q^2)/\,\mathrm{dn}\,p$, and $\mathrm{cn}^2 p\,(\mathrm{dn}^2 p + q^2) = \mathrm{dn}^2 p$, so $J_2 = \alpha = (1-k^2)\mathcal{E}(p)q > 0$. Thus all are regular. The cut time $2kp_1^1$ [3, (2.2)] selects exactly the first f_1 pair; all remaining sources are post-cut.

For fixed u, both sequences have $k_n \to 0$, since their increasing scalar left sides diverge with n at every fixed positive modulus. Using $\mathcal{D}(k) = \pi k^2/4 + O(k^4)$ and $I(u, \sqrt{1-k^2}) = u + O_u(k^2)$ yields

$$k_n^2 = \frac{2u}{\pi n} + O_u(n^{-2}), \qquad T_n^2 = 8\pi u n + O_u(1)$$

in both families; subtracting $2kF(\chi)$ in the f_1 time changes its square by $O_u(1)$. The two families, with two phases each, therefore contribute $t^2/(2\pi u) + O_u(1)$ sources in total. The C_1 pair times are $2\pi n + O_R(n^{-1})$, contributing $t/\pi + O_R(1)$. Their sum proves (109), without any ordering assumption between the two C_2 time sequences. □

Theorem 11.2 (Generic zero-heading count)**.** *Fix $xy \neq 0$, put $u = |y|/2$, $v = |x|/2$, $\rho = \sqrt{u^2+v^2}$, $\nu = v/\rho$, and $S = \rho/[2\mathcal{D}(\nu)]$. Define*

$$k_{\min}^2 = \frac{2v^2}{z + \sqrt{z^2 - 4v^2}}, \qquad z = 1 + u^2 + v^2, \tag{111}$$

$$\phi = \arctan(u/h), \qquad h = \sqrt{1-k^2},$$

$$a = \arcsin\frac{vh}{k\sqrt{u^2+h^2}}, \qquad A_L = a - \phi, \quad A_R = \pi - a - \phi,$$

$$N_j(k) = \frac{u d_j/h - F(A_j \mid k^2) + E(A_j \mid k^2)}{2\mathcal{D}(k)}, \quad d_j = \sqrt{1 - k^2\sin^2 A_j}. \tag{112}$$

Let $M = \sup_{k_{\min}\le k<1} N_L(k)$. The winding-zero fiber consists of finitely many C_2 sources, with exact count

$$N_0(x,y,0) = 1 + 2\lfloor S\rfloor + 2\lfloor M\rfloor - 2\mathbf{1}_{\{S\in\mathbb{Z}\}} - \mathbf{1}_{\{M\in\mathbb{Z}\}}. \tag{113}$$

The unique complementary cell-zero source minimizes; every other zero-winding source is post-cut.

We prove the count with the distinction between an inverse-formula branch and a connected fixed-label source branch kept explicit. In this section, a *complementary cell* means the latter branch; its label n is not generally the actual period index $\lfloor p/(2K)\rfloor$. In particular, the complementary cell-zero interval is $0 < p < p_1^1$, a proper subset of actual period cell zero.

Lemma 11.3 (Complementary scalar and physical cells)**.** *The sources on $\cos\eta = 0$ are in bijection with $N_L(k) = n$ or $N_R(k) = n$, $n \in \mathbb{Z}_{\ge 0}$, subject to $p = 2nK + F(A_j \mid k^2) > 0$ and one gluing identification at $k = k_{\min}$. Both index branches are bounded, with limits $N_L \to 1/2$, $N_R \to -1/2$ as $k \to 1$.*

Proof. In a sign-normalized cell write $s = \sin A$, $c = \cos A$, $d = \sqrt{1-k^2s^2}$ and

$$\begin{aligned} q &= 2n\mathcal{D}(k) + F(A \mid k^2) - E(A \mid k^2), \\ e = 2nE + E(A \mid k^2), &\qquad u = hq/d, \quad v = k(s + qc/d). \end{aligned} \tag{114}$$

Eliminating q gives $\sin A + (u/h)\cos A = v/k$, whose two representatives are exactly (112). The existence inequality is $k^4 - (1+u^2+v^2)k^2 + v^2 \le 0$, giving (111). Conversely these equations recover both target components. The actual phase is fixed by $s_2 = \operatorname{sgn} U$ and $\sin\eta = \operatorname{sgn}(UV)(-1)^{n+1}$, with $\psi = F(\eta \mid k^2) - p$. A simultaneous half-period shift changes the sign-normalized representative and its integer label, not the source. For $n \ge 1$ the physical positive-v interval runs from an f_1 root in $(-\pi/2, 0)$ to one in $(\pi/2, \pi)$; for $n = 0$ it begins at $A = 0$.

At $k \to 1$, $A_L = -\pi/2 + O(h)$ and $A_R = \pi/2 + O(h)$. The quantities ud_j/h remain bounded, whereas the incomplete drifts are respectively $-\mathcal{D}(k) + O(1)$ and $\mathcal{D}(k) + O(1)$. The error is bounded by rescaling the $O(h)$ endpoint layer by h in its integral. This proves the limits and, together with continuity at $k_{\min}$, boundedness. □

The real-label interpolation below is an auxiliary analytic family. For noninteger n, q, e are defined by (114); they need not equal $p - \mathcal{E}(p), \mathcal{E}(p)$ at that value of p. Write $\widehat{f} = -cq - ds$ for its boundary factor. At integer labels its zeros are the physical f_1 zeros.

Lemma 11.4 (Strict ordering for real cell labels)**.** *Allow $n \ge 1$ to be real in (114). In its positive-v source interval there is exactly one root of*

$$\alpha_1 = cd(q - e) + s(d^2 + eq).$$

At this root $A = A_n(k)$ the physical target $v_n(k)$ increases strictly from zero to infinity as k runs from zero to one. For fixed $v > 0$, the corresponding value $u_n^*(v)$ increases strictly with n. On the connected physical curve $v =$ constant this is the unique nondegenerate minimum of u.

Proof. All derivatives below keep the real label fixed unless indicated. Direct differentiation gives

$$\det\frac{\partial(u,v)}{\partial(A,k)} = \frac{k^2\alpha_1}{hd^2}, \qquad v_A = \frac{k(cd - h^2qs)}{d^3}. \tag{115}$$

The latter is positive for negative amplitude, negative after $\pi/2$, and has exactly one zero in $(0,\pi/2)$, where $q\tan A/d = 1/h^2$. At that fixed-modulus maximum, $dv_{\max}/dk = (d^2/s + k^2ce/d)/h^2 > 0$. For fixed n, its limits are zero and infinity; the latter follows already by taking $A = 0$. Hence a fixed positive v defines one connected source curve, with two amplitudes above a unique coalescence modulus.

Put $D_\alpha = cd + es$ and $\beta_1 = ce - ds$. At an α_1 root,

$$qD_\alpha = d\beta_1, \quad \widehat{f} = -de/D_\alpha, \quad \alpha_{1,A} = \frac{e^2}{dD_\alpha}, \quad cd - qs = \frac{d^2}{D_\alpha}. \tag{116}$$

In the negative quarter $\beta_1 > 0$, so any root has $D_\alpha > 0$ and crosses upward. In the positive quarter β_1 decreases strictly, since $\beta_{1,A} = -s(e - k^2sc/d) < 0$; a root can only occur before its zero and again crosses upward. After $\pi/2$ any root would cross downward, but α_1 is positive at both ends of that last physical quarter. The $\widehat{f}$ endpoint signs follow from $s\alpha_1 = eq$ there. These signs prove uniqueness. Since $\alpha_1(0) = 2n(K - 2E)$, the root has positive amplitude below the unique modulus k_0 solving $K(k_0^2) = 2E(k_0^2)$, negative amplitude above it, and zero amplitude at it. The modulus k_0 is unique because $(K - 2E)' = K_k + 2H > 0$ with opposite endpoint signs. The root is smooth, with $D_\alpha > 0$ throughout.

Set $B_\alpha = d^3 + k^2ces$. This is positive: for $s < 0$, $D_\alpha > 0$ implies $e|s| < cd$, whence $k^2ce|s| < k^2c^2d < d^3$. At the root,

$$v_A = \frac{kB_\alpha}{d^2D_\alpha} > 0.$$

For the half-time $\tau_n = k(e + q)$, the exact derivatives $q_k = k(e - sc/d)/h^2$, $e_k = -q/k$, $q_A = k^2s^2/d$, $e_A = d$ give

$$\begin{gathered} \tau_n' = \frac{dI}{h^2e^2D_\alpha}, \qquad v_n' = \frac{B_\alpha}{dD_\alpha}\tau_n', \\ I = ds\{(e^2 - d^2)^2 + 2e^2(1 - 2k^2c^2)\} + ce(1 - 4k^2s^2)(e^2 - d^2). \end{gathered} \tag{117}$$

We give the sign argument, including noninteger n. The closed forms below, namely F_R and its two boundary values, P, the Bernstein coefficients, and the factorization of I through G, are polynomial identities in the substituted variables; they were verified by computer algebra in the script `research/e33_zero_heading_count.py` of the repository, and the substitutions are collected in Appendix A.

For positive amplitude put $z = d/e$ and $\tan A = rz$. Then $e > 2$, $d^2 > \max(4z^2, c^2)$, $rz^2 < 1$, and

$$I = \frac{d^3F_R}{rz^3\sqrt{1 + r^2z^2}}, \qquad F_R = d^2\{2 + r(1 - z^2)\}^2 + 2r^2z^2 + 3rz^2 - 3r - 4.$$

This expression increases with d^2. At the active lower boundary $d^2 = c^2$, its value is

$$\frac{r(r + 1)(rz^2 - 1)\{(2r + 1)z^2 - 1\}}{1 + r^2z^2} > 0,$$

because $4z^2 \le c^2$ makes both linear factors negative. At the other active boundary put $w = z^2$, $\xi = rw \in (0, 1)$. Multiplication by w gives

$$P = 4w^2(\xi - 2)^2 + w(-8\xi^2 + 19\xi - 4) + 3\xi(2\xi - 1).$$

It is positive termwise for $\xi \ge 1/2$. For $\xi < 1/2$, the active condition is $w \ge 1/4 - \xi^2$, and

$$P(1/4 - \xi^2, \xi) = \frac{\xi(\xi - 1)(2\xi - 3)(2\xi - 1)(4\xi^2 - 4\xi - 1)}{4} > 0.$$

Moreover $P_{ww} > 0$. At the lower endpoint, the Bernstein coefficients of P_w on $0 \le 2\xi \le 1$ are

$$(4,\ 43/8,\ 31/6,\ 35/8,\ 7/2).$$

All are positive, completing the subcritical sign proof.

For negative amplitude put $\chi = -A$, $z = d/e$ and $\tan\chi = rz$. The root relation gives $0 < r < 1$ and $q/e = (1+rz^2)/(1-r) > 1$. We first establish $e > 2n - 1/2$. The complementary expression $D_0 = e\sin\chi - d\cos\chi$ is negative at the root and has derivative $D_0' = \cos\chi(e + k^2\sin\chi\cos\chi/d) > 0$. At $\chi_0 = \pi/6$ it is positive already for $n = 1$: if $w(\vartheta) = \sqrt{1-k^2\sin^2\vartheta}$ and $g = w - \cos\vartheta$, then g increases, since $w^2 - k^4\cos^2\vartheta = (1-k^2)(1+k^2\cos^2\vartheta) > 0$. Subtracting the equality at $k = 1$ gives

$$2E - E(\chi_0) - \sqrt{3}w(\chi_0) = \int_0^{\chi_0} g + 2\int_{\chi_0}^{\pi/2} g - \sqrt{3}g(\chi_0) > 0,$$

using $2\pi/3 > \sqrt{3}$. Adding $2(n-1)E\sin\chi$ proves the same comparison for every real $n \ge 1$. Thus $\chi < \pi/6$ and

$$e > 2(n-1)E + 2E - E(\pi/6) > 2(n-1) + \sqrt{3}\sqrt{1-k^2/4} > 2n - \tfrac{1}{2}.$$

The paired integrands satisfy $w(\vartheta)w(\pi/2-\vartheta) > h$, hence $w(\vartheta)+w(\pi/2-\vartheta) > h\{1/w(\vartheta)+1/w(\pi/2-\vartheta)\}$. Integrating gives $E > hK$, so $K - 2E < 0$ at $k^2 = 3/4$ and $k_0^2 > 3/4$. Above k_0, $h^2 < 1/4$ and $2h^2\mathcal{D}(k) < 3\pi/16 < 3/4$; the latter follows from $\mathcal{D}(k) \le \pi k^2/(4h)$ and $h(1-h^2) < 3/8$. Consequently

$$e(e^2-1) - h^2\{2n\mathcal{D}(k) + 3e\} > \frac{(n-1)(32n^2+8n-3)}{4} \ge 0. \tag{118}$$

The strict lower bound is obtained by substituting $e > 2n - 1/2$ in the increasing cubic $e(e^2 - 7/4) - 3n/4$.

To connect this margin to I, set

$$\begin{aligned} G &= (1-r)(1+rz^2)\{1+(2r-1)z^2\} \\ &\quad - h^2rz^2\{2-r(1-z^2)\}^2, \\ B_0 &= eq(e^2-d^2) - h^2d^2(e+q)^2. \end{aligned}$$

Exact substitution gives $I = d^3G/[z^3(1+r^2z^2)^{3/2}]$ and

$$G - \frac{(1-r)^2B_0}{e^4} = (1-r)z^2\{2r(1+rz^2) + h^2[2-r(1-z^2)]^2\} > 0.$$

Here $q > e$, $q < 2n\mathcal{D}(k)$ and $(e+q)^2/q < 2n\mathcal{D}(k) + 3e$, so (118) implies $B_0 > 0$. This proves $I > 0$ supercritically. At $k = k_0$, directly $\tau_n' = (e^2-1)/(h^2e) > 0$. Thus (117) proves $v_n' > 0$ everywhere. Its limits are zero and infinity: at the upper end $q > (2n-1)\mathcal{D}(k)$ and (116) imply $v_n \ge kq/2 \to \infty$.

For fixed v, the unique alpha intersection is therefore smooth in n. Along its increasing-amplitude branch,

$$\left.\frac{du}{dk}\right|_v = -\frac{k^2\alpha_1}{hd^2v_A}, \qquad \left.\frac{d\alpha_1}{dk}\right|_v = -\frac{\alpha_{1,A}v_n'}{v_A} < 0 \quad \text{at the root.}$$

It is a nondegenerate minimum, with

$$u'' = \frac{k^2\alpha_{1,A}v_n'}{hd^2v_A^2} > 0. \tag{119}$$

Both ends of the connected fixed-v curve have $u \to \infty$: $q \geq (2n-1)\mathcal{D}(k) \to \infty$ and bounded v force $c/d \to 0$, hence $h/d \to 1$. There is no other critical point. Finally the envelope derivative with respect to the real label is

$$\frac{\partial u_n^*(v)}{\partial n} = \frac{2\mathcal{D}(k)hd^2}{B_\alpha} > 0. \tag{120}$$

This proves strict ordering without interpolating integer samples. □

Proof of Theorem 11.2. Each positive complementary cell has two, one or zero sources according as $u > u_n^*(v)$, $u = u_n^*(v)$ or $u < u_n^*(v)$, by the unique minimum in Lemma 11.4. These thresholds diverge with n: if u stayed bounded with v fixed, phase existence would bound k away from zero, while $q \geq (2n-1)\mathcal{D}(k)$ would diverge, contradicting $|q - \rho| \leq 1$ from (114).

Cramer's rule for the inverse branches gives

$$N_j' = \frac{k\alpha_1}{2\mathcal{D}(k)h^2(cd - qs)}.$$

The denominator is positive on L and negative on R. Every alpha root for a real label at least one has $cd - qs = d^2/D_\alpha > 0$; therefore any stationary value at height at least one lies on L. By (120) there is at most one such real-label fold for fixed (u, v), and (119) makes it a strict maximum. At the inverse gluing endpoint $cd - qs = 0$ and $\alpha_1 = d^2/s > 0$, so N_L initially increases. Its limit $1/2$ then shows that any maximum of height at least one is interior. It follows that the positive-cell count is two/one/zero according as $n < M$, $n = M$, $n > M$.

The physical cell-zero interval is $0 < p < p_1^1$; all its sources are strictly pre-cut. Existence of a minimizer, the other-regime exclusions in Theorem 11.1, and the known cut locus [3] give exactly one such source: at heading zero with $x \neq 0$ the target is off the cut locus. Thus the complementary count is $1 + 2\lfloor M \rfloor - \mathbf{1}_{\{M \in \mathbb{Z}\}}$.

At a central source $p = 2nK$, (110) gives

$$q = 2n\mathcal{D}(k), \quad U = s_2 q\sqrt{1 - k^2 \sin^2 \eta}, \quad V = s_2(-1)^{n+1} kq \sin \eta.$$

Therefore $q = \rho$, with a unique modulus and two/one/zero phases according as $k > \nu$, $k = \nu$, $k < \nu$. This is the two/one/zero test $n < S$, $n = S$, $n > S$, giving central count $2\lfloor S \rfloor - \mathbf{1}_{\{S \in \mathbb{Z}\}}$. The source strata meet only at $\operatorname{sn} p = \cos \eta = 0$, where $k = \nu$ and $N_L = S$. If S is an integer, exactly one source is counted in both lists. Subtracting it proves (113). □

Proposition 11.5 (Rank at the critical overlap)**.** *The complementary equality in Theorem 11.2 is a fold of its restricted two-dimensional map for every modulus. For the full map it is a fold except when $k = k_0$ and $p = 2nK$, where the corank is two. A central coalescence is an ordinary cusp off k_0 and has corank two at k_0. At the critical target in cell n, $S = M = n$ and the exact C_2 count is $4n - 2$.*

Proof. Equation (119) gives the restricted fold, including k_0. Away from $\operatorname{sn} p = 0$, the derivative W_η in (110) is nonzero, so solving for η transports this fold to the full map. At a central source use $p = 2nK + F(A \mid k^2)$, $\eta = \pi/2$, $s_2 = 1$ and $A = 0$. With $q = 2n\mathcal{D}(k)$, $e = 2nE$, $q_k = 2nkE/h^2$, the full derivative in (A, k, η) is

$$\begin{pmatrix} 0 & k(e-q)/h & 0 \\ (-1)^{n+1}k & (-1)^{n+1}(q + kq_k) & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

The source and target coordinates are nonsingular here. Its rank is two unless $q = e$, when it is one. That equality is precisely $K = 2E$. Other signs give the same ranks by reflection.

For $q \neq e$, put $b = \eta - \pi/2$ and fix the first two target coordinates U, V at their central values. The implicit function theorem gives even analytic functions $A(b), k(b)$ with $A(0) = 0$. Differentiating U, V twice at $b = 0$ gives

$$k''(0) = -\frac{qk}{e-q}, \qquad A''(0) = \frac{qe}{h^2(e-q)}.$$

The third target coordinate therefore has the reduced expansion

$$W(b) = \frac{(-1)^n kqe}{2h^2(e-q)} b^3 + O(b^5).$$

Its cubic coefficient is nonzero. Allowing V to vary with U fixed, the same reduction gives $\partial_V \partial_b W = -1$ at the source: $A_V = 1/[(-1)^{n+1}k]$ and $W_b = (-1)^n kA + O(A^2)$. These are the nonzero cubic and unfolding derivative of the ordinary cusp criterion. At k_0 this reduction is unavailable; only the corank-two assertion is made here. At the critical target, the central equality gives $S = n$, and the restricted fold gives $M = n$. Substitution in (113) proves the final count. □

Proposition 11.6 (Zero-heading far-field law)**.** *For a fixed non-axis unit spatial direction* $\mathbf{e}$*, put* $\nu = |e_x| \in (0, 1)$*. Then*

$$N_0(R\mathbf{e}, 0) = \frac{R}{K(\nu^2) - E(\nu^2)} + O_{\mathbf{e}}(1) = \frac{4R}{L(\nu^2)} + O_{\mathbf{e}}(1).$$

The statement is not uniform toward the coordinate axes.

Proof. In (114), (u, v) is q times a unit vector plus $(0, k \sin A)$, giving $|q - \rho| \leq 1$. Since the incomplete drift has absolute value at most $2\mathcal{D}(k)$ on the chosen representatives, and $N_L(\nu) = S$,

$$S \leq M \leq \frac{\rho + 1}{2\mathcal{D}(k_{\min})} + 1.$$

For $r = k_{\min}^2$, its quadratic equation is equivalently $\rho^2(\nu^2 - r) = r(1 - r)$, so $0 < \nu^2 - r \leq 1/(4\rho^2)$. At fixed $\nu \in (0, 1)$, $\mathcal{D}(k_{\min}) = \mathcal{D}(\nu) + O_\nu(\rho^{-2})$, bounded away from zero for large ρ. Thus $M = S + O_\nu(1)$. Equation (113) with $\rho = R/2$ proves the result. The exact vertical and horizontal exceptions are in Theorem 11.1. □

12 Numerical corroboration and open questions

Numerical corroboration. The repository records symbolic checks of selected identities and numerical or interval checks of particular sources and targets. These include the exact inverse of Proposition 2.1 against the endpoint map at 180 to 360 digits on several hundred sources; twelve integrations of the Hamiltonian system without elliptic functions; and, for two targets, the complete rotating fiber certified by interval arithmetic with analytic tail exclusions, 37 sources at $(39/5, -52/5, 3/10)$ (Figures 1 and 4) and 301 sources at $(60, -80, 3/10)$. The general analytic count and image results do not depend on these two target computations. The scripts are in the repository `https://github.com/moiseevigor/se2-conjugate-locus`. Further finite checks accompany the all-radius ray count and its small-heading asymptotic, and the zero-heading central cusp reduction. They use independent defining-integral inversions, original-map evaluations and

selected Hamiltonian integrations. Their recorded sample sizes and precision do not constitute a full-domain interval certificate. The finite radial interval in Figure 3 has separate outward hyperbolic-root brackets and independent original-flow checks. Its existence boundaries are certified; its interior count follows from the analytic proof of Corollary 6.13. The wider exterior witness in Proposition 6.14 has a separate complete 256-bit count certificate with a full 384-bit alternate replay. Its pointwise count depends on those interval bounds and the analytic period exclusions. Exact rational arithmetic separately checks its domain and modulus polynomials. Eight original Hamiltonian integrations and 110-digit original-map checks corroborate the sources; these finite checks alone do not establish completeness. The angular-threshold theorem is analytic; its recorded finite checks include a separate parameter substitution, defining-integral evaluations and 32 original Hamiltonian integrations covering sign choices and several actual periods. The narrow numerical enclosure in Corollary 6.16, however, depends on its complete interval cover and alternate replay. An independent exact rational sweep rechecks coverage and rejects missing/duplicated-box controls; symbolic identities check the explicit replay derivatives. The separate localization certificate in Theorem 6.17 excludes 7465 boxes at the upper comparison level and retains 1343 inside a small rectangle. All exclusions pass an alternate 384-bit replay. A parentwise rational area and overlap audit checks coverage independently of the sweep. Uniform Hessian and rank bounds, independently recalculated by symbolic differentiation in the original elliptic coordinates, prove a unique ordinary fold at this first threshold and its local radial count window. Both implementations share Arb; Corollary 6.18 extends the count window to an explicit slope cap. Its larger-domain certificate has 3111 global exclusions and 2048 local boxes; alternate implementations replay them in 3203 and 2080 nodes respectively. Independent edge checks prove strict boundary exclusion, and fresh root inclusions verify both cap endpoints. These computations supply the quantitative extent and the endpoint enclosures; the connected-graph and convex-image arguments establish completeness throughout that range. The time-order theorem uses a separate cover of 117 positive-momentum boxes and 38 slope exclusions. A full alternate calculation in the original elliptic coordinates checks all 155 boxes in 287 nodes at 384 bits. Exact rational identities verify the simplified derivatives; sixteen original Hamiltonian integrations corroborate the signed reconstruction. These finite integrations do not replace the interval signs or the positive integral argument for the universal time comparison. The separate cross-period example has 256 parameter intervals and 257 joint and endpoint boxes. A fresh 448-bit forward-derivative implementation replays all 513 root inclusions, and exact checks verify the complete label chain and joins. Five independent polynomial identities check the differential and physical source-chart formulas. The target is now fully enumerated: analytic endpoint elimination gives two exhaustive arcs. The first has 1062 excluded intervals, replayed independently in 1274 nodes; the second has 1953, replayed with the original endpoint residual in 11841 nodes. Exact polynomial tail bounds exclude all other periods and boundary arrivals. Nine unique simple roots give the full rotating count and disjoint clock intervals give the time order. Nine original Hamiltonian integrations corroborate the signed reconstruction and actual period labels; they do not replace the exhaustive interval proof. The fixed-line count in Proposition 6.22 uses the additional monotone-modulus argument and a complete critical-source cover: 350 higher-period polynomial intervals, four endpoint tails and 285 retained-family intervals. All have separate alternate replays; the last uses 1768 interval nodes. The two larger fold boxes are independently included and their intrinsic quadratic terms are nonzero. Properness and the exact count at $u = 6$ then give the full 7/8/9/10/11 pattern. The within-family comparison in Theorem 6.23 is analytical: defining-integral monotonicity, first variation and horizontal slicing prove it for general $V > 1, W > 0$. Separate exact identities, 16 roots refined to 90 digits and four direct-

integral checks guard the reductions. A finite scan of 543 detected pairs guided the proof but provides neither root completeness nor a universal sign certificate. For Corollary 6.24, 41 rational inter-family comparisons retain the fixed-line separation. The earlier 14-interval sibling certificate, 38 root enclosures, two fold corridors and three clock-gap anchors remain independently replayed corroboration. The new checks and proof audit are recorded under `references/e130/`. The dilation results in Section 6.9 have analytical proofs based on the scalar source equation and compact exhaustion. Their computer-assisted examples use separate 320-bit and 384-bit outward certificates. Independent full-amplitude calculations at 110 and 140 digits corroborate the continuation obstruction and critical examples, respectively; they do not replace outward bounds. The critical replay checks 45 values inside the raw intervals, 20 finite-scale maxima, 16 roots and eight wrong-negative-label controls. Exact algebra checks the derivative reductions. Source definitions, domains, producer hashes and proof-status matrices are under `references/e135/`, `references/e136/` and `references/e137/`.

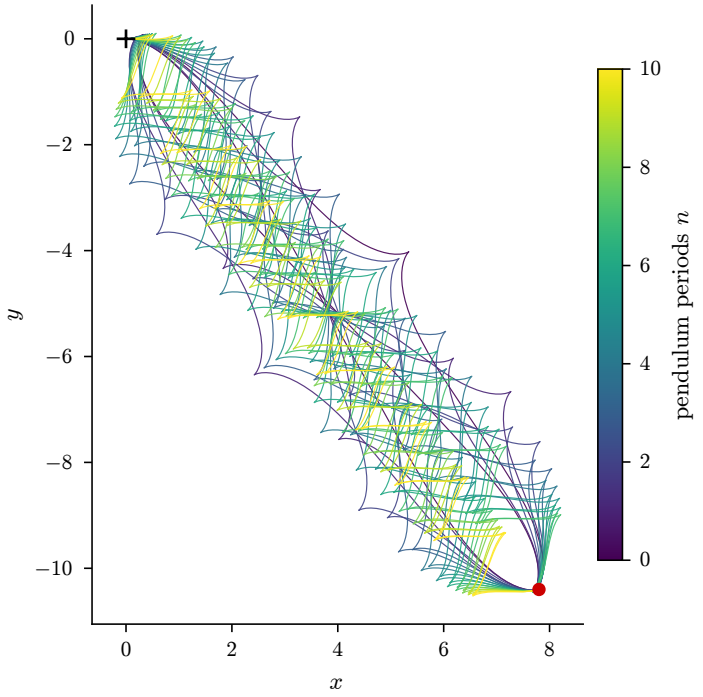


Figure 4: The 37 rotating geodesics from the identity (cross) to $(39/5, -52/5, 3/10)$ (dot), coloured by the period index n of (4), the number of pendulum periods.

Open questions. Effective values of the interior and boundary thresholds R_0 in Theorems 1.1–1.3; finite-target counts at strict heading in arbitrary spatial directions, beyond the all-radius half-heading ray, exterior-sector results, the fixed target line in Proposition 6.22, and the exact catalogues at headings zero and π; certified evaluation of further exterior angular thresholds, uniqueness of their minimizing parameters and the local singularity types beyond the first threshold at $w = 3/4$; continuation of the first fold window beyond the certified cap in Corollary 6.18; time comparisons between arbitrary branches in different positive periods at strict heading, beyond the complete fixed-line ordering of Corollary 6.24; time comparisons between separate sublevel components within one signed period family, beyond Theorem 6.23; finite-target comparisons on bounded scaling components, which Proposition 6.30 shows cannot be covered by continuation to infinity; explicit starting scales in Theorems 6.27–6.28, the simultaneous critical degeneracy $L = \kappa = 0$ (one certified instance, experiment E138 in the repository, is not treated here), and a complete critical transition diagram; effective spatial neighborhoods preserving these counts and orders; and the incidences of the higher conjugate sheets [12] with the fibers counted here.

AI assistance. This paper was developed with OpenAI Codex and Claude (Anthropic); the author is responsible for the content.

A Substitutions in the proof of Lemma 11.4

With $s = \sin A$, $c = \cos A$, $d = \sqrt{1 - k^2 s^2}$ and $h = \sqrt{1 - k^2}$, the closed forms in the proof of Lemma 11.4 are polynomial identities after the substitutions below. They were verified by computer algebra in the script `research/e33_zero_heading_count.py` of the repository, which also checks the derivative identities (115), (116), (117) and (120) modulo the relations $s^2 + c^2 = 1$, $d^2 + k^2 s^2 = 1$ and $h^2 + k^2 = 1$.

Positive amplitude. With $z = d/e$ and $\tan A = rz$,

$$s = \frac{rz}{\sqrt{1 + r^2 z^2}}, \qquad c = \frac{1}{\sqrt{1 + r^2 z^2}}, \qquad e = \frac{d}{z}, \qquad k^2 = \frac{(1 - d^2)(1 + r^2 z^2)}{r^2 z^2},$$

and $I = d^3 F_R / [r z^3 \sqrt{1 + r^2 z^2}]$. The boundary $d^2 = c^2$ is $d^2 = 1/(1 + r^2 z^2)$, on which F_R equals the displayed rational function of (r, z). On the boundary $d^2 = 4z^2$, with $w = z^2$ and $\xi = rw$, one has $w F_R = P(w, \xi)$. The Bernstein coefficients are those of $P_w(1/4 - \xi^2, \xi)$, a polynomial of degree four in ξ, on $0 \le 2\xi \le 1$.

Negative amplitude. With $\chi = -A$, $z = d/e$ and $\tan \chi = rz$,

$$s = -\frac{rz}{\sqrt{1 + r^2 z^2}}, \qquad c = \frac{1}{\sqrt{1 + r^2 z^2}}, \qquad e = \frac{d}{z}, \qquad h^2 = \frac{d^2(1 + r^2 z^2) - 1}{r^2 z^2},$$

and $I = d^3 G / [z^3 (1 + r^2 z^2)^{3/2}]$, with G evaluated at this h^2. In the comparison of G with B_0, $q/e = (1 + rz^2)/(1 - r)$ and

$$\frac{B_0}{e^4} = \frac{q}{e}(1 - z^2) - h^2 z^2 \Big(1 + \frac{q}{e}\Big)^2.$$

Finally, the lower bound in (118) is the identity $(2n - \tfrac{1}{2})\{(2n - \tfrac{1}{2})^2 - \tfrac{7}{4}\} - \tfrac{3}{4}n = (n - 1)(32n^2 + 8n - 3)/4$.

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